Market risk

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About the course

- Lectures: 16.5 h, partly online.
- Tutorials: 15 h, theoretical exercises and implementation of methods related to market risk analysis presented in lectures (bring your computer). Exercises (theoretical then practical) must be prepared by students before the related course.
- Evaluation:
  - 50%: a project by groups of two students (**the content of the project is based on questions on which you will work during the five tutorials**; the exact subject of the project will be announced at the end of the course; expected: report in LaTeX and code),
  - 50%: written exam on a date to be determined (duration: 1h30).
- TD teachers: Nicolas Pesci, Julie Gamain, and Cassandre Lefèvre.
- Questions: primarily during or right after lectures, otherwise by appointment.
- Course material: the present file is rich and contains:
  - slightly more than what I will present during the lectures,
  - useful details for a professional perspective (risk management, trading, asset management, regulator...),
  - some elements that curious minds should at least read but also many very important elements (if you want to work in the financial industry) that you must know perfectly.

So don't be afraid by the size of the file!

What is risk?

In general, it is an exposure to chance, fortuity.

- In finance, we focus on negative consequences (loss).
- What is chance (*hasard* in French is rarely translated by *hazard* in finance which is closer to danger; for example, the hazard rate may be compared to a default rate)? Providence, fortuitous event, convergence of two Cournot-style causal chains, black swan?
- How can one model chance? By random variables (possible since Kolmogorov's works of 1933)? chaotic systems?

Several perceptions of risk:

- **Trader**: loss risk of a portfolio $\longrightarrow$ but a loss is possible.
- **Insurer**: risk of the occurrence of an event (death, fire, ...) treated statistically $\longrightarrow$ a margin of error is included in the insurance premium in order to have a probability close to 1 of having a gain if the customers are sufficiently numerous and representative of the population.
- **Asset Liability Management**: risk that the asset will pay less than the cost of the liability $\longrightarrow$ a loss is not possible.

What is risk?

So a portfolio optimization in asset management looks like this:

$$\min_{\omega} \left\{\omega' \Sigma \omega\right\} \text{ provided that } \sum_i \omega_i r_i \geq r,$$

where the ωs are weights, Σ is the covariance matrix, r a target return, and r_i the expected return of the security i.
In ALM, the optimization problem rather looks like this:

$$\max_{\omega} \left\{\mathbb{P}\left(\sum_i \omega_i R_i \geq r\right)\right\},$$

where R_i is a random variable ($\mathbb{E}[R_i] = r_i$). If $R = (R_1, ..., R_n)'$ is a Gaussian vector (and not only a vector of Gaussian variables) with mean $\mu = (r_1, ..., r_n)'$, then the problem is similar to maximizing

$$\Phi\left(\frac{\omega'\mu - r}{\sqrt{\omega'\Sigma\omega}}\right),$$

where Φ is the standard Gaussian cumulative distribution function and where the covariance matrix Σ appears.

What is risk?

Different natures of risk:

- **Market risk**,
- Credit risk (including counterparty risk),
- **Liquidity risk** (considered as market risk by Basel committee),
- Operational risk: risk of losses due to bad procedures (no control, poorly specified models, errors, fraud, computer failures, ...) or external risks (flood, fire, ... but also country / political / legal risk).

Also other risks whose classification is vague: meteorological risk (for the insurer but also can induce a market risk if impact on a company's performance). Other risks that may have an indirect impact on market risk: reputation risk (of an issuer), systemic risk.

What is risk?

After Kloman (1990) :

> *Risk management is a discipline for living with the possibility that future events may cause adverse effects.*

So the risk can be useful (no return without risk), but we can also want to transfer it, especially with derivatives.

A long history

- **17th century**: use of put and call options, forward contracts, and short selling (Amsterdam, tulip mania).
- **1952**: securities are not only judged anymore on their expected return but on the risk / return ratio (portfolio theory, Markowitz). The risk is the standard deviation. Refinements of the efficient border followed with asymmetric ratios instead of the Sharpe ratio, for example, etc.
- **1970**: abolition of the Bretton-Woods system, exchange rates are no longer fixed and become a risk factor to be covered.
- **1973**: new method of valuation and replication of calls / puts with the Black-Scholes-Merton formula.
- **1973**: creation of the Chicago Options Exchange (less than 1,000 options the first trading day, more than one million every day in 1995).
- **1995**: the bankruptcy of the Barings because of Nick Leeson's straddles (short call and short put of the same strike: premium stability), reveals to all the leverage effects allowed by derivatives.
- **2007**: subprime crisis, the role of securitization (*titrisation*) in the transfer of risk.

Regulation

Basel I Agreement (1988): Regulatory capital ratio, mainly to limit the spread of credit risk to other banks. Assets are risk-weighted: 0% for cash and treasuries, 20% for mortgage-backed securities rated AAA, 50% for other mortgages (*hypothèques*), for example on home loans, 100% for corporate loans. International banks must have a capital of at least 8% of these RWA (risk-weighted assets).

Criticism: The credit risk is not taken into account, the regulator only considers the nominal of the at-risk securities.

1996 Amendment: incentive to calculate a VaR (value-at-risk, born in 1993).

Regulation

Basel II Agreement (2004): 3 pillars (I: risk measurement and minimum capital requirement; II: supervisory system and risk control; III: public display of loss data and of risk management methods).

Capital requirement: In addition to credit and counterparty risk, market risk and operational risk arise. Capital must now be greater than 8% of 85% of the credit risk, $+5\%$ of market risk, $+10\%$ of operational risk. Credit risk now includes probability of default and LGD (loss on default).

Criticism: dependence on US rating agencies (S& P, Moody's); agreement signed by many countries (including the USA) but never applied by the Americans.

Regulation

Basel III Agreement (2010): the 2007 crisis could be explained by excessive growth in off-balance sheet banking (in particular derivatives) and insufficient capital to cope with liquidity risk.

Decisions :

- Implementation of liquidity ratios: the Liquidity Coverage Ratio is supposed to allow banks to face a liquidity crisis for one month. Liquidity reserves (anything considered liquid: cash, treasuries that can be refinanced in the central bank) must be greater than leakage due to sudden illiquidity. Depending on the type of asset is therefore defined a leak/loss rate (0% for the cash, 50% for loans to individuals, between 25 and 75% for loans to private companies, 100% for interbank loans).
- Adding a cushion to the solvency ratio to take into account the off-balance sheet (*le hors bilan*).

FRTB ('Basel IV') Agreement (2019, postponed in 2022): ratios are sharper, use of the expected shortfall instead of the VaR, risk measures must include liquidity risk.

Consequences of regulation

- During the tulip crisis, prohibition of short selling has favored the growth of a bubble (only positive views can be expressed).
- In 1933, the Glass-Steagall Act prohibited US commercial banks from subscribing to any insurance policy. This has indirectly favored the emergence of investment banks.
- The Basel agreements transform the model of many banks that originate but hold less. The risk nevertheless exists and is transferred to other investors.

Risk management

- Identify, measure, and limit risk.
- Challenges for measuring risk: few data (EVT can extrapolate a 99.9% VaR, even if only 100 historical observations), measure on which scale (multifrequency aspects), aggregate risks (multidimensional aspect), which measure to choose (VaR, volatility, expected shortfall, other?).
- Incorporate risk (in particular liquidity risk) in investment decisions.
- Hedge risk with derivatives.

Notes de crédit

- Indépendance des agences de notation ?
- La probabilité de défaut historique est plus faible que celle qui est valorisée et donc observée dans le prix des obligations (prend en compte d'autres choses que le crédit comme la liquidité ? méfiance des investisseurs pour les notations ?), le ratio croissant de 1.2 pour du B à 16.8 pour du Aaa (source : Hull).
- IG de Aaa (ou AAA) à Baa (ou BBB) et HY en-dessous.
- CreditMetrics : migrations de ratings.

Notes de crédit

Table 8.2. Probabilities of migrating from one rating quality to another within one year. Source: Standard & Poor's CreditWeek (15 April 1996).

Initial rating	Rating at year-end (%)							
	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	90.81	8.33	0.68	0.06	0.12	0.00	0.00	0.00
AA	0.70	90.65	7.79	0.64	0.06	0.14	0.02	0.00
A	0.09	2.27	91.05	5.52	0.74	0.26	0.01	0.06
BBB	0.02	0.33	5.95	86.93	5.30	1.17	1.12	0.18
BB	0.03	0.14	0.67	7.73	80.53	8.84	1.00	1.06
B	0.00	0.11	0.24	0.43	6.48	83.46	4.07	5.20
CCC	0.22	0.00	0.22	1.30	2.38	11.24	64.86	19.79

Figure: Source : McNeil, Frey, Embrechts.

Notes de crédit

- Tableaux de probabilités historique par note et maturité. Probabilité cumulée suit souvent loi log-logistique : $F(x) = \frac{x^\beta}{\alpha^\beta + x^\beta}$, avec $x > 0$, $\alpha > 0$ et $\beta > 0$.
- Probabilité de défaut non-conditionnelle entre s et t : $F(t) - F(s)$.
- Probabilité de défaut conditionnelle entre s et t, loi de Bayes : proba(défaut en t et pas en s)/proba(pas défaut en s), soit $(F(t) - F(s))/(1 - F(s))$, supérieure à la proba non-conditionnelle.
- Intensité de défaut λ est la proba conditionnelle rapportée à $t - s$. Pour intervalle de temps faible, on obtient $\lambda(t)dt = dF(t)/(1 - F(t))$, donc
$$1 - F(t) = exp\left(-\int_0^t \lambda(\tau)d\tau\right).$$
- Pour un titre de T années, avec $F(t) = t/T$, le défaut est certain. La proba non-conditionnelle attribue une probabilité de $1/T$ chaque année. La proba conditionnelle est croissante et vaut pour l'année t, $(T + 1 - t)^{-1}$, qui vaut 1 en T. Un proba conditionnelle annuelle de $1/T$ équivaut à $F(1) = 1/T$, $F(2) = 2/T - 1/T^2$ et plus généralement à $F(t) = 1/T + F(t-1)(1 - 1/T)$.

Notes de crédit

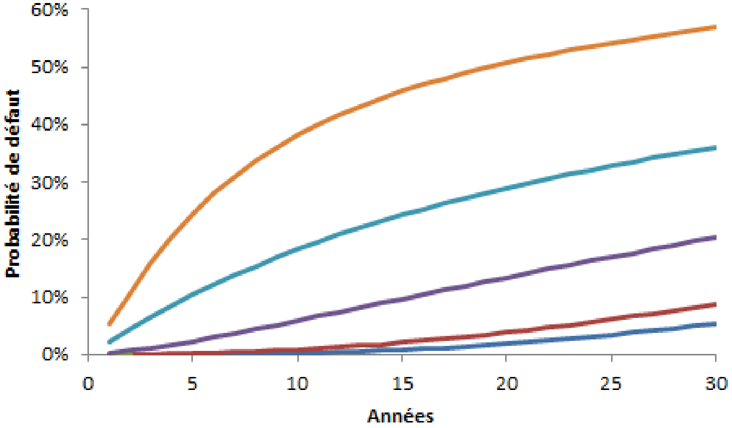


Figure: Probabilité de défaut cumulée pour notes AAA, AA, A, BBB, BB, B.

Notes de crédit

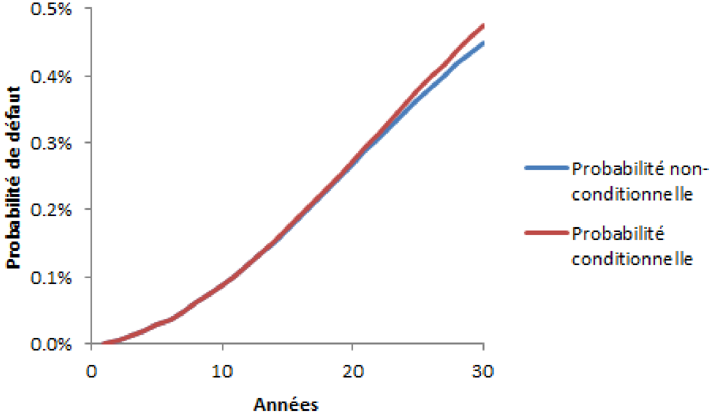


Figure: Densité de probabilité de défaut AAA.

Notes de crédit

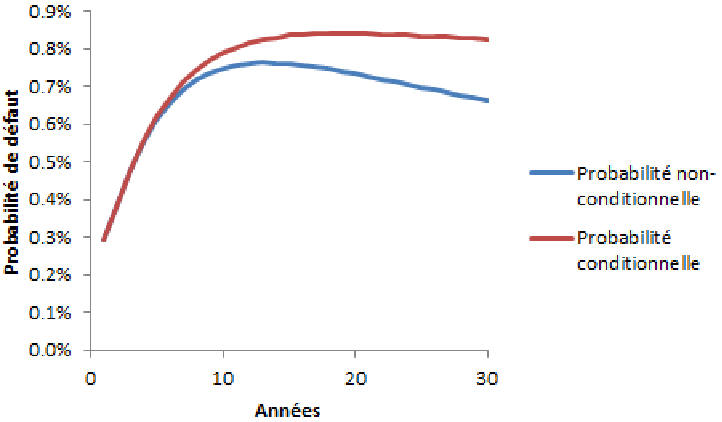


Figure: Densité de probabilité de défaut BBB.

Notes de crédit

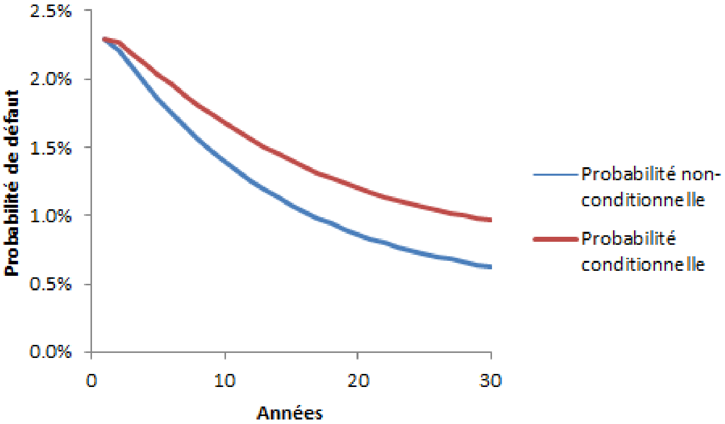


Figure: Densité de probabilité de défaut BB.

Recouvrement

- Actions en justice pour recouvrement.
- Délai de recouvrement.
- Impact de la séniorité (1982-2003) :
  - senior garantie : 51.6%
  - senior non garantie : 36.1%
  - senior subordonnée : 32.5%
  - subordonnée : 31.1%
  - junior subordonnée : 24.5%
- Impact note de crédit (1982-2003) : $R = 50.3\% - 6.3 \times P$, où R est le taux de recouvrement moyen et P la proba de défaut moyenne.

Valorisation d'une obligation

- Deux versions : risque de crédit au numérateur ou au dénominateur :

$$price_t = \sum_{i=1}^{n} \frac{\mathbb{E}_t[cashflow_{t_i}]}{(1 + r_{t_i})^{t_i - t}}$$

ou

$$price_t = \sum_{i=1}^{n} \frac{cashflow_{t_i}}{(1 + r_{t_i} + s)^{t_i - t}}$$

- L'intensité de défaut moyenne est égale à $s/(1 - R)$, où s est le spread et R le taux de recouvrement.

Définition du risque opérationnel

Dans le document de Bâle II : "*Operational risk is defined as the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events. This definition includes legal risk, but excludes strategic and reputational risk.* "

Exemples : fraudes internes ou externes, pertes à cause d'erreurs ou de défaillances informatiques, erreurs dans le règlement des transactions, litiges et pertes dues à évènements externes (inondation, incendie, tremblement de terre, braquage).

Les mauvais choix de management (de portefeuille ou de l'institution) n'en font pas partie.

Un cas frappant, la Barings, cumule insuffisance des processus de contrôle, fraude interne et évènement externe (tremblement de terre de Kobe, lequel provoque un décalage du marché).

Mesure du risque opérationnel

Deux approches sont possibles :

- Une mesure historique avec les techniques propres aux séries temporelles, ce qui est préconisé pour tout ce qui est exogène aux portefeuilles : fraudes (*a priori* impossible à prévoir), risques climatiques (risque plus prévisible mais demande des compétences non-financières).
- Une description du mécanisme (plus spécifiquement pour le risque de modèle) avec une mesure qui n'est pas nécessairement historique et dépend par exemple des la natures des actifs gérés.

Mesure du risque opérationnel

Dans les accords de Bâle II, il y a trois options pour calculer le risque opérationnel selon l'exposition de chaque banque à ce type de risque (en backtest, la VaR de risque opérationnel à 99.9% doit être vérifiée à un horizon un an) :

- **Basic Indicator Approach** : proportion des revenus bruts positifs moyens des trois dernières années.
- **Standardized Approach** : proche de l'indicateur de base mais avec disjonction en *business lines*.
- **Advanced Measurement Approaches** : pour les banques ayant une activité internationale. Le comité exige une historique de pertes internes de 5 ans minimum (3 ans si méthode nouvellement mise en place).

Basic Indicator and Standardized Approaches

- **Basic Indicator Approach** : Capital requis en année t est proportion $\alpha = 15\%$ des revenus bruts positifs moyens des trois dernières années :

$$\frac{1}{Z_t}\sum_{i=1}^{3}\alpha \max(R^{t-i}, 0),$$

où R^{t-i} est le revenu brut de l'institution en année $t-i$ et $Z_t = \sum_{i=1}^{3} \mathbb{1}_{\{R^{t-i}>0\}}$.

- **Standardized Approach** : 8 *business lines* avec une proportion de capital requis β_j propre à chacune :

$$\frac{1}{3}\sum_{i=1}^{3}\max\left(\sum_{j=1}^{8}\beta_j R_j^{t-i}, 0\right).$$

La division par 3 plutôt que Z_t, ainsi que le fait que des performances négatives sur une ligne de β_j élevé peut compenser les gains d'autres lignes, est une incitation à passer à cette approche.

Standardized Approache

Business lines :

$j = 1$ *corporate finance* : $\beta_1 = 18\%$;

$j = 2$ *trading and sales* : $\beta_2 = 18\%$;

$j = 3$ *retail banking* : $\beta_3 = 12\%$;

$j = 4$ *commercial banking* : $\beta_4 = 15\%$;

$j = 5$ *payment and settlement* : $\beta_5 = 18\%$;

$j = 6$ *agency services* : $\beta_6 = 15\%$;

$j = 7$ *asset management* : $\beta_7 = 12\%$;

$j = 8$ *retail brokerage* : $\beta_8 = 12\%$.

Advanced Measurement Approaches

Dans le document de Bâle II : "*Given the continuing evolution of analytical approaches for operational risk, the Committee is not specifying the approach or distributional assumptions used to generate the operational risk measure for regulatory capital purposes. However,* **a bank must be able to demonstrate that its approach captures potentially severe tail loss events***. Whatever approach is used, a bank must demonstrate that its operational risk measure meets a soundness standard comparable to that of the internal ratings-based approach for credit risk (comparable to a one year holding period and the 99.9 percent confidence interval).*"

De plus, les pertes doivent êtres réparties en 8 lignes, comme dans l'approche standard, et pour chaque ligne en **7 types de pertes** : internal fraud; external fraud; employment practices & workplace safety; clients, products & business practices; damage to physical assets; business disruption & system failures; execution, delivery & process management.

Comité impose historique haute fréquence pour événements internes et plus basse fréquence pour événements externes, avec corrélation entre types de pertes (en l'absence de corrélation, simple somme, ce qui correspond à borne haute de Fréchet-Hoeffding, cas comonotonique : hypothèse la plus conservatrice).

Advanced Measurement Approaches

- Distribution jointe en général inconnue.
- On filtre les pertes inférieures à un seuil.
- La perte suit une variable aléatoire et la fréquence aussi.

Pour chaque business line, on est donc ramené au calcul de quelque chose comme

$$VaR_{0.999}\left(\sum_{k=1}^{N} X_k^j\right),$$

où le nombre d'occurrences N est aléatoire et la k-ème perte dans la business line j, X_k^j (la sévérité), aussi.

Sur historiques 1992-2001, McNeil, Frey et Embrechts citent les faits stylisés suivants :

- sévérité des pertes distribuée selon loi à queue épaisse (EVT, paramètre de queue ξ de 0.85 à 1.39 selon *business lines*) ;
- les pertes arrivent à des temps aléatoire ;
- la fréquence des pertes peut varier substantiellement dans le temps (même si cela est partiellement dû au fait que le monitoring est plus complet ces dernières années)

Par ailleurs, effets cycliques (fraude liée à conjoncture économique ?) et corrélation à variables économiques (erreur de back-office liée aux volumes échangés), mais on suppose que toutes les sévérités sont iid.

Advanced Measurement Approaches

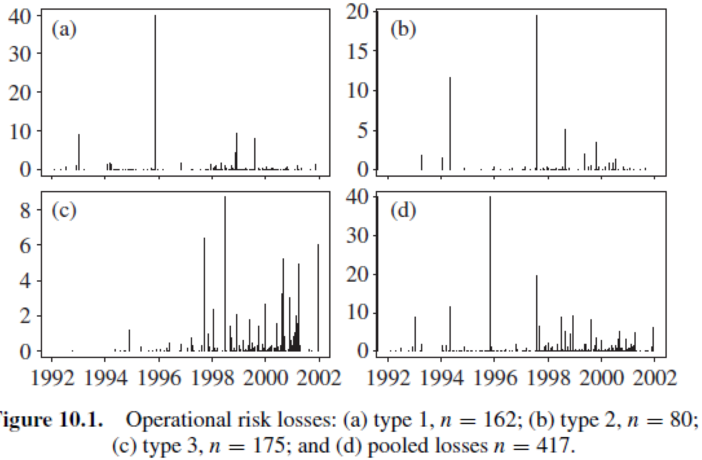


Figure: 3 types de pertes et dernier graphique somme. Source : McNeil, Frey et Embrechts.

Un cas particulier du risque opérationnel : le risque de modèle

Le risque de modèle peut provenir de plusieurs sources :

- le modèle utilisé est mal spécifié : les résidus sont distribués différemment de ce qui est supposé (utiliser modèles non-paramétriques, mais risque d'overfitting), un paramètre supplémentaire est nécessaire (mouvement brownien vs mouvement brownien fractionnaire), un facteur de risque n'est pas identifié ;
- il y a une incertitude dans l'estimation : présence de bruit (débruitage partiel), variation des paramètres dans le temps et impossibilité de prévoir leur valeur future ;
- même bien estimé, la sortie du modèle peut être incertaine : intervalles de confiance d'un Monte-Carlo, approximations numériques d'un schéma aux différences finies.

Hedging market risk using futures

We can use futures to hedge a risk. For example:

- if we own a barrel of oil at $S_0 =$19USD in t_0, we are exposed to a price variation in t_1: we may gain or lose $S_1 - S_0$ at t_1.
- if we just buy in t_0 a futures contract on the barrel maturing at t_1 with a futures price $F_0 =$19.25USD, we receive the barrel at the spot price S_1 (because the price of the futures contract converges towards the spot price, otherwise arbitrage) in t_1.
- If we own the barrel and sell a futures contract at time t_0, then at maturity t_1, the futures price F_1 equals the spot price S_1. The net payoff on the futures position is $F_1 - F_0 = S_1 - F_0$. Combined with owning the barrel, whose value at t_1 is S_1, the total value is $S_1 - (S_1 - F_0) = F_0$, which means we have effectively locked in the price $F_0 = 19.25\,\text{USD}$.

If the spot in t_1 is worth 15USD or 21USD, we will still be paid 19.25USD.

Drawbacks:

- Futures (listed in an organized market) generate daily flows (margin calls) and therefore require the immobilization of cash, while forwards (over-the-counter transactions) only generate a terminal flow but are more exposed to counterparty risk.
- The theoretical price of a future is $S_0 e^{r(t_1 - t_0)}$ (say 19.3USD). But hedging a risk is not free, there is an implicit cost (here $19.3 - 19.25$).

Hedging market risk using futures

Time	Underlying price	Cash flows for a futures with physical delivery	Cash flows for a futures with cash settlement
t_0	100	-10 (initial margin)	-10 (initial margin)
t_1	105	+5 (variation margin)	+5 (variation margin)
t_2	100	-5 (variation margin)	-5 (variation margin)
t_3	110	+10 (variation margin)	+10 (variation margin)
t_4	120	+10 (variation margin)	+10 (variation margin)
At maturity	120	-120 (pay spot price) + 10 (margin returned)	0 + 10 (margin returned)
Total cash flows	-	-100	+20

Table: Cash flows comparison for futures contracts (we are a buyer) with physical delivery versus cash settlement, including margin calls (and initial margin [here 10% of the spot: kind of security deposit] returned at the end). If physical delivery, we can sell immediately at the spot (120), thus leading to the same net result as a cash-settled futures: +20.

Hedging market risk using futures

The theoretical price of the futures is the expected value of the discounted spot price. In a world with zero interest rates (which doesn't really change much because you have to borrow at this rate to invest, which prevents you from making an arbitrage) where the market is efficient (no arbitrage opportunity: the expected value of the spot in t_1 is the spot in t_0), $F_0 = S_0$ (with a rate r and a maturity T, the futures price is $F_0 = S_0 e^{rT}$). This allows you to roll hedges. Indeed, the liquidity of futures is only ensured for low maturities, for example 3 months. Example of hedging the exchange rate risk (over-the-counter, because an organized market has never established itself):

- in January, we buy a 100GBP bond (obtained by converting 125EUR at our disposal) and we hedge the exchange rate risk with a forward exchange contract in euros for 100GBP=125EUR. Concretely:
  - we commit to giving 100GBP in April in exchange for 125EUR;
  - we spend 100GBP for the bond, which will pay 100GBP plus a coupon (we only hedge the capital) in October.
- in April, we roll the hedge: we receive 125EUR by the forward exchange that we exchange for 96.15GBP (new exchange rate: 100GBP=130EUR) and we give 100GBP; we then commit to giving 96.15GBP in July in exchange for 125EUR;

Hedging market risk using futures

- in July, we roll the hadge: we receive 125EUR by the forward exchange that we exchange for 113.64GBP (new exchange rate: 100GBP = 110EUR) and we give 96.15GBP; we then commit to giving 113.64GBP in October in exchange for 125EUR;
- in October, we receive 125EUR from the forward exchange and we give 113.64GBP; we also receive 100GBP and a coupon from the bond.

In summary, the synthetic flows (excluding coupons) are as follows (beware of the need for cash each quarter):

- -125EUR in January
- -3.85GBP in April
- +17.49GBP in July
- -13.64GBP + 125EUR in October

The sum of all these flows is indeed zero.

Hedging market risk using futures

Deviations from this ideal case exist, this is the basis risk, the basis being the spread between spot and future $S_t - F_t$:

- the asset is sometimes different from the underlying hedged (for example: index future vs. single-name security);
- date mismatch between maturity of the futures and desired transaction, the date of which may be unknown at t_0 (we receive the futures value before the ideal date or we must sell the contract on the transaction date, and the basis is added positively or negatively to our P&L);
- date mismatches can appear even if the transaction date is known at t_0 because contracts with the right maturity may not exist.

Hedging market risk using futures

If the goal is simply to minimize the variance of the hedged asset price (relevant in presence of basis risk), we can hedge only a portion h of the asset. We denote

- ΔS the random variation of the spot price during the hedging period;
- ΔF the random variation of the futures price during the hedging period;
- σ_S the standard deviation of ΔS;
- σ_F the standard deviation of ΔF;
- ρ the correlation between ΔS and ΔF.

We assume that we can historically estimate σ_S, σ_F and ρ. At the end of the hedging period, the profit is $\Delta S - h\Delta F$, whose variance is

$$v = \sigma_S^2 + h^2\sigma_F^2 - 2\rho h\sigma_S\sigma_F$$

which is minimal for h equal to $h^\star$ with

$$h^\star = \rho\frac{\sigma_S}{\sigma_F}.$$

So, by hedging with this ratio, the variance goes from σ_S^2 to $(1-\rho^2)\sigma_S^2$.

Hedging exchange rate risk with a swap

Possible hedge with a currency swap:

- At $t = 0$, we give 100 EUR and receive $100S_0$ GBP where S_0 is the exchange rate at $t = 0$.
- At $t = 1$, we pay the British domestic rate, say $2\% \times 100S_0$ GBP and we receive the European rate, say $1\% \times 100$ EUR.
- At $t = 2$ we pay the British rate (different from $t = 1$ if variable rate) and the principal of $100S_0$ GBP against the European rate and the principal of 100 EUR;

If we bought a British bond paying a variable coupon in GBP and we enter into the swap described, we have transformed the bond into a security in EUR paying a variable coupon indexed to the European rate.
Swaps are over-the-counter transactions, so there is a counterparty risk that can be taken into account in the valuation. Without this, the price is zero at $t = 0$ and is worth at t the difference in the price of the two bonds constituting the swap, one of which is converted into the currency of the other at the spot exchange rate S_t.

VaR or sensitivity?

- The Greeks are used to express the sensitivity of a portfolio to different risk factors.
- These factors are limited for a stock option (the stock price, the strike price, the time before maturity, the volatility of the underlying stock, the interest rate, the dividends).
- But for an entire bank, they can be numerous. This is why we often prefer VaR, which is more synthetic but has a different philosophy since it replaces the sensitivity to risk factors with an assumption on their behavior.
- Be careful, sensitivity is a local linear measure and can poorly describe certain non-linear phenomena when a shock occurs.
- Antifragility concept (Taleb):
  - fragile portfolio: evolves negatively with shock;
  - robust portfolio: insensitive to shock;
  - "resilient" portfolio: evolves negatively with shock then returns to its pre-shock value;
  - antifragile portfolio: evolves positively with shock.

Hedging equity risk for options

Option replication with a portfolio of value V_t invested in δ_t stocks of price S_t, while the rest invested in a zero-coupon bond at rate r_t. Endogenous value variation:

$$dV_t = \delta_t dS_t + (V - \delta_t S_t) r_t dt.$$

V_t is a function of t and S_t: $V(t, S_t)$. If S_t is an Ito process, i.e.

$$dS_t = \mu_t dt + \sigma_t dW_t,$$

with W_t a Brownian motion (special case: Black-Scholes model with $\mu_t = \mu S_t$ and $\sigma_t = \sigma S_t$, which leads to $S_t = S_0 \exp\left(\sigma B_t + \mu t - \frac{1}{2}\sigma^2 t\right)$), we can apply Ito's lemma if V is $\mathcal{C}^2(\mathbb{R}_+ \times \mathbb{R}, \mathbb{R})$:

$$d(V(t, S_t)) = \frac{\partial V}{\partial t}(t, S_t)dt + \frac{\partial V}{\partial x}(t, S_t)dS_t + \frac{1}{2}\frac{\partial^2 V}{\partial x^2}(t, S_t)\sigma_t^2 dt.$$

Thus, the important result of Black, Scholes and Merton concerning the hedging of equity risk is that $\delta_t = \frac{\partial V}{\partial x}(t, S_t)$: if we have sold an option and want to hedge the equity risk of our position, we must buy a quantity $\frac{\partial V}{\partial x}(t, S_t)$ of shares.

Hedging equity risk for options

In the particular case of a European call with S_t a geometric Brownian motion and with a series of assumptions (no arbitrage opportunity, continuous time, short sales allowed, no transaction costs, no dividends, and perfectly divisible assets), we obtain the call price as the expected discounted payoff under risk-neutral probability:

$$C(S_0, K, r, T, \sigma) = S_0\mathcal{N}(d_1) - Ke^{-rT}\mathcal{N}(d_2),$$

with $d_1 = \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T\right]$ and $d_2 = d_1 - \sigma\sqrt{T}$. The Greeks (sensitivity to risk factors, which can possibly be hedged by a replicating portfolio in which the weights invested in risk factors are equal to these Greeks) are then:

- $\delta = \partial C/\partial S = \mathcal{N}(d_1)$;
- $\Gamma = \partial^2 C/\partial S^2 = \mathcal{N}'(d_1)/S_0\sigma\sqrt{T} > 0$;
- etc. with theta $\Theta = -\partial C/\partial T$ (thus derivative on remaining life), the vega $\mathcal{V} = \partial C/\partial\sigma$, the rho $\rho = \partial C/\partial r$ and $\partial C/\partial K$.

Comparing the dt parts of Ito's formula applied to C and the dV_t formula of the replicating portfolio (so $V_t = C_t$), we have the PDE:

$$\frac{1}{2}\sigma^2 S_0^2\Gamma + rS_0\delta - rC + \Theta = 0.$$

Hedging of a European call

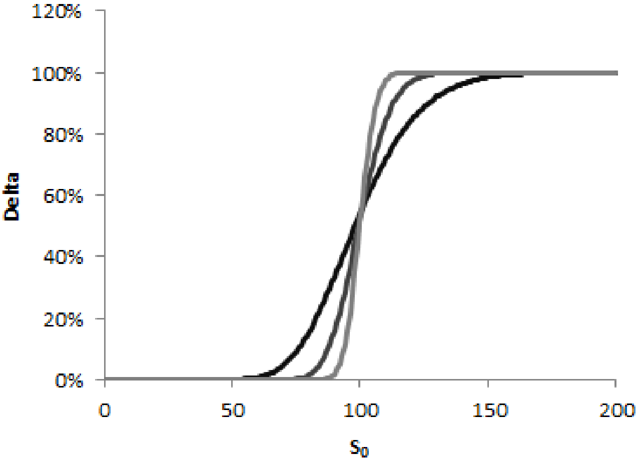


Figure: Delta for $\sigma\sqrt{T} = 0.05$ (light grey), then 0.1, then 0.2 (black). $K = 100$, $r = 0$.

Hedging of a European call

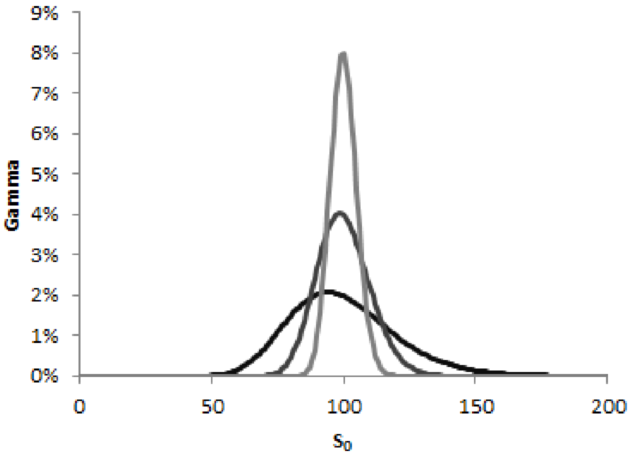


Figure: Gamma for $\sigma\sqrt{T} = 0.05$ (light grey), then 0.1, then 0.2 (black). $K = 100$, $r = 0$.

Hedging of a European call

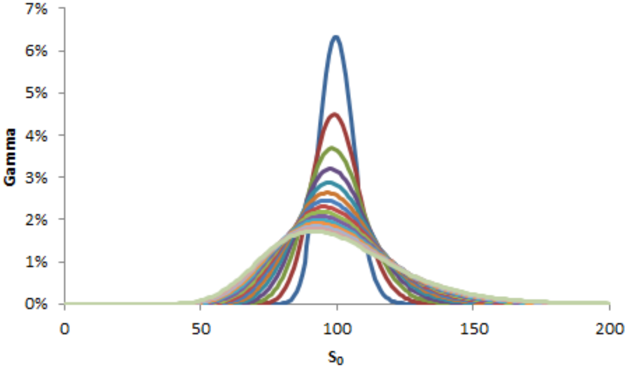


Figure: Gamma fo $\sigma = 0.2$, $K = 100$, $r = 0$. T is between 0.1 (highest value when $S_0 = 100$) and 1.5.

Hedging of a European call

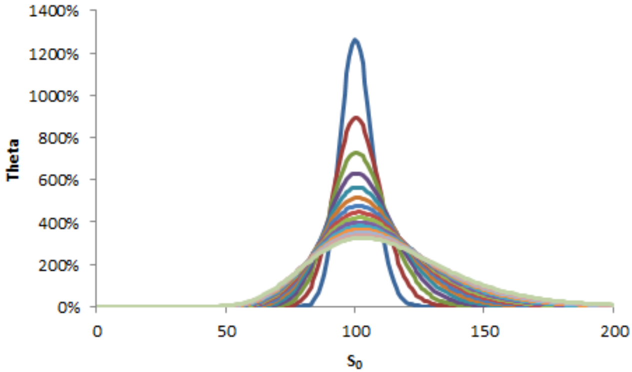


Figure: Theta for $\sigma = 0.2$, $K = 100$, $r = 0$. T is between 0.1 (highest value when $S_0 = 100$) and 1.5. Obtained by PDE.

Sensitivity to rate variations

The price of a bond is written as

$$P = \sum_{i=1}^{n} c_i e^{-y t_i},$$

Where the c_i are the future cash flows and y is the actuarial rate of the bond in continuous compounding. The **(McCauley) duration** is the average date on which the value of the bond is paid back (time weighted by proportion of value):

$$D = \frac{\sum_{i=1}^{n} t_i c_i e^{-y t_i}}{P}.$$

If the actuarial rate changes by Δy, the price changes by ΔP, which is, by linear approximation, $\Delta P \approx \frac{\partial P}{\partial y} \Delta y$, so:

$$\frac{\Delta P}{P} \approx -D \Delta y.$$

Sensitivity to rate variations

- If the actuarial rate is compounded m times per year, the price is:

$$P = \sum_{i=1}^{n} \frac{c_i}{(1+y/m)^{mt_i}}.$$

Therefore $\frac{\partial P}{\partial y}$ becomes equal to $-(1+y/m)^{-1}\sum_{i=1}^{n} t_i \frac{c_i}{(1+y/m)^{mt_i}}$ and:

$$\frac{\Delta P}{P} \approx -D^\star \Delta y,$$

where $D^\star = D/(1+y/m)$ is the **modified duration**, with limit D when $m \to \infty$.

- We can also adjust the sensitivity with the **convexity**, C :

$$C = \frac{1}{P}\frac{\partial^2 P}{\partial y^2} = \frac{\sum_{i=1}^{n} t_i^2 c_i e^{-yt_i}}{P}.$$

Then, by a Taylor expansion of order 2:

$$\frac{\Delta P}{P} \approx -D\Delta y + \frac{1}{2}C(\Delta y)^2.$$

Sensitivity to rate variations

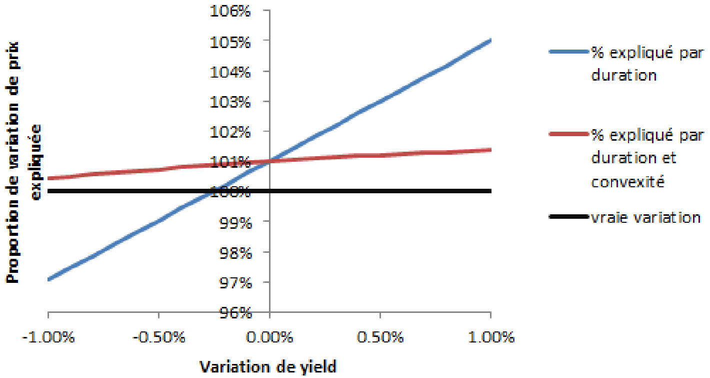


Figure: 7-year bond, fixed coupon 1% per year, initial price 100 (therefore initial yield of 1%, duration 6.80 years, convexity 47.03 years^2). Curves not defined for a yield variation of 0; extension with value 100% at 0 gives discontinuous curves.

Sensitivity to rate variations

If we decompose the actuarial rate into risk-free rate and spread (containing credit risk, liquidity risk, etc.), we can decompose into rate variation and spread variation, thus defining a rate duration (and convexity) and a spread duration (and convexity). This leads to a few remarks:

- We can only hedge the rate risk for example and remain exposed to the spread risk.
- The hedge of the rate risk is done by selling *pure* rate products (e.g. Schatz [2 years], Bobl [5 years], Bund [10 years]) of the same duration and convexity.
- Such a hedge actually only covers against translations of the yield curve: in fact, we write, in t_i, $r_i + s$ instead of the constant y, so the rate duration is understood as

$$-\frac{1}{P}\frac{\partial}{\partial\varepsilon}\left(\sum_{i=1}^{n} c_i(1+(r_i+\varepsilon)+s)^{-t_i}\right)\Bigg|_{\varepsilon=0}.$$

  One solution is **gap management** which consists of hedging against different rate variations on several predefined maturity segments (with swaps, futures, etc.).

Sensitivity to rate variations

- ...
- For a floating-rate bond, $c_i = f_i$, forward rate, equal by absence of arbitrage opportunity to
$$f_i = -1 + (1 + r_i)^i / (1 + r_{i-1})^{i-1}.$$
We show that a translation of rates has an identical impact on f_i. Thus, a translation of rates will have an identical impact on the price return: the sensitivity of the price to a translation of rates is zero. Therefore the rate duration is zero.
- The relative variation of spreads seems *more invariant* than its absolute variation, which led to the introduction of the DTS (duration times spread), with the approximation (if rate risk is covered elsewhere): $(\Delta P/P) = -DTS(\Delta s/s)$.

How to define a risk measure?

- $X : \Omega \longrightarrow \mathbb{R}$ is a random **loss**.
- We focus on finite losses, that is the set $\mathcal{X}$ of random variables X such that $\|X\|_\infty < \infty$, where $\|X\|_\infty = \sup_{\omega\in\Omega}|X(\omega)|$.
- $\rho : \mathcal{X} \longrightarrow \mathbb{R}$ will be a capital requirement (or risk measure) associated to the random loss X.
- $F(x) = \mathbb{P}(X \leq x)$ is the cumulative distribution function of X.
- F is not necessarily invertible. Why? We will use its generalized inverse:

$$F^{-1}(r) = \inf\{x \in \mathbb{R}, F(x) \geq r\}.$$

  $F^{-1}(r)$ is the quantile of X of probability r.

Definition

ρ *is a* **monetary risk measure** *if:*

1. *it is* **monotonic***:* $X \leq Y \Rightarrow \rho(X) \leq \rho(Y)$,
2. *it is* **invariant by translation***:* $\forall m \in \mathbb{R}$, $\rho(X + m) = \rho(X) + m$.

Volatility is **not** a monetary risk measure.

What are the properties of a risk measure?

Proposition

If ρ is a monetary risk measure, it is Lipschitz:

$$|\rho(X) - \rho(Y)| \leq \|X - Y\|_\infty.$$

Proof:

- By definition of an infinite norm: $X - Y \leq \|X - Y\|_\infty$.
- By monotony, we get $\rho(X) \leq \rho(Y + \|X - Y\|_\infty)$.
- By invariance by translation, it becomes $\rho(X) \leq \rho(Y) + \|X - Y\|_\infty$.
- So, $\rho(X) - \rho(Y) \leq \|X - Y\|_\infty$ and, by symmetry in X and Y, we also have $\rho(Y) - \rho(X) \leq \|X - Y\|_\infty$.

The financial intuition behind

- The difference of risk measure between two assets is lower than the maximal difference of loss the two assets can simultaneously encounter.
- For example, if stock 1 has a VaR of 1% and stock 2 a VaR of 0.8%, the maximal loss of a portfolio composed of stock 1 (long) minus stock 2 (short) is above 0.2%.
- What about the risk measure of the portfolio, $\rho(X - Y)$, compared to $\rho(X) - \rho(Y)$?

Coherent risk measure

Definition

The risk measure μ is **coherent** *if it satisfies all the following properties:*

- **Normality**: $\mu(0) = 0$ *(particular case of positive homogeneity).*
- **Monotonicity**: *if profits $X_1 \leq X_2$ almost surely, then $\mu(X_1) \geq \mu(X_2)$.*
- **Sub-additivity** *(diversification principle): $\mu(X_1 + X_2) \leq \mu(X_1) + \mu(X_2)$.*
- **Positive homogeneity**: *if $\lambda \geq 0$, then $\mu(\lambda X) = \lambda\mu(X)$.*
- **Invariance by translation**: *If p is a deterministic profit, then $\mu(X + p) = \mu(X) - p$.*

- In other words, a coherent risk measure is a monetary risk measure with sub-additivity and positive homogeneity.
- The notion of coherence may be soften by the introduction of convexity instead of sub-additivity and positive homogeneity: if $\lambda \in [0, 1]$, then $\mu(\lambda X_1 + (1 - \lambda)X_2) \leq \lambda\mu(X_1) + (1 - \lambda)\mu(X_2)$.
- In particular, if μ is monetary and positively homogeneous, then μ convex $\Leftrightarrow$ μ sub-additive.

Spectral risk measure

Rationale: A spectral risk measure is a weighted average of the possible payoffs of a portfolio, with a larger weight for less favourable payoffs. Thereafter, the weight is ϕ.

Definition

Let X be the payoff of a portfolio and F_X its cdf. A spectral measure M_ϕ is a function in $\mathbb{R}$ such that there exists a function $\phi : [0,1] \to [0,1]$ such that:

$$M_\phi(X) = -\int_0^1 \phi(p) F_X^{-1}(p)dp,$$

where

1. *ϕ is non-increasing,*
2. *ϕ is right-continuous,*
3. *ϕ is integrable and $\int_0^1 \phi(p)dp = 1$*

- We interpret ϕ as a weight due to assumption 3, overweighting bad outcomes due to assumption 1.
- The VaR is not spectral due to assumption 2 (see its distortion measure, in next slides).

Spectral risk measure

- If we consider the distribution of the losses, F_{-X}, which is such that $F_X^{-1}(p) = -F_{-X}^{-1}(1-p)$, then $M_\phi(X) = \int_0^1 \phi(1-p)F_{-X}^{-1}(p)dp$, after a change of variable.
- The expected value is a spectral risk measure, with $\phi = 1$.

Proposition

Every spectral measure is monetary, sub-additive and positively homogeneous. In other words, it is a coherent risk measure.

For discrete observations of price returns, where each outcome is considered as equiprobable, we define a spectral measure as

$$M_\phi(X) = -\frac{1}{N}\sum_{s=1}^{N} \phi_s X_{s:N},$$

where

- $X_{1:N} \leq ... \leq X_{N:N}$ are the order statistics,
- $\forall s,\ \phi_s \geq 0$,
- $\sum_{s=1}^{N} \phi_s = 1$,
- ϕ_s is non-increasing.

Distortion risk measure

A distortion risk measure is a risk measure defined as a weighted sum of payoffs (or returns) of a portfolio, *whatever the weight*. In particular, a spectral risk measure is a distortion risk measure.

Definition

Let μ be a distortion function: $\mu : [0,1] \to [0,1]$ and μ is non-decreasing and surjective, so that $\mu(0) = 0$ and $\mu(1) = 1$.
The corresponding **distortion risk measure**, *for gains X of cdf F_X is*
$R_\mu(X) = \int_0^1 F_{-X}^{-1}(1-p)d\mu(p) = -\int_0^1 F_X^{-1}(p)d\mu(p)$.

Proposition

Let M_ϕ be a spectral risk measure and R_μ a distortion risk measure, with μ differentiable. Then:

$$M_\phi = R_\mu \Leftrightarrow \phi = \mu'.$$

In such a case and if ϕ is differentiable, then $\mu'' \leq 0$.

Theorem

The distortion μ is concave (overweight of greater losses) $\Leftrightarrow$ R_μ is a coherent risk measure.

Distortion risk measure

Usual risk measures:

- **Value at risk** with confidence α: distortion $\mu(x) = \mathbb{1}_{[1-\alpha,1]}(x)$, not differentiable, not concave: not spectral.
- **Expected shortfall** with confidence α: distortion $\mu(x) = \mathbb{1}_{[1-\alpha,1]}(x) + \mathbb{1}_{[0,1-\alpha)}(x)\frac{x}{1-\alpha}$, not differentiable but concave: coherent, and even spectral.
- **Expected loss**: distortion $\mu(x) = x$, so that $\mu' = 1 > 0$: spectral.

What about the properties of usual risk measures?

The VaR:

- is monetary,
- is positively homogeneous,
- is **not** necessarily convex: take losses X and Y iid Bernoulli(p); then

$$VaR_\alpha\left(\frac{X+Y}{2}\right) = \begin{cases} 1 & \text{if } \alpha \geq 1-p^2 \\ 0 & \text{if } \alpha \leq (1-p)^2 \\ 1/2 & \text{else,} \end{cases}$$

  whereas

$$\frac{1}{2}\left(VaR_\alpha(X) + VaR_\alpha(Y)\right) = \begin{cases} 1 & \text{if } \alpha \geq p \\ 0 & \text{else.} \end{cases}$$

- is therefore neither sub-additive nor coherent.

Expected shortfall

The expected shortfall is coherent and even spectral. That is why the regulator finally prefers this risk measure.

Inverse distribution function

Definition

Let X be a random variable (say a price return) and F be its cumulative distribution function: $F(x) = \mathbb{P}(X \leq x)$. The **generalized inverse** *of F is*

$$F^{-1}(p) = \inf\{x \in \mathbb{R}, F(x) \geq p\}.$$

- The generalized inverse is also called the quantile.
- If F is *continuous* and *strictly increasing*, the generalized inverse is the standard inverse.
- F^{-1} is nondecreasing.
- $F^{-1}(F(x)) \leq x$ and $F(F^{-1}(p)) \geq p$.
- If a random variable U is uniform in $[0, 1]$, then $F^{-1}(U)$ is distributed according to F. This is used to generate pseudo-random numbers in F.

Value at risk

Definition

Let X be a random variable depicting a gain. Then, the value at risk with confidence α is:

$$VaR_\alpha(X) = -F_X^{-1}(1-\alpha).$$

Alternatively, if Y is a loss ($Y = -X$), $VaR_\alpha(Y) = F_Y^{-1}(\alpha)$.

- In other words, the VaR at 99% is the quantile at 99% of the losses or the opposite of the quantile at 1% of the gains.
- The VaR is the maximum expected loss for a given confidence level.
- For successive gains $X_1, ..., X_n$, we can either assume that they are identically distributed (and even independent) or not, for example with conditional cdf $F(X_i | X_{i-1}, ..., X_{i-d})$ or with any model of dynamic.

Estimating Value at risk (1/2)

Several notions of VaR depending on their estimation method.

1. **Historical VaR**: based directly on past values (simulation directly in the historical distribution, which is supposed not to change in the close future). Drawbacks: equal weight to each observation and static (possibility to weight data to have something more dynamic with more weight on recent observations), often iid assumption.
   - **Empirical VaR**: empirical quantile, but need big amount of data (should have a number of observations $>> 1/(1-\alpha)$).
   - **VaR based on an estimated distribution**: estimation of the whole distribution, when less data than needed for empirical quantiles (e.g., 100 data for $\alpha = 99.9\%$), and then quantile on the estimated distribution. Distribution to be estimated can be **parametric** or **non-parametric** (would be better), like a kernel-based distribution.
   - **EVT VaR**: much more accurate for tail behaviour than a Kernel approach, it leverages on the fact that the distribution of extreme events should be close to a known parametrized distribution, whatever the distribution of the returns (restricts the difficulty of the problem).
2. ...

Estimating Value at risk (2/2)

1. ...
2. **Monte-Carlo VaR**: based on a dynamic, a model estimated on historical data.
   - **parametric VaR**: parametric model such as
     - iid Gaussian (or other) returns: sounds arbitrary, not that efficient and does not need Monte-Carlo (closed formulas). But could be used to model factors (and then Monte-Carlo approach useful). Difficulty to identify factors (econometric approach).
     - parametric dynamic, such as GARCH, which is more accurate.
   - **non-parametric VaR**: already proposed non-parametric models, here Monte Carlo is a method and not a model, it can then use any non-parametric model.

Non-parametric distribution

An estimator $\hat{f}$ of a density should check: $\forall x \in \mathbb{R}, \hat{f}(x) \geq 0$, and $\int_{-\infty}^{+\infty} \hat{f}(x)dx = 1$. From a series of observed price returns $X_1, ..., X_n$, the probability density could be estimated by:

- A **histogram** (not continuous, not invariant by translation), with a discretization $a_1 < ... < a_m$: if $x \in [a_i, a_{i+1})$, then

$$\hat{f}(x) = \frac{1}{n(a_{i+1} - a_i)} \sum_{j=1}^{n} \mathbb{1}_{[a_i, a_{i+1})}(X_j).$$

- A **kernel density**, where the Kernel K, a density with cdf $\mathcal{K}$ can be, for example, $K(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$, with a smoothing parameter $h > 0$ indicating the smoothness/robustness:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right) \text{ and } \hat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{K}\left(\frac{x - X_i}{h}\right).$$

  Drawback: fixed-size resolution (h), thus spurious effects for tails and smoothing of essential elements in the main part of the distribution.

- An **adaptive (or variable) kernel density**:

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{hd_{i,k}} K\left(\frac{x - X_i}{hd_{i,k}}\right) \text{ and } \hat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{K}\left(\frac{x - X_i}{hd_{i,k}}\right),$$

  where $d_{i,k}$ is the Euclidean distance between X_i and its k-th nearest neighbour among the $n - 1$ other observations, $X_1, ..., X_{i-1}, X_{i+1}, ..., X_n$.

Non-parametric distribution

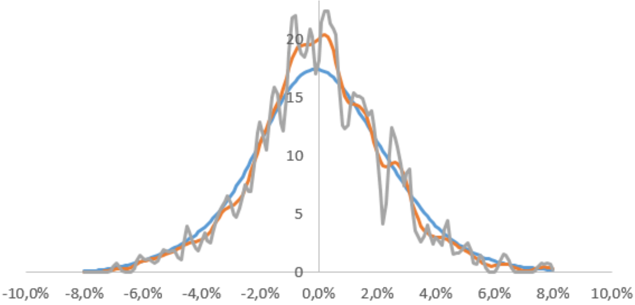


Figure: Kernel density of historical daily price returns (Natixis stock, 2015-2016) for $h = 0.01$ (smooth one), $h = 0.003$, and $h = 0.001$ (roughest one).

VaR criticisms

VaR is more robust than volatility and is a market standard, but:

- In general VaR is not a coherent risk measure because it is not subadditive (does not reflect diversification well). This can lead a bank to accounting tricks: suppose a bank's supervisor requires capital to put apart per trading desk, corresponding to the 95% VaR of each desk. How does the bank define its desks to minimize the capital requirements given that there are desk pairs for which $VaR_\alpha(Y_1 + Y_2)$ can be larger than $VaR_\alpha(Y_1) + VaR_\alpha(Y_2)$?
- VaR does not give any indication of the potential size of the loss above it (it is only a quantile); one solution is to observe several quantiles (spectral VaR).

The **expected shortfall** does not have these two drawbacks (with the assumption of an absolutely continuous loss distribution as far as coherence is concerned).

Expected shortfall

Let X be the gain of the portfolio, $\alpha \in [0,1]$ a confidence level. We define several risk measures depicting the average of the $1-\alpha$ worst losses:

- the **worst conditional expectation**: $WCE_\alpha(X) = \sup\{\mathbb{E}[-X|A], \mathbb{P}(A) \geq 1-\alpha\}$,
- the **tail conditional expectation**: $TCE_\alpha = \mathbb{E}\left[-X|X \leq -VaR_\alpha(X)\right]$,
- the **average VaR** (AVaR), or **expected shortfall** (ES), or **conditional VaR**: $ES_\alpha(X) = \frac{1}{1-\alpha}\int_\alpha^1 VaR_r(X)dr = \frac{1}{1-\alpha}\int_\alpha^1 F_{-X}^{-1}(r)dr$.

Theorem

We have: $AVaR_\alpha(X) \geq WCE_\alpha(X) \geq TCE_\alpha(X) \geq VaR_\alpha(X)$. Moreover, if the distribution of X is without atoms ($\forall \varepsilon > 0$, $\exists (A_i)$, a finite partition of the set Ω such that $\mathbb{P}(A_i) \leq \varepsilon$): $AVaR_\alpha(X) = WCE_\alpha(X) = TCE_\alpha(X)$.

Theorem

If $(\Omega, \mathcal{F}, \mathbb{P})$ is without atoms, then ES_α is the smallest coherent risk measure, continuous by below (that is for $X_n \nearrow X$, $\rho(X_n) \to \rho(X)$), and invariant by the distribution (ρ is invariant by distribution if $\rho(X) = \rho(Y)$ when X and Y have the same distribution), which dominates VaR_α.

Risk measures based on moments

- Variance and volatility (square root of variance) are not monetary risk measures because they are neither monotonic (X uniform in $[0,1]$, Y constant equal to 2, then $X < Y$ but $\rho(X) > \rho(Y)$) nor invariant by translation ($vol(X+1) = vol(X) \neq vol(X) + 1$).
- Nevertheless, massively used to monitor the *risk* of a portfolio.
- Could use other moments too for asymmetry (skewness) or tail behaviour (kurtosis).

About cumulative volatility

We want to know the contribution of one asset to the global volatility, with a linear combination.

- $Var(X+Y) = Var(X) + Var(Y)$ if zero correlation (not independence), but not linear for volatility $Vol(X+Y) = \sqrt{Vol(X)^2 + Vol(Y)^2}$.
- In general, if we note $Vol_X = \sigma_X$ and ρ the correlation, $\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y$ and thus $\sigma_{X+Y} = \frac{\sigma_X^2 + \rho\sigma_X\sigma_Y}{\sigma_{X+Y}} + \frac{\sigma_Y^2 + \rho\sigma_X\sigma_Y}{\sigma_{X+Y}}$.
- For n assets, $\sigma^2_{\sum_{i=1}^n X_i} = \sum_{i=1}^n \sigma^2_{X_i} + \sum_{i \neq j} \rho_{i,j}\sigma_{X_i}\sigma_{X_j}$. Therefore the volatility can be written as a linear combination of n volatilities relative to each asset: $\sigma_{\sum_{i=1}^n X_i} = \sum_{i=1}^n \left[\frac{\sigma^2_{X_i} + \sum_{j \neq i} \rho_{i,j}\sigma_{X_i}\sigma_{X_j}}{\sigma_{\sum_{j=1}^n X_j}} \right]$.

Deviation risk measure

A generalization of volatility, quantifies deviation and not downside risk (differences with coherent risk measure, in addition to monetary and monotonicity, which disappears, are in bold):

1. Normality: $\mu(0) = 0$ (particular case of positive homogeneity).
2. **Positivity**: $\mu(X) > 0$ if X is not constant, $\mu(X) = 0$ else.
3. Sub-additivity (diversification principle): $\mu(X_1 + X_2) \leq \mu(X_1) + \mu(X_2)$.
4. Positive homogeneity: if $\lambda \geq 0$, then $\mu(\lambda X) = \lambda\mu(X)$.
5. **Invariance by translation**: If p is a deterministic profit, then $\mu(X + p) = \mu(X)$.

Examples:

- volatility (standard deviation): $\sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2]}$,
- lower and upper semi-deviation: $\sigma_-(X) = \sqrt{E[(X - EX)_-{}^2]}$ and $\sigma_+(X) = \sqrt{E[(X - EX)_+{}^2]}$, where $[X]_- = \max\{0, -X\}$ and $[X]_+ = \max\{0, X\}$,
- mean absolute deviation: $MAD(X) = E(|X - EX|)$.

Motivation: St. Petersburg paradox

A game proposed by Bernoulli (1713)

A fair coin is tossed. If heads appears, the coin is tossed again until tails appears. The player wins 2 euros if tails appears on the first toss, 4 euros on the second, ... 2^n euros on the n-th toss. How much would you pay to take part to this game?

- We know that the price is the expected value of the discounted future cash flows. What is it here?
- The paradox is that no one would pay such a price.
- How to go beyond the paradox?
  1. The law of great numbers does not apply with infinite expectancies.
  2. The utility function biases our perception of the gain. We sometimes price using this utility function.
  3. The risk aversion is not considered in the lottery.

Defining the entropic risk measure

The exponential or entropic risk measure $e_\gamma(X)$, for a risk aversion $\gamma > 0$ and for a given random variable X representing a profit, say a price return of a financial asset between two dates, is defined by:

$$e_\gamma(X) = \frac{1}{\gamma} \ln \mathbb{E}[e^{-\gamma X}] = \frac{1}{\gamma} \ln M_X(-\gamma), \tag{1}$$

where M_X is the moment-generating function of X. Symmetrically, if a random variable Y represents a risk, say a loss, the corresponding profit X is equal to $-Y$ and the corresponding entropic risk measure is $e_\gamma(-Y)$.

The entropic risk measure $e_\gamma(Y)$ of a random variable Y representing a risk is also the insurance premium against the risk Y in an exponential utility framework. For the risk Y and a utility function u, the premium P is the capital required such that the risk Y is properly hedged, that is to say the expected utility of the risk hedged by its premium is zero:

$$\mathbb{E}[u(P - Y)] = 0.$$

In the case of the entropic risk measure, $e_\gamma(-Y)$ is the premium for an exponential utility function of risk aversion γ:

$$u : x \mapsto \frac{1}{\gamma}\left(1 - e^{-\gamma x}\right).$$

Properties of the entropic risk measure

1. Since the entropic risk measure is related to the moment-generating function of the profits, it is based on all its moments. This is an evident advantage over the simplistic volatility as a risk measure.
2. The entropic risk measure is convex. This implies that diversification does not increase the risk: when two portfolios are merged, the resulting risk is not higher than the sum of the risk of each portfolio. Other popular risk measures, like the value-at-risk (VaR), do not have this important property.
3. Neither the entropic risk measure nor the VaR are coherent risk measures. In particular, the entropic risk measure lacks positive homogeneity. A risk measure ρ is positively homogeneous if $\forall \lambda \geq 0, \rho(\lambda X) = \lambda \rho(X)$. The VaR and the expected shortfall, which is convex as well and thus coherent, are positively homogeneous. The example of a Gaussian profit X of mean μ and variance σ^2 shows that the entropy is not positively homogeneous. Indeed, the entropic risk measure is then equal to $-\mu + \sigma^2\gamma/2$. But if the variable X is doubled, the entropy is not doubled since it is equal to $-2\mu + 2\sigma^2\gamma$. However, positive homogeneity of risk measures is not a realistic property since in practice the risk of a position is often non-linearly related to the held amount, due to liquidity risk.

Scaling rule of the entropic risk measure

- Scaling rule (when horizon multiplied by t) of the VaR for i.i.d. Gaussian returns: $\sqrt{t}$.
- Alternative example of dependent Gaussian: scaling in t^H for a H-fractional Brownian motion.
- Alternative example of i.i.d. non-Gaussian: scaling in t for Cauchy process.

Instead of that, the entropic risk measure has a unique scaling rule for i.i.d. returns, whether they are Gaussian or not: the log-return until time t is the sum of t i.i.d. returns of horizon 1, $X_1, ..., X_t$, and $e_\gamma\left(\sum_{i=1}^t X_i\right) = \sum_{i=1}^t e_\gamma(X_i) = te_\gamma(X_1)$, so that the scaling factor is t.

Risk aversion

The risk measure is increasing with the risk aversion γ. However, determining the risk aversion of an agent may seem as arbitrary as choosing a confidence level α for a VaR. In fact, both parameters can be linked. For example, assuming Gaussian price returns X of mean μ and variance σ^2, the corresponding entropic risk measure and VaR are respectively $-\mu + \sigma^2\gamma/2$ and $-\mu + \sigma G^{-1}(1-\alpha)$. Both are equal if $\gamma = 2G^{-1}(1-\alpha)/\sigma$. We can therefore define a risk aversion for a type of asset. For instance, concerning a stock market for which we assume a long-term annual volatility of 20%, values of the VaR parameter α equal to 5%, 1% and 0.01% are respectively associated to a risk aversion γ equal to 16.45, 23.26 and 30.90.

Estimating the risk aversion

- Price of a stock is discounted sum of future dividends: does not take into account variations of preference over different horizons. On the opposite, options exist for different horizons: they are well-suited to estimate risk aversion.
- Estimate mean of a distribution on stocks and all the density on options: $q(S_T) = e^{r(T-t)} \left. \frac{\partial^2 C(t,S_t,K,T)}{\partial K^2} \right|_{K=S_T}$.
- This distribution is estimated at a single date, not on historical data: it is therefore much more responsive to changing market expectations.
- ...

Estimating the risk aversion

- ...
- Unfortunately, the option-implied density is risk-neutral, which may not be consistent with the actual forecast of the most representative investor (the market). The difference between risk-neutral (q) and physical (p) distributions is due to risk aversion (λ for representative utility function U) or equivalently to the **stochastic discount factor** (aka **pricing kernel**: $\zeta(S_T)$):

$$\frac{p(S_T)}{q(S_T)} = \lambda \frac{U'(S_T)}{U'(S_t)} = \zeta(S_T).$$

- If two among p, q, U are known, the third one is inferred. In practice, only q is known. Solutions: assume stationarity of p or stationarity of U (parametric function, depending on λ, which in this case is constant).
- The risk aversion is then the λ maximizing the forecast ability over historical data.

Stochastic discount factor

- For assets $i \in \{1, ..., n\}$ of pay-off π_i, initial price p_i and return $R_i = \pi_i / p_i$, the stochastic discount factor ζ is such that, $\forall i \in \{1, ..., n\}, \mathbb{E}(\zeta \pi_i) = p_i$, in which $\mathbb{E}$ is expressed in the risk-neutral distribution.
- As a consequence $\forall i, j$, $\mathbb{E}[\zeta(R_i - R_j)] = 0$.
- In the risk-neutral distribution, the risk-free rate R_f is deterministic. Therefore, $\mathbb{E}(\zeta) = 1/R_f$.
- As $\text{cov}(X, Y) = \mathbb{E}[X, Y] - \mathbb{E}[X]\mathbb{E}[Y]$, we have $1 = \text{cov}(\zeta, R_i) + \mathbb{E}(\zeta)\mathbb{E}(R_i)$. In particular:

  $$\mathbb{E}(R_i) - R_f = -R_f \text{cov}(\zeta, R_i),$$

  which is the risk premium of asset i.

Statistical modelling

1. From data, find relevant empirical observations to be reproduced by the model.
2. **Model specification**: consider various specifications to build a class of appropriate models (selection of explanatory variables and of various parametric [or nonparametric] specifications).
3. **Statistical inference**: estimate the parameters for each specification by choosing the most appropriate method: maximum-likelihood, (generalized) moment method...
4. **Model selection**: among all the models, select the best one (information criteria, best out-of-sample performance).
5. **Model validation**: the *best* model is not always a *good* model... If model not validated, go back to step 2 or even step 1.
6. Using the model: Depending on the use (forecast of price returns, of VaR, etc.), we will focus on specific properties of observations (step 1, 2, 3 (censored estimation?), 4 and 5).

Backtesting risk measures

What is an appropriate risk measure?

- **Coverage**: *ex ante* risk = *ex post* risk. We must historically observe 5% (or a quantity not significantly different from 5%) of crossings of the $VaR_{95\%}$.
- **Independence**: If the violations of the VaR limit form a cluster, the risk measurement is not good. Example: a constant VaR and the presence of a crisis in the dataset.

The right variable to observe is therefore the hit function, which is an indicator:

$$I_t(\alpha) = \begin{cases} 1 & \text{if } r_t < -VaR_{t|t-1}(1-\alpha) \\ 0 & \text{otherwise.} \end{cases}$$

The $(I_t(\alpha))_t$ must be iid Bernoulli random variables with parameter α.

Other approaches (out of the scope of the course)

Predictability
Back-testing
Statistical bootstrap
Stress tests

Sharpe ratio

Definition

Let $r_1, ..., r_n$ be price returns (supposed centred; otherwise the Sharpe ratio is often considered for returns in excess of a risk-free rate).
The **Sharpe ratio** *(William Sharpe, 1966) is the ratio of the average return over its standard deviation:*

$$Sharpe = \frac{\frac{1}{n}\sum_{i=1}^{n} r_i}{\sqrt{\frac{1}{n}\sum_{i=1}^{n} r_i^2}}.$$

The **Information ratio** *is the Sharpe ratio of the returns in excess of a benchmark (excess return over tracking error volatility):*

$$IR = \frac{\frac{1}{n}\sum_{i=1}^{n} r_i - r_i^B}{\sqrt{\frac{1}{n}\sum_{i=1}^{n} (r_i - r_i^B)^2}}.$$

Sharpe ratio

The higher the Sharpe, the better. The portfolio manager can achieve this by increasing the return or decreasing the volatility (considered here as the risk measure).

Some drawbacks

- The ratio may be negative, but in this case it is difficult to interpret it and the portfolio manager could ameliorate it by increasing the risk (perverse effect).
- Returns and volatility are implicitly considered as having the same magnitude but the comparison of Sharpe ratios between different strategies is not as obvious:
  - it should depend on the type of asset (what meaning for money-market with almost zero volatility?)
  - it should depend on the rebalancing frequency and thus on the skill of the portfolio manager (fundamental law of active management, by Grinold and Kahn).
- A positive jump implies a rise of volatility and thus an increased risk and a lower Sharpe, whereas this jump is something positive. Possibility to separate *positive and negative volatility*?
- Estimation difficulties: very unstable, depends on time scale, etc.

Approach based on the decomposition of the volatility

Asymmetric ratios: losses and gains are not supposed to be equivalent.

Rationale

- The risk does not stem from positive jumps of the price.
- We prefer to penalize the *volatility* of a subset of returns.

Use $v^- = \sqrt{\frac{1}{Card\{r_i<0, i\in[\![1,n]\!]\}}\sum_{i=1}^n (\min(r_i,0))^2}$ rather than $v = \sqrt{\frac{1}{n}\sum_{i=1}^n (r_i-\bar{r})^2}$.

- The traditional volatility v measures the dispersion of the price returns around the average return,
- whereas v^- does not measure a dispersion (the dispersion around the average of negative returns is meaningless) but the average amplitude of price decreases.

Corresponding risk-adjusted performance ratios:

- **Sortino ratio**, for a minimal acceptable return $r^\star$ (say equal to zero): $S(r^\star) = \frac{\mathbb{E}[r-r^\star]}{\sqrt{\mathbb{E}[\min(r-r^\star,0)^2]}}$,
- **Upside potential ratio** (Sortino, van der Meer, Plantinga, 1999), upside performance relative to downside risk: $U(r^\star) = \frac{\mathbb{E}[\max(r-r^\star,0)]}{\sqrt{\mathbb{E}[\min(r-r^\star,0)^2]}}$.

Asymmetric ratio without volatility

The asymmetry between gains and losses can also be measured by a metric based on the partition between gains and losses, weighted by the cumulative distribution function of the returns, F. Known as the **Omega ratio** (Keating and Shadwick, 2002):

$$\Omega(r^\star) = \frac{\int_{r^\star}^{+\infty} (1 - F(r))\, dr}{\int_{-\infty}^{r^\star} F(r) dr}.$$

In particular, if $r^\star = 0$, after integrating by parts, we get:

$$\Omega(0) = \frac{\mathbb{E}[\max(r, 0)]}{\mathbb{E}[-\min(r, 0)]}.$$

A generalization

The risk measure focuses on:

- the partial first moment, for Ω,
- the second moment, for the Sharpe ratio,
- the partial second moment, for S and U.

Could be generalized to other (partial) moments. In particular, asymmetry is better described by skewness.

Definition

The n-th lower partial moment function of a random variable of cumulative distribution function F is $\mathcal{M}_n(\tau) = \int_{-\infty}^{\tau} (\tau - r)^n dF(r)$.

The **Kappa ratio** (van Harlow, 1991) is:

$$K_n(r^\star) = \frac{\mathbb{E}[r - r^\star]}{\sqrt[n]{\mathcal{M}_n(r^\star)}}.$$

As particular cases:

- $S(r^\star) = K_2(r^\star)$,
- $\Omega(r^\star) = 1 + K_1(r^\star)$ (by integrations by parts).

Replacing partial moments by maximum drawdown

Definition

For a series of prices $P_1, ..., P_n$, the **maximum drawdown** *is the maximal loss (a positive metric) encountered at one of the dates of the observed time interval, regarding another date in the same interval:*

$$MDD = \max_{1 \leq i < j \leq n} -\left(\frac{P_j - P_i}{P_i}\right).$$

Also known as peak to valley loss.

How to calculate it rapidly?

- For $j = 2$ to n, calculate the drawdown at the current date j from the previous peak: $m_j = -\left(\frac{P_j}{\max_{1 \leq i < j} P_i} - 1\right)$.
- The maximum drawdown is the maximum of these drawdowns: $MDD = \max_{2 \leq j \leq n} m_j$.

Related metrics:

- the **maximum drawdown duration**: $j - i$ such that $MDD = -\left(\frac{P_j - P_i}{P_i}\right)$,
- the **recovery duration** (if it exists): if $[\![i, j]\!]$ is the MDD time interval, the recovery duration is $k - j$ such that k is the lowest integer greater than j such that $P_k > P_i$ (then $\frac{P_k - P_j}{P_j} > MDD$, even if $P_k = P_i$).

Replacing partial moments by maximum drawdown

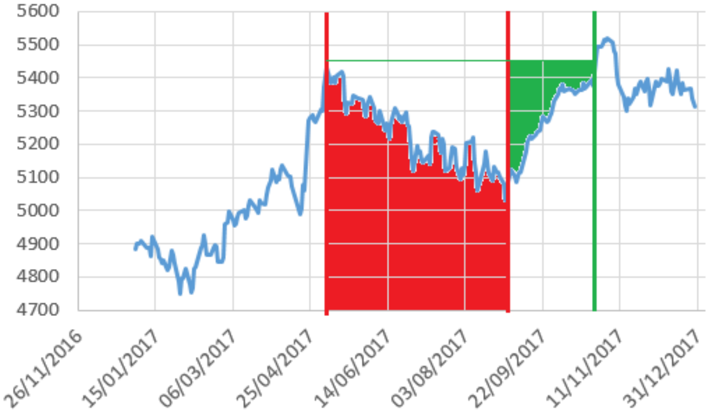


Figure: Maximum drawdown in 2017 for the CAC40 index: 7.37% between 8th May and 29th August (81 trading days), for a recovery above the price of 5442.10 the 27th October (43 trading days).

Replacing partial moments by maximum drawdown

The **Sterling ratio** is an adaptation of Sharpe or Sortino ratios.

Warning concerning the time scale!

- In Sharpe (or Sortino), it is easy to calculate metrics at the numerator and the denominator on the same time scale, for example average daily return and daily volatility or lower partial volatility.
- Calculating daily MDD is meaningless (unless working in high frequency, and in this case the daily MDD is a proxy of the lower partial realized volatility).
- We prefer decomposing the observation time interval in adjacent subsets of same duration (for example calendar years) and calculate the average of the MDD of each subset.
- To be consistent, the numerator must be an average of the return on the same subsets.

For example, the yearly Sterling ratio, calculated on N years, is:

$$Sterling = \frac{\frac{1}{N}\sum_{i=1}^{N} R_i}{\frac{1}{N}\sum_{i=1}^{N} MDD_i},$$

where $R_i = \frac{P_{256i}}{P_{256(i-1)}} - 1$ and $MDD_i = \max_{256(i-1)\leq m<n\leq 256i} - \left(\frac{P_n - P_m}{P_n}\right)$.

Basic properties of the ratios

- Some ratios are always positive (Upside potential, $\Omega(0)$), others not necessarily (Sharpe, Sortino, $\Omega(r^\star)$ if $r^\star \neq 0$, Kappa, Sterling).
- If the price follows a geometric Brownian motion (not very realistic...), theoretically:
  - $Sharpe = S(0) = 0$.
  - The denominator of Sortino and Sharpe converges toward σ, but more rapidly for the volatility than for the lower partial volatility (less observations).
  - The numerator of $U(0)$ is $\int_0^{+\infty} rdg(r)/\int_0^{+\infty} dg(r) = \int_{-\infty}^{+\infty} |r|dg(r) = \sigma\sqrt{2/\pi}$, because for $Z \sim \mathcal{N}(0,1)$, $\mathbb{E}\left[|Z|^k\right] = \frac{2^{k/2}\Gamma\left(\frac{k+1}{2}\right)}{\Gamma\left(\frac{1}{2}\right)}$. Then, $U(0) = \sqrt{\frac{2}{\pi}}$.
  - $\Omega(0) = 1$.
- Risk metrics robust to a positive jump (the jump is not to be considered as risky): the lower partial moments but not the MDD and certainly not the volatility. Ratios, as taking into account positive performance, are never robust to such a jump.

Poisson process

We denote by $\mathcal{P}(\lambda)$ the Poisson distribution, of Siméon-Denis Poisson (1781-1840), with parameter λ. A random variable X with values in $\mathbb{N}$ following this distribution admits as density, for $k \in \mathbb{N}$:

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}.$$

Proposition

The expectation and variance of a Poisson process are equal to λ.

Proposition

The characteristic function of a Poisson variable of parameter λ is:

$$\phi_X(t) = \exp\left(\lambda(e^{it} - 1)\right).$$

Poisson process

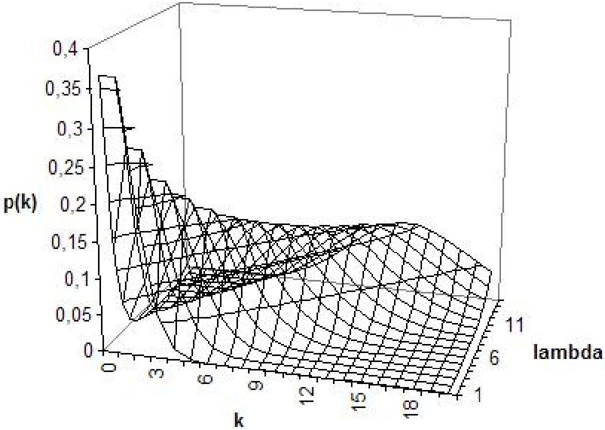


Figure: Probability density of the Poisson distribution for a parameter λ between 1 and 15.

Poisson process

Definition

Let m be a measure on $\mathbb{R}$. A Poisson process N of intensity m is a stochastic process with independent and homogeneous increments with Poisson distribution of intensity m: for all $n > 0$, $0 \leq t_1 \leq ... \leq t_n$, N_{t_1}, $N_{t_2} - N_{t_1}$, ..., $N_{t_n} - N_{t_{n-1}}$ are independent, and for all real numbers s and t positive or zero, $N_{t+s} - N_t$ has distribution $\mathcal{P}(m(s))$.

Proposition

Let X be a Poisson random variable with parameter λ_X, and Y be a Poisson random variable with parameter λ_Y. Then, $X + Y$ is a Poisson distribution with parameter $\lambda_X + \lambda_Y$.

Note that we can show that the time between two jumps of a Poisson process (N_t) with parameter λ follows an exponential distribution with parameter λ, that is, for n a positive integer, if $T_n = \inf\{t \geq 0, N_t \geq n\}$, and $S_n = T_n - T_{n-1}$, then $\mathbb{P}(S_n \leq t) = 1 - e^{-\lambda t}$.

Poisson process

- A Poisson process N_t of intensity λ is not a martingale, but it can be transformed in a martingale thanks to the **compensated Poisson process**: $\tilde{N}_t = N_t - \lambda t$.
- The process $(N_t - \lambda t)^2 - \lambda t$ is a martingale too.
- The Brownian motion and the Poisson process belong to a wider family of processes: the Lévy processes (càdlàg processes with independent and stationary increments).
- Example of Lévy process, combining a Brownian motion W_t, a drift, and jumps (number of jumps defined by a Poisson process N_t and magnitude by iid random variables Z_i):

$$X_t = \gamma t + W_t + \sum_{i=1}^{N_t} Z_i.$$

  If the magnitude of the jump is a constant Z, then the process becomes $\gamma t + W_t + Z N_t$.

Comparison of the ratios

We simulate a price with three distinct models:

- a geometric Brownian motion $P_t^1 = P_0 exp\left(-\frac{\sigma^2}{2}t + \sigma W_t\right)$, with $P_0^1 = 100$,
- a geometric Brownian motion with jumps (modelled by a Poisson process N_t directly in the price): $P_t^2 = P_t^1 + ZN_t$, with Z the size of the jumps,
- the same with a compensated Poisson process $\tilde{N}_t$ for the jumps, so that the price is a martingale: $P_t^3 = P_t^1 + Z\tilde{N}_t$.

Empirically, we show that partial vol is more robust to positive jumps and that U and Ω are more stable ratios. MDD is not very significant.

Comparison of the ratios

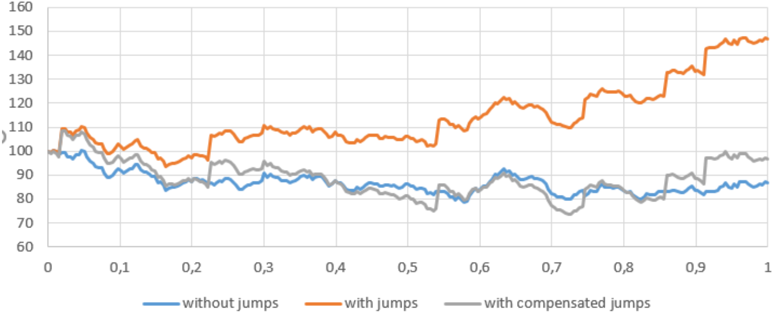


Figure: Example of trajectory. Positive jumps ($Z = 10$), $\lambda = 5$ jumps per year.

Comparison of the ratios

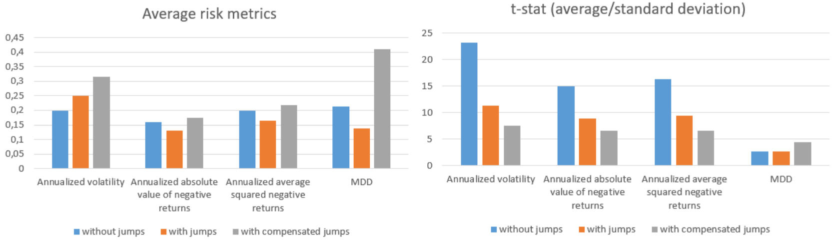


Figure: Positive jumps ($Z = 10$), $\lambda = 5$ jumps per year, average for $1,000$ trajectories. MDD low t-stat (not very significant), Volatility not robust to positive jumps, contrarily to partial volatility.

Comparison of the ratios

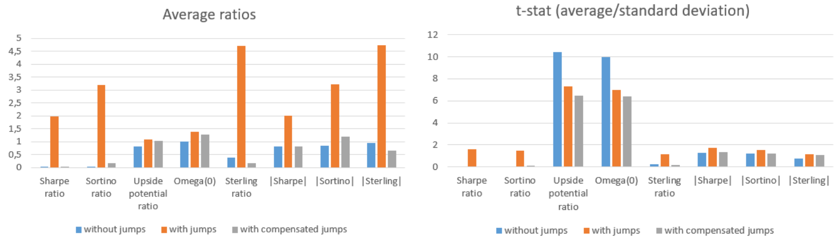


Figure: Positive jumps ($Z = 10$), $\lambda = 5$ jumps per year, average for $1,000$ trajectories.

Comparison of the ratios

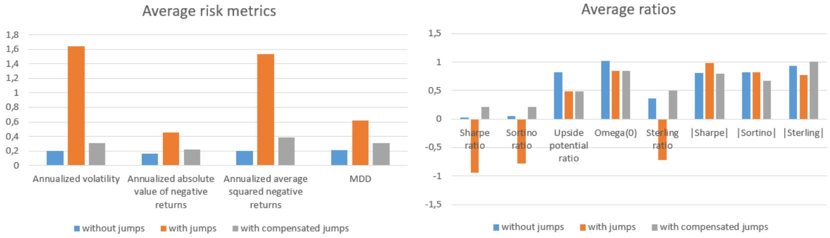


Figure: **Negative** jumps ($Z = -10$), $\lambda = 5$ jumps per year, average for $1,000$ trajectories.

Liquidity: confronting many viewpoints

The bid-ask spread (difference of price between seller and buyer) can be split into two from the **market maker**'s perspective:

- **Liquidity cost**: the real spread that compensates the market maker for his role as guarantor of market liquidity and which corresponds to the technology and inventory costs incurred by the market maker.
- **Market maker risk**: the information asymmetry that compensates market makers for the losses they might incur when they make a bet without knowing its ins and outs: since they cannot distinguish between informed and uninformed bets, they can be the "opposite part" of an informed order because of their role as guarantor of market liquidity.

Liquidity: confronting many viewpoints

According to Amihud and Mendelson (2006), there would be three components in liquidity costs:

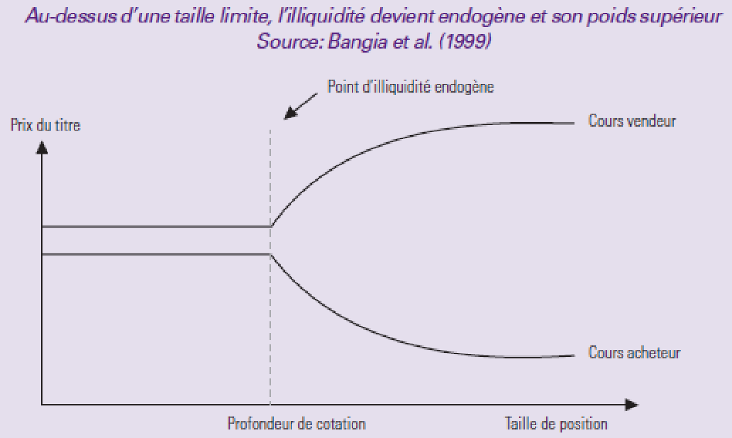
- **Direct trading costs**: deterministic transaction costs that include brokerage commissions, transaction taxes and exchange fees.
- **The impact of orders on the market price**: all the more important as the order is substantial, because a large order will involve several counterparts and therefore go far in the order book.
- **The cost induced by a delay in execution**: to limit the impact of a large order, traders can do block orders, which means that some blocks are exchanged with a certain delay, which exposes more to the possibility of market movements.

Liquidity: confronting many viewpoints

According to Harris (1990), the notion of a liquid market is defined as a market on which a large quantity can be bought and sold at any time, quickly and at low cost. It is measured by four pillars:

- **Width**: cost incurred by a simultaneous purchase/sale, which corresponds to the sum of direct costs and impact in Amihud and Mendelson.
- **Depth**: the quantity of securities that can be traded without impact (only cost: bid/ask spread) because absorbed without difficulty by the market maker (see following graph).
- **Immediacy**, which determines the speed with which positions can be traded and which corresponds to the delay between the placing of the order and its settlement.
- **Resilience**, which designates the market's capacity to absorb random shocks, such as non-informative orders.

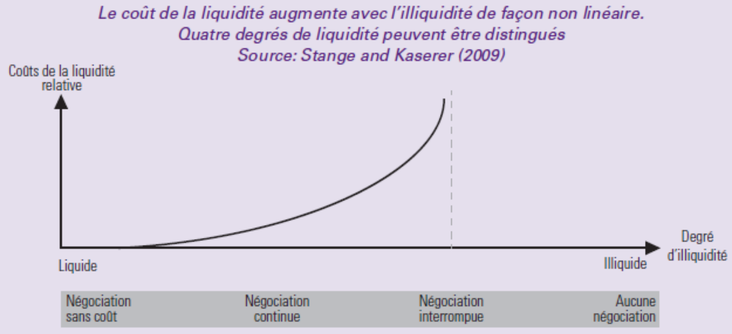
Liquidity: confronting many viewpoints



Liquidity: confronting many viewpoints

According to Stange and Kaserer (2009), liquidity costs can be defined continuously and thus form four degrees of liquidity:

- **Cost-free trading** (e.g. cash).
- **Continuous trading**: orders executed at a given cost.
- **Interrupted trading**: orders executed punctually.
- **No trading**: the market is completely illiquid, prices are not available and are recovered by adapted techniques



Some remarks

- Liquidity risk cannot be diversified: no hedging possible, no liquidity derivatives, etc.
- What is the price of an asset when it is not traded? Can we use a comparison with a liquid universe? (example of yield curves for corporate bonds)
- Liquidity premium: positive link between yield and illiquidity (Amihud and Mendelson) but concave (yields increase less for very illiquid assets).
- For corporate bonds, illiquidity increases when the credit rating deteriorates (risk premium of 0.6% in IG [investment-grade bonds], 1.5% in HY [high-yield bonds]).

Liquidity measures on stock markets

- Liquidity in this market is provided mainly by high-frequency trading. The comparison between trading volume and quantity to be traded gives a good idea of liquidity.
- In Kyle's model (1985), illiquidity is measured as the average ratio of the daily return in absolute value to the volume traded that day (a sort of average impact).
- Another measure, the turnover ratio, which is the ratio of the volume traded for a security to the number of remaining securities for the same company. Ratio negatively correlated with illiquidity costs according to Amihud and Mendelson.
- Other factors that can be considered: capitalization, traded volume, etc.

Liquidity measures on bond markets

- Bonds have a maturity: the liquidity for a security with a residual maturity of 10 years is not the same as for 1 year.
- Other characteristics to be taken into account: issue size (500 M€ for a company *vs* 20 G€ for a sovereign issue, for example), yield to maturity, duration, spread, subordination...
- A liquidity measurement based on the volume traded can be misleading because a traded bond is not necessarily liquid (e.g.: forced sale, IG securities becoming HY). Conversely, non-traded bonds are not necessarily illiquid.
- Measuring liquidity if the spreads between ask and bid prices are observed is easy, otherwise it is tricky... Factor model based on the factors mentioned above. At the end of 2008 and the beginning of 2009, perpetuals were valued at around 25%, it was not uncommon to see a 5% spread between the bid price and the ask price.

Liquidity-adjusted VaR

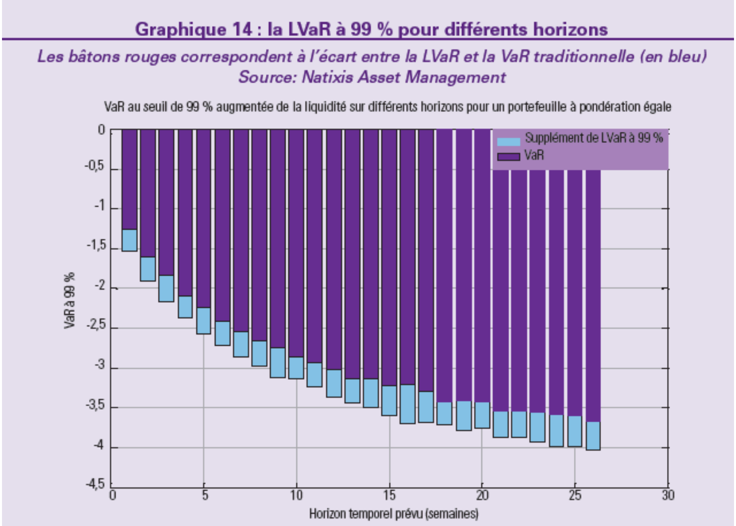
L-VaR can be obtained in several ways:

- If liquidity causes delays in the liquidation of a portfolio, we can consider that the one-day L-VaR is equal to the classic two-day VaR, for example.
- If liquidity is understood as an impact, it must be integrated into the yield used to calculate the VaR: this L-VaR will therefore be adapted to the amount held in the portfolio for the security studied.
- Bangia, Diebold, Schuermann, Stroughair approximation: integration of half of the bid/ask spread, S_t, which corresponds to the liquidation hypothesis. The price at $t+1$ is $P_{mid,t+1} = P_{mid,t}\exp(r_{t+1}) - S_{t+1}/2$, where r_{t+1}, the yield, and S_{t+1} are random. If they are Gaussian, perfectly correlated, and the yield is centered, then the L-VaR of level α (e.g. 90%) is written as the sum of a standard component and a liquidity component:

$$P_{mid,t}(1-\exp(Q_{1-\alpha}\sigma_r)) + (\mu_S + Q_\alpha \sigma_S)/2,$$

  with Q_α the quantile of probability α of a standard Gaussian distribution.
- If the strategy is to keep the securities in the portfolio, there is no reason to use this adjustmentα.

L-VaR of Bangia *et alii*



Permanent impact model: Almgren and Chriss (1/5)

Almgren and Chriss: optimal liquidation framework (metaorder: take time to limit impact *vs* hurry to limit market risk).

- Quantity X to be liquidated in tranches $n_1, n_2, ...$, with $x_k = X - \sum_{j=1}^{k} n_j$.
- Discrete-time trading, time step τ.
- Price dynamics according to arithmetic random walk: $S_k = S_{k-1} + \sigma\sqrt{\tau}\varepsilon_k - \tau g(n_k/\tau)$, where g is the permanent impact function, depending on the liquidation speed n_k/τ between times t_{k-1} and t_k.
- Let's assume that observed price is different from S (underlying dynamics) due to transitory impact (trader drains order book between t_{k-1} and t_k, but it instantly replenishes in t_k): $\bar{S}_k = S_k - h(n_k/\tau)$.
- Price obtained for liquidation (initial frictionless price + illiquidity losses):

$$\sum_{k=1}^{N} n_k \bar{S}_k = XS_0 + \sum_{k=1}^{N} \left[\left(\sigma\sqrt{\tau}\varepsilon_k - \tau g\left(\frac{n_k}{\tau}\right)\right) x_k - n_k h\left(\frac{n_k}{\tau}\right)\right].$$

Permanent impact model: Almgren and Chriss (2/5)

- Problem: minimize expected value and variance of the loss with ε_k the only random variables: $E(x) + \lambda V(x)$, with:

$$\begin{cases} E(x) &= \sum_{k=1}^{N} \left[\tau x_k g\left(\frac{n_k}{\tau}\right) + n_k h\left(\frac{n_k}{\tau}\right) \right] \\ V(x) &= \sigma^2 \sum_{k=1}^{N} \tau x_k^2. \end{cases}$$

- Almgren and Chriss hypothesis: linear impact:
  - $g(v) = \gamma v$, which leads to a total cost

$$\begin{aligned} \sum_{k=1}^{N} \tau x_k g\left(\frac{n_k}{\tau}\right) &= \gamma \sum_{k=1}^{N} x_k (x_{k-1} - x_k) \\ &= \frac{1}{2}\gamma \sum_{k=1}^{N} x_{k-1}^2 - x_k^2 - (x_k - x_{k-1})^2) \\ &= \frac{1}{2}\gamma X^2 - \frac{1}{2}\gamma \sum_{k=1}^{N} n_k^2. \end{aligned}$$

  - $h(n_k/\tau) = \xi sgn(n_k) + \eta n_k/\tau$, with ξ being half the bid-ask spread. This model for h is often called the quadratic cost model because the associated cost is $n_k h(n_k/\tau) = \xi |n_k| + \eta n_k^2/\tau$.

Permanent impact model: Almgren and Chriss (3/5)

- Minimum impact by taking constant speed: $n_k = X/N$, but variance can be high.
- Minimum variance by liquidating everything at the first instant: $n_1 = X$ and $n_2 = ... = n_N = 0$. Then, $V = 0$ and $E = \xi X + \eta X^2/\tau$.
- Between the two: an efficient frontier for several levels of risk aversion corresponding to the Lagrangian λ. Solved by Euler-Lagrange:

$$x_k = \frac{sinh(K(T-k))}{sinh(KT)} X,$$

where $K \overset{\tau \to 0}{\sim} \sqrt{\frac{\lambda\sigma^2}{\eta}} + \mathcal{O}(\tau)$.

Permanent impact model: Almgren and Chriss (4/5)

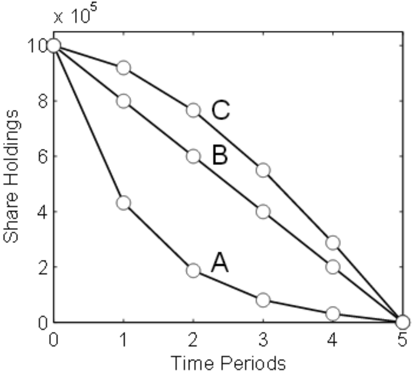


Figure 2: *Optimal trajectories.* The trajectories corresponding to the points shown in Figure 1. (A) $\lambda = 2 \times 10^{-6}$, (B) $\lambda = 0$, (C) $\lambda = -2 \times 10^{-7}$.

Figure: x for various risk aversions. Source : Almgren et Chriss.

Permanent impact model: Almgren and Chriss (5/5)

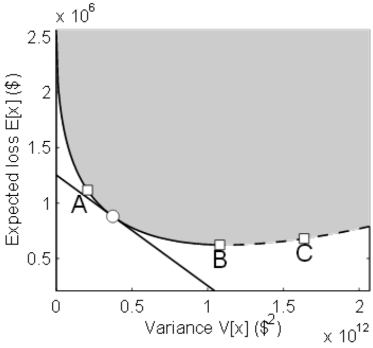


Figure 1: *The efficient frontier.* The parameters are as in Table 1. The shaded region is the set of variances and expectations attainable by some time-dependent strategy. The solid curve is the efficient frontier; the dashed curve is strategies that have higher variance for the same expected costs. Point B is the “naïve” strategy, minimizing expected cost without regard to variance. The straight line illustrates selection of a specific optimal strategy for $\lambda = 10^{-6}$. Points A,B,C are strategies illustrated in Figure 2.

Figure: Source : Almgren et Chriss.

Transitory impact model

Backward definition of the price: the price is the sum of past impacts (Bouchaud):

$$p_t = p_{-\infty} + \sum_{s=-\infty}^{t-1} G(t-s)\varepsilon_s S_s V_s^r,$$

where:

- S is the bid-ask spread;
- ε is -1 or 1 depending on whether the transaction is a buy (price = ask) or a sell (price = bid) on the market;
- V is the volume of the transaction, with r close to zero to have a concave function of the volume;
- G is a function worth 0 on $\mathbb{R}^-$, which can be interpreted as the impact of a single order. A statistical study on long-term correlation can allow this function to be fixed, in the form, for example, of a decreasing power function.

Liquidity and crisis

- In a bubble, market liquidity and prices grow together.
- This mechanism is unstable, with high demand and low supply.
- When market liquidity stops growing, we can enter a liquidity spiral, involving the two types of what we call *liquidity* :
  - market liquidity ;
  - financing.
- Contagion to an entire market can occur through financing risk.

Liquidity spiral

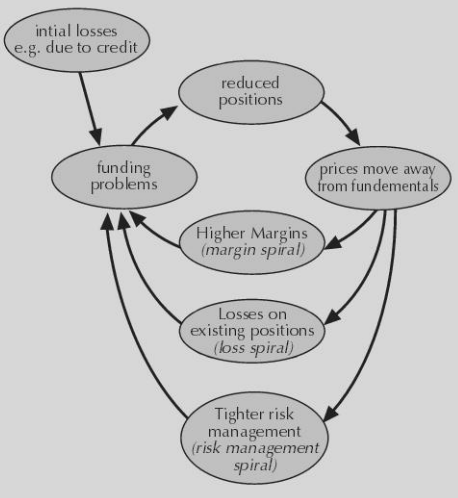


Figure: Liquidity spiral. Source : Pedersen.

Liquidity spiral

Initial volatility 1%, a shock on date 100 makes it rise to 1.3%. Then liquidity spiral induced by 10 agents (portfolio managers) who have an admissible range of volatility and a target of volatility, in this range, that they aim when they are forced to reallocate.
Price impact: Bouchaud model. Parameters:

- V, scaling factor in the price impact model;
- γ, in the function of the transitory price impact model $G : t \geq 0 \mapsto 1/t^{\gamma}$;
- r, volume exponent in price impact model;
- λ, exponential weighting parameter for the volatility (the larger λ, the less weight the old values have).

Liquidity spiral

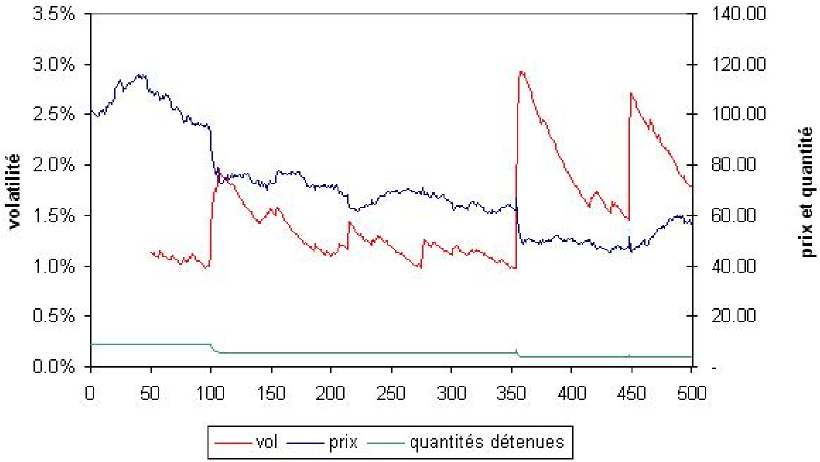


Figure: $V = 5$, $\gamma = 0.2$, $r = 0.25$, $\lambda = 0.03$ (half life in 24 days).

Liquidity spiral

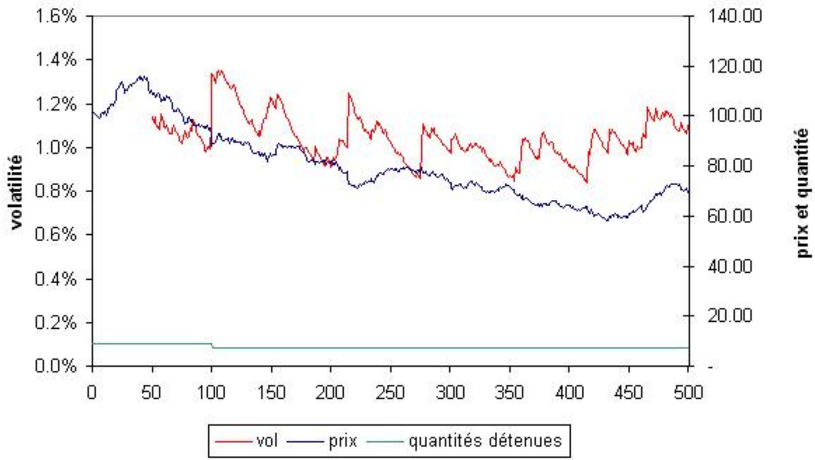


Figure: $V = 0$, $\gamma = 0.2$, $r = 0.25$, $\lambda = 0.03$ (half life in 24 days).

Liquidity spiral

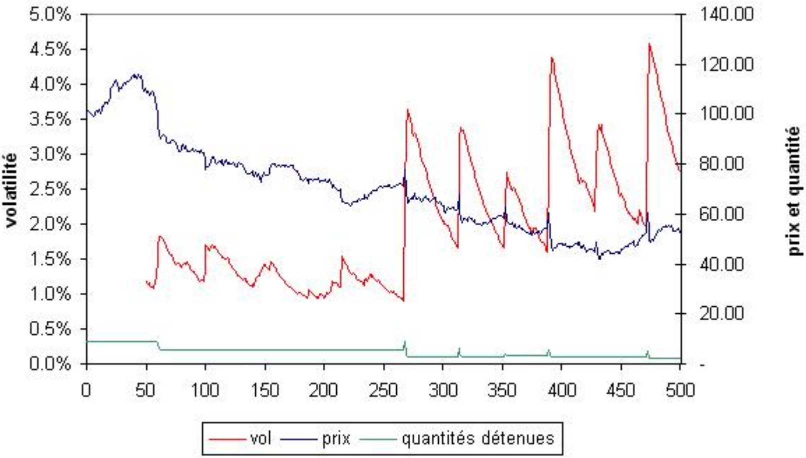


Figure: $V = 5$, $\gamma = 0.2$, $r = 0.25$, $\lambda = 0.05$ (half life in 15 days).

Liquidity spiral

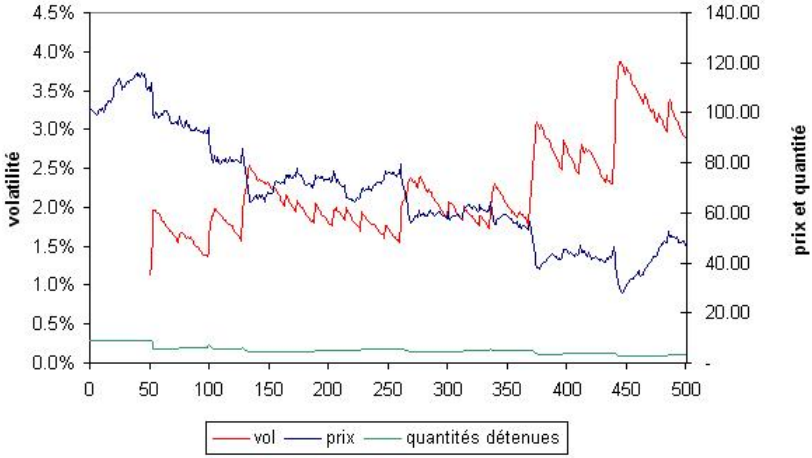


Figure: $V = 5$, $\gamma = 0.2$, $r = 0.25$, $\lambda = 0.03$ (half life in 24 days). Minimal volatility of 0.8% instead of 0.5%.

Liquidity spiral

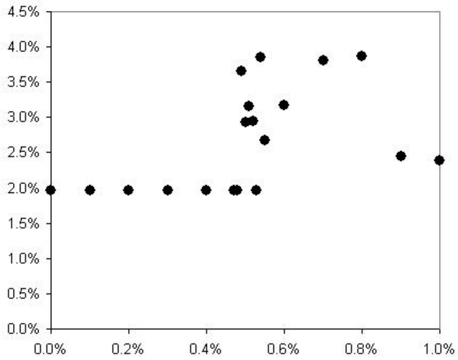


Figure: $V = 5$, $\gamma = 0.2$, $r = 0.25$, $\lambda = 0.03$ (half life in 24 days). Maximal volatility reached for the lower bound of the range of admissible volatility between 0 and 1%.

Liquidity spiral

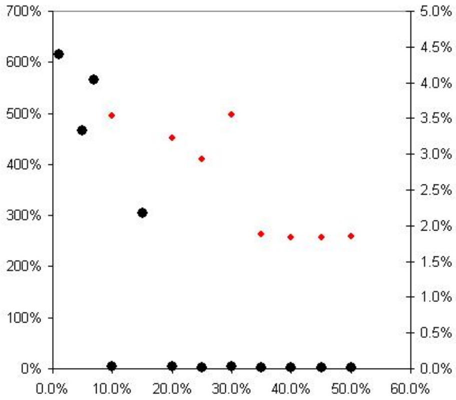


Figure: $V = 5$, $\gamma = 0.2$, $\lambda = 0.03$ (half life in 24 days). Maximal volatility reached for several values of r. In red, right axis, zoom.

Convergence of sequences of random variables

Definition

$(X_n)_{n\in\mathbb{N}}$ **converges $\mathbb{P}$-almost surely** *towards X smeans that there exists a subset $N \in \Omega$ of measure zero under the probability measure $\mathbb{P}$ such that:*

$$\forall \omega \in \Omega \setminus N, lim_{n\to\infty} X_n(\omega) = X(\omega).$$

Definition

$(X_n)_{n\in\mathbb{N}}$ **converges in $\mathbb{P}$-probability** *towards X means that:*

$$\forall \epsilon > 0, lim_{n\to\infty} \mathbb{P}\left(|X_n - X| > \epsilon\right) = 0.$$

Definition

For, $p \in [1, \infty]$, $(X_n)_{n\in\mathbb{N}}$ **converges according to the norm L^p** *towards X means that:*

$$lim_{n\to\infty} \|X_n - X\|^p = 0,$$

et, si $p < \infty$:

$$lim_{n\to\infty} \mathbb{E}\left(|X_n - X|^p\right) = 0.$$

Convergence of sequences of random variables

Proposition

1. *Almost sure convergence, probability convergence, and L^p-norm convergence are stable under linear combinations.*
2. *Almost sure convergence and probability convergence are stable under multiplication.*

Convergence of sequences of random variables

Proposition

1. *If $(X_n)_{n\in\mathbb{N}}$ converges in $\mathbb{P}$-probability towards X, then there exists a subsequence of $(X_n)_{n\in\mathbb{N}}$ converging to X $\mathbb{P}$-almost surely.*
2. *If $(X_n)_{n\in\mathbb{N}}$ converges according to the norm L^p to X, then there exists a subsequence of $(X_n)_{n\in\mathbb{N}}$ converging to X $\mathbb{P}$-almost surely.*

Proposition

If $(X_n)_{n\in\mathbb{N}}$ converges $\mathbb{P}$-almost surely or in expectation to X, then $(X_n)_{n\in\mathbb{N}}$ converges to X in $\mathbb{P}$-probability.

Convergence of sequences of random variables

Definition

$(X_n)_{n\in\mathbb{N}}$ **converges in distribution** *towards* X *means that, for any function* f *from* $\mathbb{R}$ *to* $\mathbb{R}$*, continuous and bounded:*

$$lim_{n\to\infty}\mathbb{E}\left[f\left(X_n\right)\right] = \mathbb{E}\left[f\left(X\right)\right].$$

Convergence of sequences of random variables

Theorem

There is equivalence between the following assertions:

1. $(X_n)_{n\in\mathbb{N}}$ *converges in distribution towards* X.
2. *For every closed set* F *of* $\mathbb{R}$, $\limsup_{n\to\infty}\mathbb{P}(X_n \in F) \leq \mathbb{P}(X \in F)$.
3. *For every open set* O *of* $\mathbb{R}$, $\liminf_{n\to\infty}\mathbb{P}(X_n \in O) \geq \mathbb{P}(X \in O)$.

Theorem

For any integer n, *let* F_{X_n} *and* F_X *be the distribution functions of respectively* X_n *and* X:
$F_{X_n}(x) = \mathbb{P}(X_n \leq x)$ *and* $F_X(x) = \mathbb{P}(X \leq x)$.
We then have equivalence between the two following assertions:

1. $(X_n)_{n\in\mathbb{N}}$ *converges in distribution towards* X.
2. $\forall x \in \mathbb{R}$, *so that* F_X *is continuous in* x, $F_{X_n}(x)$ *converges towards* $F_X(x)$.

Independence

Events

Definition

The events $A, B \in \mathcal{F}$ *are independents means that* $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Definition

The family of events $(A_i)_{i \in I} \in \mathcal{F}^I$*, where* $I \subseteq \mathbb{N}$*, is independent means that for any subset* J *of* I *comprising a finite number of elements,* $\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i)$.

Random variables

Definition

The random variables $X_1 : (\Omega, \mathcal{F}) \to (E_1, \mathcal{E}_1)$ *and* $X_2 : (\Omega, \mathcal{F}) \to (E_2, \mathcal{E}_2)$ *are independent means that* $\forall A_1 \in \mathcal{E}_1,\ A_2 \in \mathcal{E}_2,\ \mathbb{P}(X_1 \in A_1, X_2 \in A_2) = \mathbb{P}(X_1 \in A_1)\mathbb{P}(X_2 \in A_2)$.

Definition

The family of random variables $(X_i)_{i \in I}$*, where* $\forall i \in I, X_i : (\Omega, \mathcal{F}) \to (E_i, \mathcal{E}_i)$ *and* $I \subseteq \mathbb{N}$*, is independent means that for any subset* J *of* I *comprising a finite number of elements,* $\mathbb{P}\left(\bigcap_{i \in J} X_i \in A_i\right) = \prod_{i \in J} \mathbb{P}(X_i \in A_i)$.

Independence

Proposition

We have equivalence between the following two assertions:

1. *The random variables* $X_1 : (\Omega, \mathcal{F}) \to (E_1, \mathcal{E}_1)$ *and* $X_2 : (\Omega, \mathcal{F}) \to (E_2, \mathcal{E}_2)$ *are independent for the probability measure* $\mathbb{P}$*.*
2. - $\forall f_1 : E_1 \to \mathbb{R}$ $\mathcal{E}_1$*-measurable and bounded, or positive;*
   - *et* $\forall f_2 : E_2 \to \mathbb{R}$ $\mathcal{E}_2$*-measurable and bounded, or positive;*

   one has the identity $\mathbb{E}\left(f_1(X_1) f_2(X_2)\right) = \mathbb{E}\left(f_1(X_1)\right) \mathbb{E}\left(f_2(X_2)\right)$*.*

Characteristic function

Definition

The **characteristic function** *of the real random variable X is the function ϕ_X, defined from $\mathbb{R}$ to $\mathbb{C}$ by the relation:*

$$\phi_X(t) = \mathbb{E}\left[e^{itX}\right].$$

Proposition

The mapping $F_X \mapsto \phi_X$, where F_X is the cdf of the random variable X is injective. Thus, the characteristic function characterizes the distribution of a random variable: if two random variables have the same characteristic function, then they have the same distribution.

Definition

The **Laplace transform** *of the real random variable X is the function L_X, defined from $\mathbb{C}$ to $\mathbb{C}$ by the relation:*

$$L_X(a) = \mathbb{E}\left[e^{aX}\right].$$

Some inequalities

Theorem (Jensen's inequality)

Let f be a convex function on $]a,b[$ and X be a random variable of finite expectation, with values in $]a,b[$. Then:

$$f(\mathbb{E}(X)) \leq \mathbb{E}(f(X)).$$

Theorem (Markov inequality)

Let X be a real random variable that is almost surely positive or zero, and a be a strictly positive real number. Then:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}(X)}{a}.$$

Theorem (Bienaymé-Tchébychev inequality)

Let X be a random variable with finite expectation μ and variance σ^2 – by Jensen's inequality, finite variance implies finite expectation. Then, for all $\alpha > 0$, we have:

$$\mathbb{P}(|X - \mu| \geq \alpha) \leq \frac{\sigma^2}{\alpha^2}.$$

Distribution of the sum

Theorem (Weak law of large numbers)

If the expectation and variance of the random variable X_1 are finite, then we have the convergence in probability of the arithmetic mean of the i.i.d. X_i towards the expectation of X_1, therefore:

$$\frac{S_n - n\mathbb{E}(X_1)}{n} \longrightarrow_{\mathbb{P},n\to\infty} 0.$$

Distribution of the sum

Theorem (Kolmogorov's strong law of large numbers)

If the expectation of the random variable X_1 is finite, then, we have the almost sure convergence of the arithmetic mean of the i.i.d. X_i towards the expectation of X_1, therefore:

$$\frac{S_n - n\mathbb{E}(X_1)}{n} \longrightarrow_{p.s.,n\to\infty} 0.$$

Theorem (Central limit theorem)

If the expectation and variance of the i.i.d. random variables X_i are finite, then we have the law of deviations from the mean of S_n :

$$\frac{S_n - n\mathbb{E}(X_1)}{\sqrt{nVar(X_1)}} \longrightarrow_{law,n\to\infty} \mathcal{N}(0,1),$$

where $\mathcal{N}(\mu,\sigma^2)$ is the normal law of expectation μ and variance σ^2.

Distribution of the sum

Definition

A probability law F is **stable** *if for all random variables X, Y, and Z with distribution function F and for all $\alpha, \beta \geq 0$, there exist γ and δ such that:*

$$\alpha X + \beta Y \stackrel{d}{=} \gamma Z + \delta,$$

where $\stackrel{d}{=}$ stands for the equality in distribution.

Theorem

Let a_n and $b_n > 0$ be the terms of a sequence such that, when $n \to \infty$:

$$\frac{S_n - a_n}{b_n} \xrightarrow{d} G,$$

where G is a non-degenerate distribution function.
Then G is of the same type as a stable distribution.

Distribution of the sum

Proposition

The characteristic function of a stable distribution with parameters α, β, μ, γ (with $\gamma > 0$) is necessarily of the following form:

- *for a 2-stable distribution (special case of the following case) we find a Gaussian distribution, with γ half the variance:*

$$\phi_X(t) = \exp(-\gamma t^2) \;;$$

- *for $\alpha \in [0, 2]$ but non-integer and $\beta \in [-1, 1]$:*

$$\phi_X(t) = \exp\left(i\mu t - \gamma|t|^\alpha \left[1 - i\beta sign(t) tan\left[\frac{\pi\alpha}{2}\right]\right]\right) \;;$$

- *for $\alpha = 1$ and $\beta \in [-1, 1]$, we find a Cauchy distribution:*

$$\phi_X(t) = \exp\left(i\mu t - \gamma|t|\left[1 + i\beta sign(t)\frac{2}{\pi}\log|t|\right]\right).$$

Nature of the limit distribution

Let $(X_i)_{i\in\mathbb{N}^+}$ be a sequence of independent, identically distributed random variables, with F as the distribution function. For a sample composed of the first n terms of this sequence, where $n \in \mathbb{N}$, we define the **order statistic** $(X_{i:n})_{i\in\mathbb{N}^+\cap[1,n]}$ by:

$$\min\{X_1,...,X_n\} = X_{1:n} \leq X_{2:n} \leq ... \leq X_{n-1:n} \leq X_{n:n} = \max\{X_1,...,X_n\} \ ;$$

this is a reordering of the sample.

Definition

The random variables X and Y are said to be of the same type if there exists $a \in \mathbb{R}$ and $b > 0$, such that:

$$X \overset{d}{=} bY + a.$$

Nature of the limit distribution

Proposition

Let F be a cdf, $(a_n)_{n\in\mathbb{N}}$ and $(a'_n)_{n\in\mathbb{N}}$ sequences of real numbers, and $(b_n)_{n\in\mathbb{N}}$ and $(b'_n)_{n\in\mathbb{N}}$ sequences of strictly positive real numbers, G and G' non-degenerate distribution functions, with the following convergences:

$$\begin{cases} \lim_{n\to\infty} F^n(a_n + b_n x) = G(x) \\ \lim_{n\to\infty} F^n(a'_n + b'_n x) = G'(x) \end{cases},$$

in all the x for which we have continuity.
Then, there exists a and b real numbers, with $b > 0$, such that:

$$\begin{cases} \lim_{n\to\infty} \frac{b'_n}{b_n} = b \\ \lim_{n\to\infty} \frac{a'_n}{a_n} = a \end{cases},$$

and G and G' are of the same type.

Nature of the limit distribution

Proposition

The following functional equation:

$$\forall x, y > 0,\ h(xy) = h(x)h(y)$$

admits the general solution:

$$\exists \theta \in \mathbb{R}, \forall x > 0,\ h(x) = x^{\theta}.$$

Nature of the limit distribution

Theorem (Fischer-Tippett theorem)

Let $(X_i)_{i\in\mathbb{N}^+}$ a sequence of i.i.d. random variables with F as the cdf. Suppose that there exists a normalization sequence (a_n, b_n) such that $b_n^{-1}(X_{n:n} - a_n)$ converges in distribution to a non-degenerate distribution with cdf G.
Then, G is of the same type as one of the following three distributions:

1. *Fréchet distribution, whose cdf, for $\xi > 0$, is:*

$$\Phi_\xi(x) = \begin{cases} 0, \ \textit{if} \ x \leq 0 \\ e^{-x^{-\xi}}, \ \textit{else}. \end{cases}$$

2. *Weibull distribution, whose cdf, for $\xi > 0$, is:*

$$\Psi_\xi(x) = \begin{cases} e^{-(-x)^{\xi}}, \ \textit{if} \ x \leq 0 \\ 1, \ \textit{else}. \end{cases}$$

3. *Gumbel distribution, whose cdf is:*

$$\Lambda(x) = e^{-e^{-x}}.$$

Nature of the limit distribution

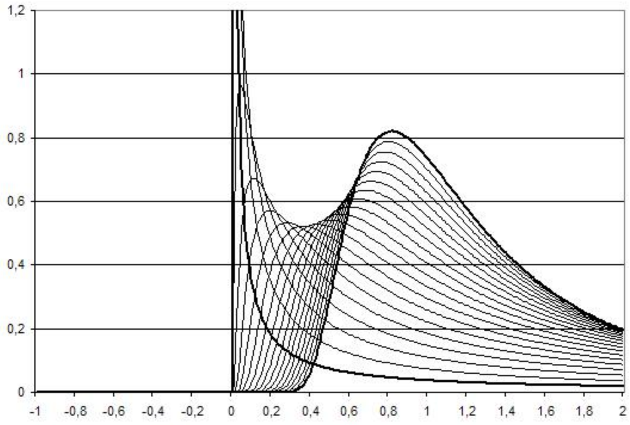


Figure: Fréchet pdf for various values of ξ, between 0.1 and 2.

Nature of the limit distribution

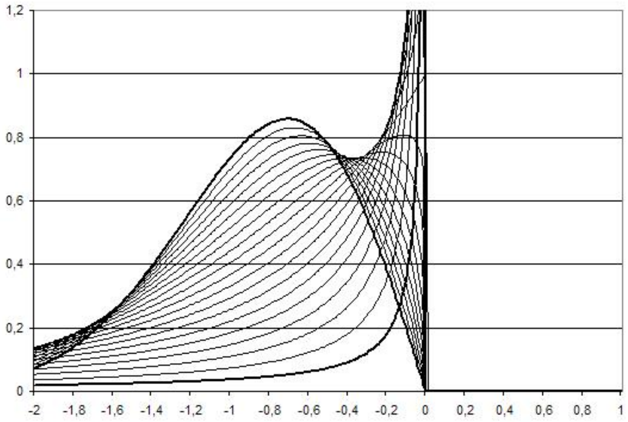


Figure: Weibull pdf for various values of ξ, between 0.1 and 2.

Nature of the limit distribution

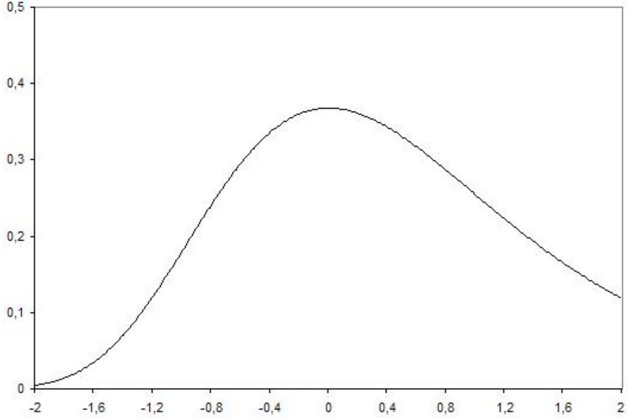


Figure: Gumbel pdf.

Nature of the limit distribution

Note that we can replace the three limiting laws we mentioned by a single law, called the generalized extreme values (GEV) distribution, introduced by Jenkinson and von Mises, which we write in the following simplified form:

$$G_\xi(x) = \begin{cases} \exp\left(-(1+\xi x)_+^{-1/\xi}\right) & \text{if } \xi \neq 0 \\ \exp\left(-e^{-x}\right) & \text{if } \xi = 0. \end{cases}$$

Remark

Then:

- *if $\xi > 0$, the GEV is of Fréchet kind,*
- *if $\xi = 0$, the GEV is of Gumbel kind,*
- *if $\xi < 0$, the GEV is of Weibull kind.*

Nature of the limit distribution

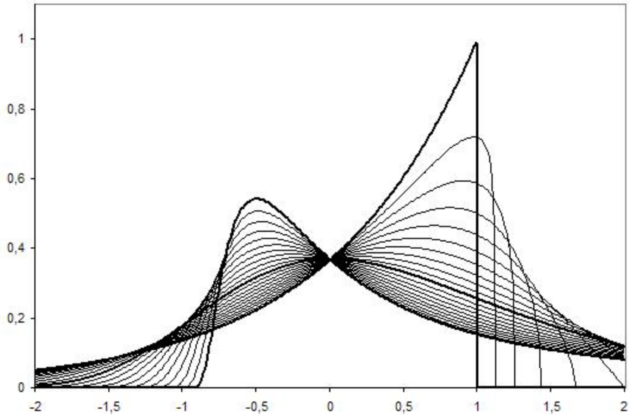


Figure: GEV pdf for various values of the parameter ξ, between -1 and 1.

Existence of the limit distribution

Definition

Let $\xi \in \mathbb{R}$. If G_ξ is the limit cdf of the maximum of a sequence of i.i.d. random variables $(X_i)_{i\in\mathbb{N}}$ of cdf F, then we say that F belongs to the **(max-)domain of attraction** *of a GEV law of parameter ξ. We also write:*

$$F \in MDA(G_\xi).$$

Definition

Let X be a random variable with an absolutely continuous pdf f and a cdf F. We call **reciprocal hazard function** *the function h defined for all x by:*

$$h(x) = \frac{1 - F(x)}{f(x)}.$$

Existence of the limit distribution

Theorem (Theorem of von Mises)

If for $x^M = \sup\{x|F(x) < 1\}$, the upper bound of the support of F, there exists $\xi \in \mathbb{R}$:

$$\lim_{x \to x^M} h'(x) = \xi,$$

then F belongs to the max-domain of attraction of the GEV distribution with parameter ξ. Moreover, the parameters a_n and b_n of Fischer-Tippett theorem are defined by:

$$\begin{cases} 1 - F(a_n) = \frac{1}{n} \\ b_n = h(a_n). \end{cases}$$

Existence of the limit distribution

Theorem

A cdf F belongs to the domain of attraction of the GEV distribution with parameter $\xi \in \mathbb{R}$ if and only if, for all $x, y > 0$ with $y \neq 1$, we have:

$$\lim_{t\to\infty} \frac{U(tx) - U(t)}{U(ty) - U(t)} = \begin{cases} \frac{x^\xi - 1}{y^\xi - 1} & \text{if } \xi \neq 0 \\ \frac{\log(x)}{\log(y)} & \text{if } \xi = 0, \end{cases}$$

where U is defined, for $t > 1$, by:

$$U(t) = F^{-1}\left(1 - \frac{1}{t}\right),$$

with F^{-1} the generalized inverse of F.

Existence of the limit distribution

- The **Fréchet** max-domain of attraction groups together the **thick-tailed** distributions, often polynomial, but also the Cauchy distribution, Student distribution, etc.
- The **Gumbel** max-domain of attraction groups together the **thin-tailed** distributions, such as the exponential and other close distributions: normal, log-normal, gamma, etc.
- The **Weibull** max-domain of attraction groups together the distributions with **finite support**, such as the uniform distribution or the beta distribution.

Existence of the limit distribution

Definition

A mapping $u : \mathbb{R}^+ \to \mathbb{R}$ *is said to be of* **regular variations (at infinity)** *if there exists a mapping* v *such that, for all* $x > 0$:

$$\lim_{t \to \infty} \frac{u(tx)}{u(t)} = v(x).$$

Definition

A positive mapping $u : \mathbb{R}^+ \to \mathbb{R}^+$ *is said to be of* **regular variation of index** α *if, for all* $x > 0$:

$$\lim_{t \to \infty} \frac{u(tx)}{u(t)} = x^{\alpha}.$$

In particular, if $\alpha = 0$*, the mapping is said to be of* **slow variation**.

Definition

A slow-variation function $u : \mathbb{R}^+ \to \mathbb{R}^+$ *is said to be of* **regular variation at the second order** *if there exists a positive function* ρ *such that, for all* $x > 0$:

$$\lim_{t \to \infty} \frac{u(t + x\rho(t))}{u(t)} = e^{x}.$$

Existence of the limit distribution

Theorem (Fréchet-Gnedenko theorem)

A cdf F belongs to the max-domain of attraction of **Fréchet** *with GEV parameter $\xi > 0$ if and only if $1 - F$ is of regular variation with index $-1/\xi$.*

Theorem (Gnedenko theorem)

A cdf F belongs to the max-domain of attraction of **Weibull** *with GEV parameter $\xi < 0$ if and only if $x^M = \sup\{x|F(x) < 1\} < \infty$ and $x \mapsto 1 - F\left(x^M - \frac{1}{x}\right)$ is of regular variation of index $1/\xi$.*

Theorem

A cdf F belongs to the max-domain of attraction of **Gumbel** *if and only if the function $1 - F$ is slowly varying at the second order.*

Theoretical examples

1. Uniform distribution: for all $x \in [0, \tau]$, $F(x) = x/\tau$.
2. Exponential distribution: for all $x \geq 0$, $F(x) = 1 - e^{-\lambda x}$.

Nature of the limit distribution

Definition

The **Generalized Pareto distribution (GPD)** *is defined by the cdf:*

$$H_\xi(x) = \begin{cases} 1-(1+\xi x)^{-1/\xi} & \text{if } \xi \neq 0 \\ 1-\exp(-x) & \text{else,} \end{cases}$$

where $x \geq 0$ *if* $\xi \geq 0$, *and* $0 \leq x < -1/\xi$ *if* $\xi \leq 0$. *The case* $\xi = 0$ *is obtained by extending by continuity.*

We can also generalize our standard form by replacing x by $(x-\mu)/\sigma$.

Nature of the limit distribution

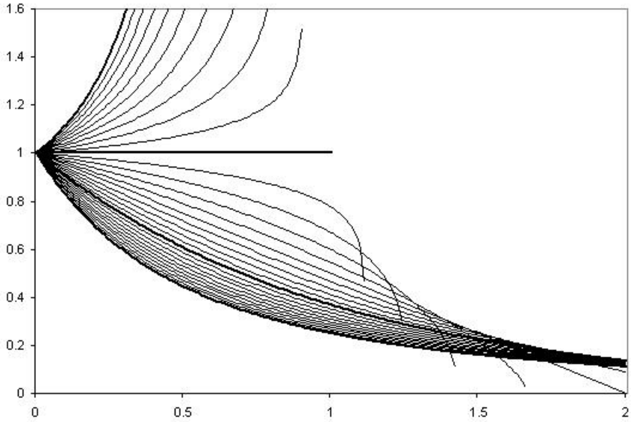


Figure: Probability density function of the GPD for various values of the parameter ξ, between -2 and 1.

Nature of the limit distribution

Definition

The **mean excess function** *of a random variable is the function defined by the relation:*

$$e(u) = \mathbb{E}(X - u | X > u).$$

Theorem (Pickands-Balkema-de Haan theorem)

Let F be a cdf. We have equivalence between

1. *F belongs to the max-domain of attraction of a GEV distribution.*
2. *The distribution of excesses is of type GPD with parameter ξ, that is:*

$$\lim_{u \to x^M} \sup_{0 < x < x^M - u} |\mathbb{P}(X - u \leq x | X > u) - H_\xi(x/\sigma(u))| = 0,$$

with σ a positive function of the threshold u.

Once again, we can evaluate the GPD for the theoretical examples of a uniform or an exponential distribution of the i.i.d. random variables.

Pickands estimator

Theorem

Let (X_n) be a sequence of i.i.d. random variables, whose cdf F belongs to the max-domain of attraction of a GEV distribution of parameter $\xi \in \mathbb{R}$. Let k be a function $\mathbb{N} \to \mathbb{N}$. If

$$\lim_{n\to\infty} k(n) = \infty$$

and

$$\lim_{n\to\infty} \frac{k(n)}{n} = 0,$$

then, **the Pickands estimator***, defined by:*

$$\xi^P_{k(n),n} = \frac{1}{\log(2)} \log\left(\frac{X_{n-k(n)+1:n} - X_{n-2k(n)+1:n}}{X_{n-2k(n)+1:n} - X_{n-4k(n)+1:n}}\right)$$

converges **in probability** *to ξ.*
Moreover, if

$$\lim_{n\to\infty} \frac{k(n)}{\log(\log(n))} = \infty,$$

then the convergence of $\xi^P_{k(n),n}$ towards ξ is **almost sure** *and not only in probability.*

Hill estimator

Theorem

Let (X_n) be a sequence of i.i.d. random variables, whose cdf F belongs to the max-domain of attraction of a GEV distribution of parameter $\xi > 0$. Let k be a function $\mathbb{N} \to \mathbb{N}$. If

$$\lim_{n\to\infty} k(n) = \infty$$

and

$$\lim_{n\to\infty} \frac{k(n)}{n} = 0,$$

then, **the Hill estimator**, *defined by:*

$$\xi^H_{k(n),n} = \frac{1}{k(n)} \sum_{i=n-k(n)+1}^{n} \log\left(\frac{X_{i:n}}{X_{n-k(n)+1:n}}\right)$$

converges **in probability** *to ξ.*
Moreover, if

$$\lim_{n\to\infty} \frac{k(n)}{\log(\log(n))} = \infty,$$

then the convergence of $\xi^P_{k(n),n}$ towards ξ is **almost sure** *and not only in probability.*

Value at risk – theoretical

Definition (reminder)

Let X be a random variable (a gain) with distribution function F. The value at risk (VaR) is the function defined by:

$$VaR(\alpha) = -F^{-1}(1-\alpha) = -\inf\left\{x|F(x) \geq 1-\alpha\right\}.$$

Note that we have an upper bound on the VaR thanks to the Bienaymé-Tchébychev inequality, from which we obtain (if $VaR(\alpha) + \mu \geq 0$) :

$$VaR(\alpha) \leq \frac{\sigma}{\sqrt{1-\alpha}} - \mu,$$

where σ^2 is the variance of X and μ its expected value.

Value at risk – empirical

Theorem (Lindeberg-Lévy theorem)

Let $p \in [0,1]$. For a loss X, the quantile $X_{n-\lfloor np \rfloor+1:n}$ converges in probability towards $VaR(p)$.

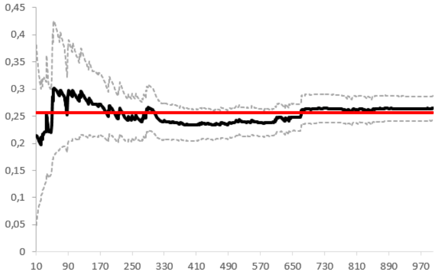
Theorem

Let $p \in]0,1[$, X_i be i.i.d. random variables with a pdf f that is continuous in x_p and such that $f(x_p) > 0$. Then, for $k(n) = np + o(\sqrt{n})$, we have the following convergence in distribution:

$$\sqrt{n}\left(X_{k(n):n} - x_p\right) \xrightarrow[n\to\infty]{d} \mathcal{N}\left(0, \frac{p(1-p)}{f(x_p)^2}\right).$$

The consequence is that, for $\alpha > 0$, the random interval $\left[X_{k(n):n} \pm \frac{q_\alpha \sqrt{p(1-p)}}{f(X_{k(n):n})\sqrt{n}}\right]$, where q_α is the quantile of order $1-\alpha/2$ of the distribution $\mathcal{N}(0,1)$, is a confidence interval for x_p, with an asymptotic level $1-\alpha$.

Value at risk – empirical



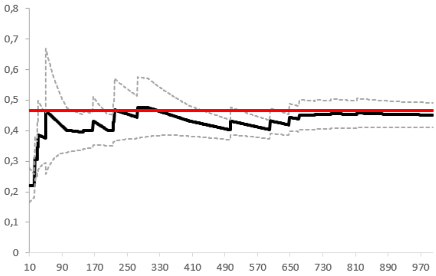


Figure: Empirical VaR at 90% (left) and 99% (right), with 95% confidence intervals (grey) and theoretical value (red), as a function of the number of observations. These observations are Gaussian random variables with standard deviation 0.2 (same numbers in the two cases).

Value at risk – EVT

Let M_n be the maximum of n i.i.d. random variables (losses) with distribution function F, such that there exists a_n, b_n such that $a_n^{-1}(M_n - b_n))$ converges in law to a GEV with parameter ξ. For k fixed (true in particular for $k = 1$) and n large, we have:

$$\mathbb{P}(a_{n/k}^{-1}(M_{n/k} - b_{n/k})) \leq x) = F(xa_{n/k} + b_{n/k})^{n/k} \approx G_\xi(x).$$

Therefore, the VaR of probability p is here such that:

$$\begin{array}{rcl} p & = & F(VaR(p)) \\ & \approx & G_\xi\left(\frac{VaR(p)-b_{n/k}}{a_{n/k}}\right)^{k/n} \\ & = & \exp\left\{-\frac{k}{n}\left(1+\xi\frac{VaR(p)-b_{n/k}}{a_{n/k}}\right)^{-1/\xi}\right\} \\ & \approx & 1-\frac{k}{n}\left(1+\xi\frac{VaR(p)-b_{n/k}}{a_{n/k}}\right)^{-1/\xi}. \end{array}$$

So the VaR is $VaR(p) = \frac{\left(\frac{k}{n(1-p)}\right)^{\xi}-1}{\xi} a_{n/k} + b_{n/k}$, where it remains to choose the $a_{n/k}$ and $b_{n/k}$ in accordance with the estimators of ξ (Hill ou Pickands).

Value at risk – EVT

Still working on variables representing losses, we obtain (depending on k):

Pickands estimator

$$VaR(p) = \frac{\left(\frac{k}{n(1-p)}\right)^{\xi^P} - 1}{1 - 2^{-\xi^P}} (X_{n-k+1:n} - X_{n-2k+1:n}) + X_{n-k+1:n},$$

where ξ^P is the Pickands estimator of the GEV parameter.

Hill estimator

$$VaR(p) = \left(\frac{k}{n(1-p)}\right)^{\xi^H} X_{n-k+1:n},$$

where ξ^H is the Hill estimator of the GEV parameter.

Value at risk – EVT

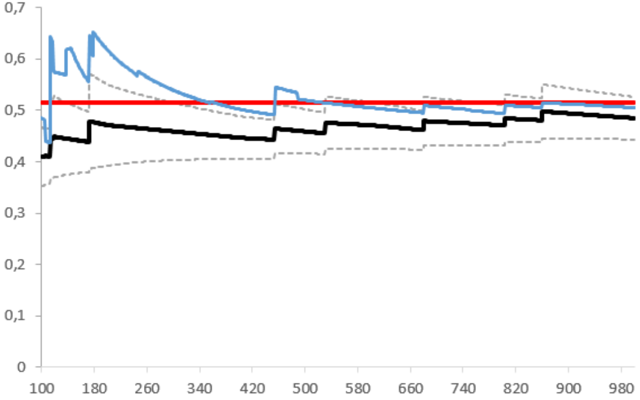


Figure: VaR at 99.5%, empirical (black) with its 95% confidence intervals (grey) and theoretical value (red), and EVT VaR (Pickands, blue) as a function of the number of observations. These observations are Cauchy random variables, $F(x) = \frac{1}{\pi}\arctan\left(\frac{x}{a}\right) + \frac{1}{2}$, with scale parameter 0.2. More rapid convergence for EVT-VaR. Relevant when in the confidence interval of the empirical one (more careful to confront them).

Practical examples

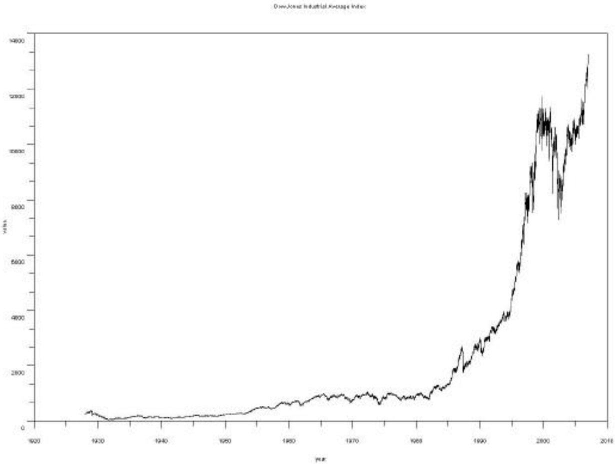


Figure: Dow Jones index between 1928 and 2007.

Practical examples

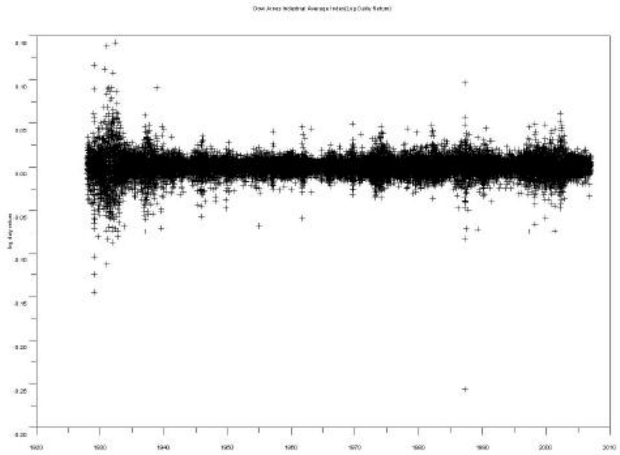


Figure: Daily logarithmic return of the Dow Jones index between 1928 and 2007.

Practical examples

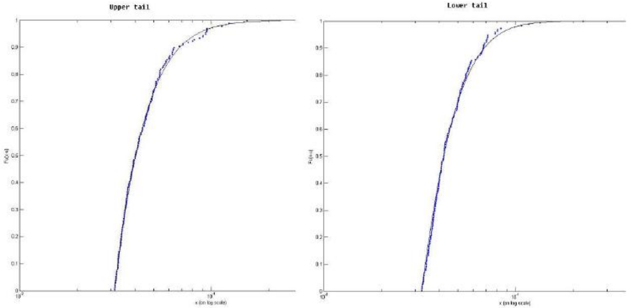


Figure: Empirical distribution of the estimated tail of gains and tail of losses distributions for daily logarithmic returns of the Dow Jones index between 1928 and 2007.

Practical examples

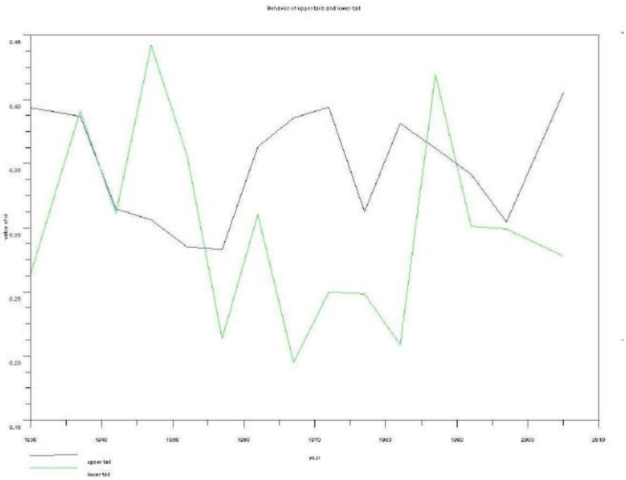


Figure: Hill estimators for the gain tail and loss tail for daily log returns of the Dow Jones index between 1928 and 2007.

Practical examples

Action (indice)	paramètres	Variation des paramètres	Seuil	Probabilité de ne pas franchir le seuil	Max
1.Dow Jones	$\hat{\xi}$ =0.1967 β=0.0114	Δ$\hat{\xi}$ =0.0693 Δβ=0.0010	0.0257	98.48%	0.1427
2.Nasdaq	0.1300 0.0120	0.0935 0.0015	0.0257	98.25%	0.1325
3.Coca-Cola	0.2153 0.0106	0.1098 0.0016	0.0366	98.12%	0.1796
4.Gen. Elect	0.2499 0.0102	0.1544 0.0018	0.0366	98.08%	0.1174
5.Microsoft	0.1446 0.0155	0.0687 0.0014	0.0366	94.40%	0.1788
6.Morgan Stanley	0.0483 0.0171	0.0684 0.0016	0.0366	93.76%	0.1488

Figure: Estimators for the tail of gains.

Practical examples

Action(indice)	Paramètres	Variation des paramètres	Seuil	Probabilité de ne pas franchir le seuil	Max
1	$\hat{\xi}$ =0.1710 β =0.0126	$\Delta\hat{\xi}$ =0.0652 $\Delta\beta$ =0.0011	-0.0288	98.58%	-0.2563
2	0.2213 0.0097	0.0989 0.0012	-0.0289	98.14%	-0.1204
3	0.3902 0.0106	0.1460 0.0018	-0.0336	98.12%	-0.2836
4	0.2038 0.0119	0.1040 0.0016	-0.0336	98.84%	-0.1921
5	0.3341 0.0142	0.0849 0.0015	-0.0336	94.98%	-0.3583
6	0.1297 0,0159	0.0894 0.0018	-0.0335	93.54%	-0.1403

Figure: Estimators for the tail of losses.

Practical examples

Action	p=0.01	P=0.005	p=0.001	p=0.0005
3	-0.0412	-0.0519	-0.0916	-0.1180
4	-0.0436	-0.0539	-0.0844	-0.1010
5	-0.0640	-0.0831	-0.1486	-0.1897
6	-0.0671	-0.0818	-0.1215	-0.1413

Figure: The *VaR* for some probabilities.

Practical examples

Action	s=-0.05	s=-0.10	s=-0.20	s=-0.30
3	0.56%	0.0786%	0.0122%	0.0042%
4	0.64%	0.0520%	0.0029%	0.0005%
5	1.90%	0.3000%	0.0430%	0.0133%
6	2.45%	0.2300%	0.0087%	0.0009%

Figure: The probabilities of excess for some thresholds.

Practical examples

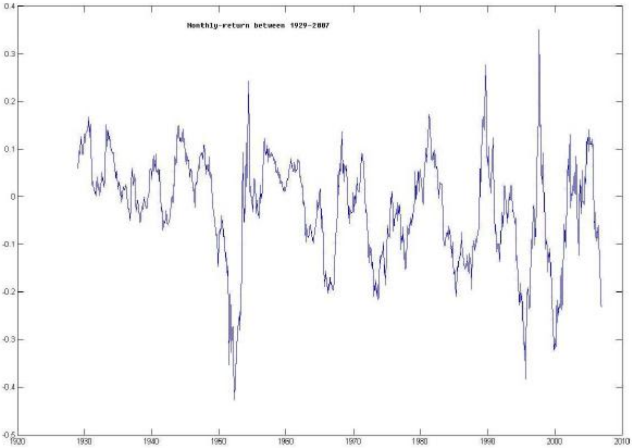


Figure: Monthly returns for the Dow Jones Index between 1928 and 2007.

Practical examples

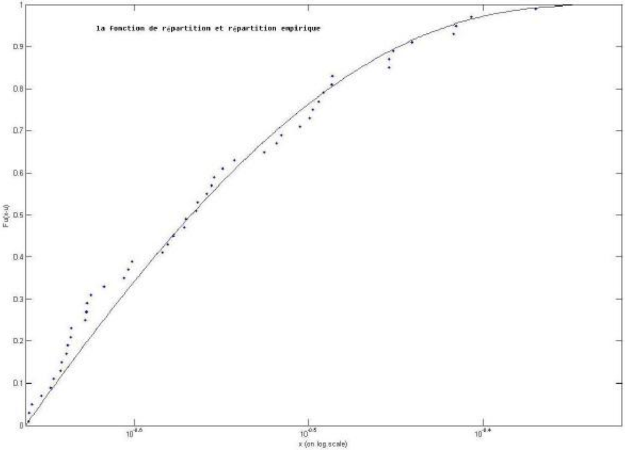


Figure: Estimation of the distribution function of monthly returns for the Dow Jones index between 1928 and 2007.

Practical examples

Seuil	-0.10	-0.14	-0.18	-0.20	-0.25	-0.30	-0.50
probabilité	38.52%	26.93%	18.40%	15.13%	9.2%	5.53%	0.76%

Figure: Excess probabilities for some thresholds for monthly returns for the Dow Jones index between 1928 and 2007.

Practical examples

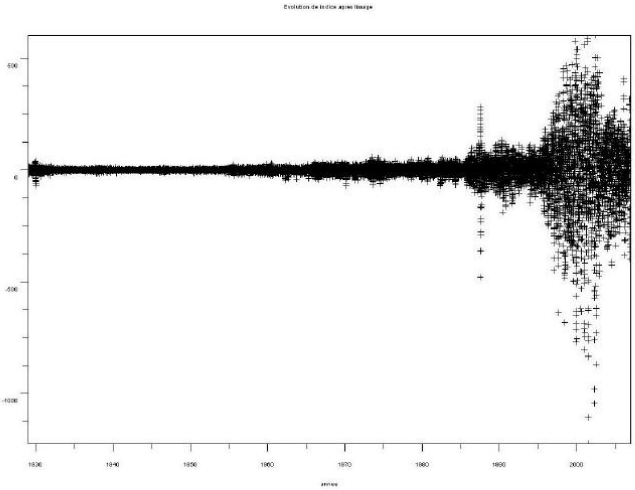


Figure: The Dow Jones Index between 1928 and 2007, with its trend removed.

Practical examples

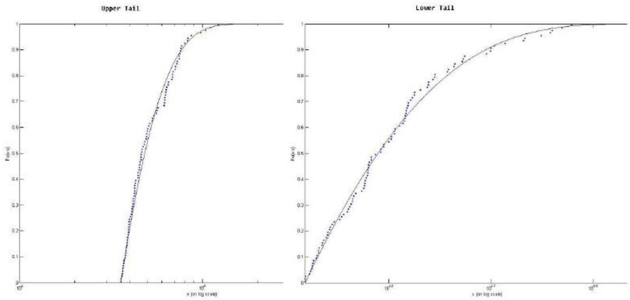


Figure: Estimation of the distribution function of gains and losses of the Dow Jones index between 1928 and 2007, from which its trend has been removed.

Serial dependence

- We assume i.i.d. random variables in EVT. We want to get rid of independence, but not accept any dependence (the case $X_i = X_1$ is trivial for example, because the maximum is distributed as X_1, so it does not admit a limit distribution).
- Constraint on the dependence: two extreme events become approximately independent if they are separated by enough time (no long-term dependence). This is a weak **mixing condition**.
- This is expressed by the condition called $D(u_n)$ (Leadbetter, 1983): for $n \in \mathbb{N}$, a threshold $u_n \in \mathbb{R}$, a time gap $l_n \in \mathbb{N}$, and for times $1 \leq i_1 < \dots < i_p < j_1 < \dots < j_q \leq n$ with $j_1 - i_p \geq l_n$, the condition is written

$$\left| \begin{array}{l} \mathbb{P}\left(X_{i_1} \leq u_n, \dots, X_{i_p} \leq u_n, X_{j_1} \leq u_n, \dots, X_{j_q} \leq u_n\right) \\ -\mathbb{P}\left(X_{i_1} \leq u_n, \dots, X_{i_p} \leq u_n\right) \mathbb{P}\left(X_{j_1} \leq u_n, \dots, X_{j_q} \leq u_n\right) \end{array} \right| < \alpha_{n,l_n},$$

  with $\alpha_{n,l_n} \overset{n\to\infty}{\longrightarrow} 0$ if $l_n/n \overset{n\to\infty}{\longrightarrow} 0$.
- $D(u_n)$ is verified for Gaussian process if correlation decreases faster than the logarithm of the lag (therefore valid AR models since geometric decrease).

Serial dependence

Theorem (Leadbetter's theorem)

Let $X_1, ..., X_n$ be identically distributed. If there exists $b_n > 0$ and a_n such that $b_n^{-1}(X_{n:n} - a_n)$ converges in law to G non-degenerate and **if** $D(u_n)$ **is true with** $u_n = b_n x + a_n$ **for each** x **such that** $G(x) > 0$*, then G is as in the Fischer-Tippett theorem (i.e. of the same type as a GEV distribution).*

Theorem (Leadbetter's extremal index theorem)

Let $X_1, ..., X_n$ be identically distributed. Let $\hat{X}_1, ..., \hat{X}_n$ distributed like X_1 with an additional hypothesis of independence between $\hat{X}_1, ..., \hat{X}_n$. If there exists $b_n > 0$ and a_n such that $b_n^{-1}(\hat{X}_{n:n} - a_n)$ converges in distribution to G non-degenerate and **if** $D(u_n)$ **is true for** $X_1, ..., X_n$ **with** $u_n = b_n x + a_n$ **for each** x **such that** $G(x) > 0$*, then $b_n^{-1}(X_{n:n} - a_n)$ converges in distribution to G^θ, where $\theta \in [0, 1]$.*

Serial dependence

θ is the extremal index. It is equal to 1 if the series is independent. The greater the dependence, the lower is θ. Estimation by de-clustering.

- **Blocks declustering**: we choose blocks of arbitrary size b and we partition $\{1, ..., n\}$ into $k = \lfloor n/b \rfloor$. We count the number of clusters, therefore of blocks for which we have at least one variable X above a threshold.
- **Runs declustering**: the same thing with a sliding window of arbitrary size r.

Thus, if we denote $M_{i,j} = \max\{X_{i+1}, ..., X_j\}$, the two de-clusterings lead to the following estimators, knowing that θ is linked to the ratio of the number of extreme clusters to the number of extremes (if the variables are independent, each extreme constitutes a cluster by itself, so $\theta = 1$; if each extreme is followed by another extreme, $\theta = 1/2$):

$$\hat{\theta}_n^B(u; b) = \frac{\sum_{i=1}^{k} \mathbb{1}(M_{(i-1)b,ib} > u)}{\sum_{i=1}^{kb} \mathbb{1}(X_i > u)} \text{ and } \hat{\theta}_n^R(u; r) = \frac{\sum_{i=1}^{n-r} \mathbb{1}(X_i > u, M_{i,i+r} \leq u)}{\sum_{i=1}^{n-r} \mathbb{1}(X_i > u)}.$$

Some questions regarding multifrequency

The goal of this chapter is to present some aspects of financial risk encountered at several time scales and their interdependence: monthly, daily, hourly, *tick-by-tick* quotes... Here is the kind of question that interests us:

- Is it relevant to annualize a risk measure by multiplying by the square root of the time ratio?
- How can a shock have different repercussions on the price of an asset depending on the risk horizon?
- To what extent would an asset manager have an interest in looking at high-frequency quotes to make decisions over a longer horizon?
- How to define the correlation of two securities in high frequency when the observations (= transactions) do not take place at the same times?
- Can we have a regime change on *intraday* data without it affecting the nature of the daily data?

To answer these questions, two main tools are adapted to the multifrequency world: wavelets and fractals.

Annualizing in a few words

A few questions:

- What is a volatility?
- Why do we need annualized volatility? (think about pricing projects in C++: a reference time scale must be set)
- How to annualize a volatility?

Annualizing in a few words

How to annualize a volatility? A practitioner's rule: the annual volatility is $\sqrt{12}$ the monthly volatility or $\sqrt{52}$ the weekly volatility. Assumptions:

- Gaussian returns,
- identically distributed returns,
- independent returns.

More realistic: a scaling in δ^H instead of $\sqrt{\delta}$. $H \in [0, 1]$ is the Hurst exponent. This defines the fractional Brownian motion:

$$\mathbb{E}\{B_H(t)B_H(s)\} = \frac{\sigma^2}{2}(|t|^{2H} + |s|^{2H} - |t-s|^{2H}).$$

Increments of B_H of duration τ are stationary: centred Gaussian of variance $\sigma^2|\tau|^{2H}$.

- if $H = 1/2$, the fBm is identical to a standard Brownian motion,
- if $H > 1/2$, the increments are persistent,
- if $H < 1/2$, the increments are anti-persistent.

Annualizing a volatility

First approach: standard deviation of a sample of returns

- We observe low-frequency prices (*e.g.* annually), $P_0, P_1, ..., P_T$. We therefore have T **low-frequency** log-returns $\log(P_i/P_{i-1})_{i\in\{1,\dots,T\}}$. For a zero mean, the sample variance is then:

$$Var^{BF} = \frac{1}{T}\sum_{i=1}^{T}\log\left(\frac{P_i}{P_{i-1}}\right)^2.$$

- We observe the same prices at a higher frequency (*e.g.* daily), $P_0, P_{1/n}, P_{2/n}, ..., P_T$. We therefore have $n \times T$ **high-frequency** log-returns $\log(P_{i/n}/P_{(i-1)/n})_{i\in\{1,\dots,nT\}}$. For a zero mean, the sample variance is then:

$$Var^{HF} = \frac{1}{nT}\sum_{i=1}^{nT}\log\left(\frac{P_{i/n}}{P_{(i-1)/n}}\right)^2.$$

Annualizing a volatility

For a geometric Brownian motion without trend $dP_t = \sigma P_t dW_t$, we have:

$$\mathbb{E}(Var^{BF}) = \frac{1}{T}\sum_{i=1}^{T}\sigma^2\mathbb{E}([W_i - W_{i-1}]^2) = \sigma^2$$

and

$$\mathbb{E}(Var^{HF}) = \frac{1}{nT}\sum_{i=1}^{nT}\sigma^2\mathbb{E}([W_{i/n} - W_{(i-1)/n}]^2) = \frac{1}{nT}\sum_{i=1}^{nT}\sigma^2\frac{1}{n} = \frac{\sigma^2}{n}.$$

So, if (in a somewhat abusive way, cf Jensen's inequality, but is volatility a standard deviation or the square root of a variance?) $Vol = \sqrt{\mathbb{E}Var}$:

$$\boxed{Vol^{BF} = Vol^{HF}\sqrt{n}}.$$

Annualizing a volatility

Second approach: estimation of the quadratic variation

For a geometric Brownian motion $dP_t = \mu P_t dt + \sigma P_t dW_t$, the quadratic variation $d[\log P]$ can be written as a limit (in the sense of a convergence in probability), if it exists:

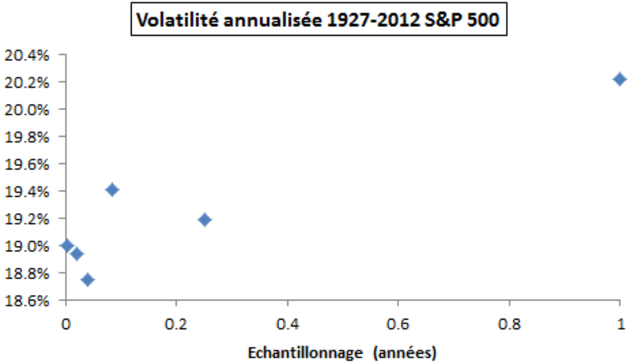
$$\lim_{m\to\infty} \sum_{i=1}^{mT} \log\left(\frac{P_{i/m}}{P_{(i-1)/m}}\right)^2 = d[\log P]_T = \sigma^2 d[W]_T = \sigma^2 T.$$

Taking $m = 1$ and $m = n$ provides two estimators of σ, denoted $\hat{\sigma}_{BF}$ and $\hat{\sigma}_{HF}$:

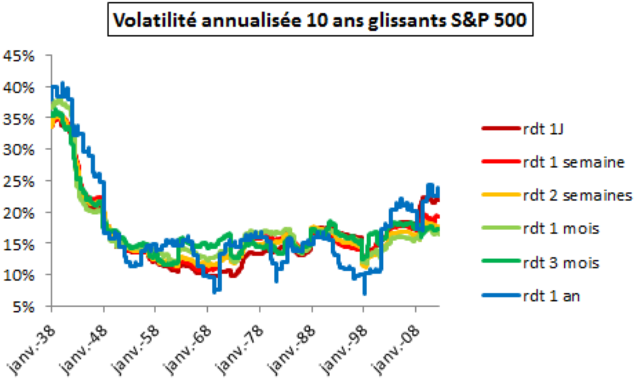
$$\boxed{Vol^{BF} = \hat{\sigma}_{BF}} \text{ and } \boxed{Vol^{HF}\sqrt{n} = \hat{\sigma}_{HF}}.$$

We can only hope that $\hat{\sigma}_{BF}$ and $\hat{\sigma}_{HF}$ are close to each other!

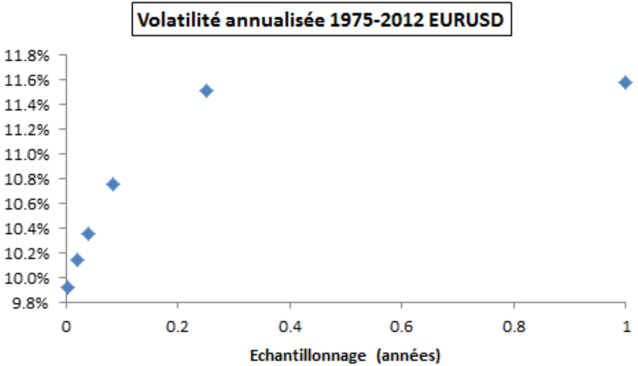
Annualizing a volatility



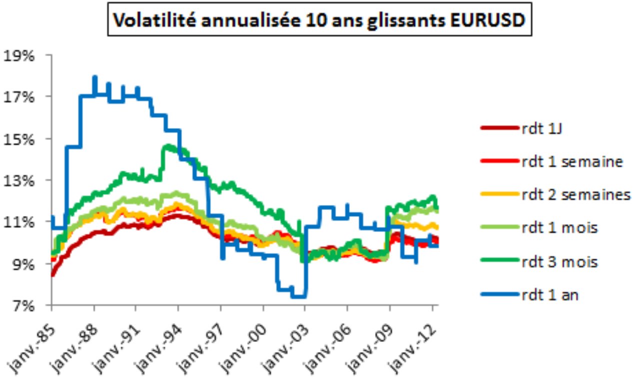
Annualizing a volatility



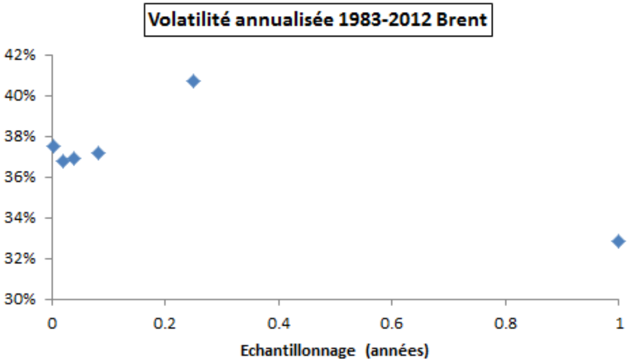
Annualizing a volatility



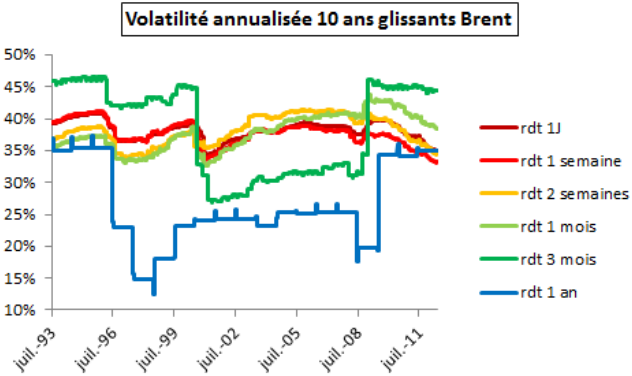
Annualizing a volatility



Annualizing a volatility



Annualizing a volatility



Annualizing a volatility - Confidence intervals

Volatility is therefore estimated from discrete data; a confidence interval can be given.

- For the trend μ, the confidence interval is given by a Student distribution.
- For σ^2, we use a limit theorem (central limit theorem), which, for a finite number of data, therefore gives an approximation of the confidence interval.

Central Limit Theorem

If the expectation and variance of iid random variables $(X_i)_i$ are finite, then we have the law of $S_n = \sum_{i=1}^n X_i$:

$$\frac{\frac{1}{n}S_n - \mathbb{E}(X_1)}{\sqrt{Var(X_1)/n}} \longrightarrow_{law, n\to\infty} \mathcal{N}(0,1),$$

where $\mathcal{N}(m, s^2)$ is the Gaussian distribution of mean m and variance s^2.

Annualizing a volatility - Confidence intervals

If $X_{i/\tau} = \log(P_{i/\tau}/P_{(i-1)/\tau})^2$ is a quadratic price return, with a sampling of τ observations per time unit, with $dP_t = \sigma P_t dW_t$, then, for a sufficiently large number n of observations, the estimator $\hat{\sigma}^2_{n,\tau} = \frac{\tau}{n}\sum_{i=1}^n X_{i/\tau}$ converges in distribution to $\mathcal{N}(\sigma^2, s^2)$, with:

$$\begin{aligned}
s^2 &= Var(X_{1/\tau}\tau)/n \\
&= \tau^2 Var((\sigma W_{1/\tau})^2)/n \\
&= \sigma^4 Var(W_1^2)/n \\
&= \sigma^4 \mathbb{E}((W_1^2 - \mathbb{E}(W_1^2))^2)/n \\
&= \sigma^4 \mathbb{E}((W_1^2 - 1)^2)/n \\
&= \sigma^4 [\mathbb{E}(W_1^4) - 2\mathbb{E}(W_1^2) + 1]/n \\
&= \sigma^4 [\kappa - 1]/n \\
&= \boxed{2\sigma^4/n},
\end{aligned}$$

where the kurtosis of a variable Y with mean m_Y and standard deviation s_Y, $\kappa = \mathbb{E}[((Y - m_Y)/s_Y)^4]$, is 3 for a Gaussian r.v.

Annualizing a volatility - Confidence intervals

- The variance of the squared volatility estimator does not depend on the sampling! ... as long as the number of observations n is equal for all samplings and is large enough to validate the CLT.
- In practice, for a sample of duration T years, we estimate the volatility from $n = T \times \tau$ observations. Then the variance of the squared volatility estimator, $2\sigma^4/(T\tau)$, decreases inversely to the sampling τ.
- Under the assumption of this Gaussian dynamic, the estimator at the highest frequency is therefore the most reliable.

Annualizing a volatility - Confidence intervals

The estimator $\hat{\sigma}_{n,\tau}$ of volatility converges in law toward a Gaussian distribution, after the Delta method.

Delta method

Let $(X_n)_{n\geq 0}$ such that

$$\sqrt{n}(X_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

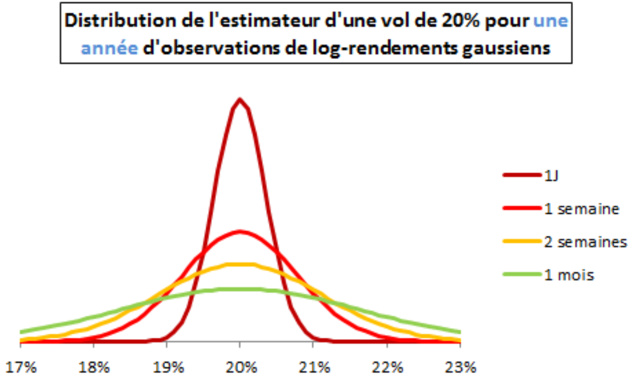
Then, for any function g continuously differentiable in θ and with $g'(\theta) \neq 0$, we have (Taylor expansion with ξ_n in the interval whose bounds are θ and X_i):

$$\sqrt{n}(g(X_n) - g(\theta)) = \sqrt{n}(X_n - \theta)g'(\xi_n) \xrightarrow{d} \mathcal{N}(0, \sigma^2 g'(\theta)^2).$$

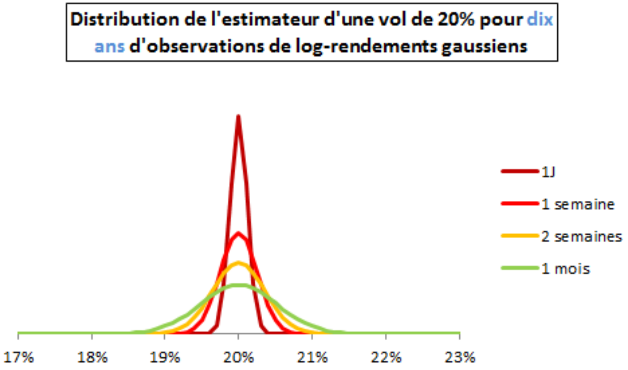
Application with $g : x \mapsto \sqrt{|x|}$:

$$\sqrt{n}(\hat{\sigma}_{n,\tau} - \sigma) \xrightarrow{d} \mathcal{N}\left(0, \frac{\sigma^2}{2}\right).$$

Annualizing a volatility - Confidence intervals

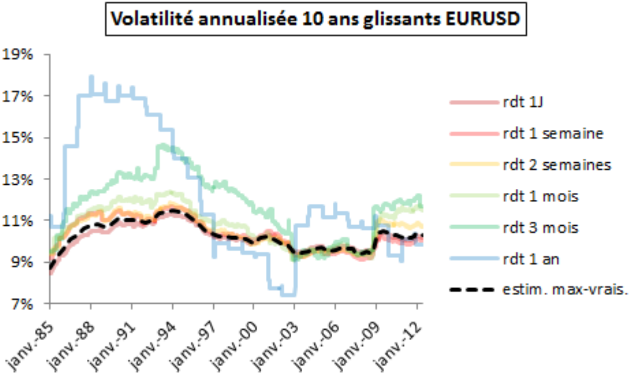


Annualizing a volatility - Confidence intervals



Annualizing a volatility - Confidence intervals

Idea: estimate the volatility of the diffusion by maximum likelihood of the volatilities calculated on different samples.



Annualizing a volatility - Confidence intervals

EURUSD, annualized volatility 1975-2012: **10.05%** based on max-likelihood estimation.

Sampling	Annualized Vol	$\mathbb{P}$ int. symmetric confidence
1 day	9.92%	92.7%
1 week	10.14%	43.1%
2 week	10.36%	81.7%
1 month	10.76%	96.6%
3 months	11.51%	98.7%
12 months	11.57%	80.5%

The confidence interval is calculated for a lognormal diffusion volatility equal to the volatility estimated by max-likelihood.

Annualizing a volatility - Confidence intervals

Simulated example of lognormal diffusion (with 20% volatility), annualized volatility over a 35-year sample: **19.88%** based on max-likelihood estimation.

Sampling	Annualized Vol	$\mathbb{P}$ int. symmetric confidence
1 day	19.96%	37.8%
1 week	19.81%	17.3%
2 week	19.76%	21.1%
1 month	19.29%	63.4%
3 months	19.33%	37.3%
12 months	20.24%	12.1%

Annualizing a volatility - Confidence intervals

S&P 500, annualized volatility 1927-2012: **18.99%** based on max-likelihood estimation.

Sampling	Annualized Vol	$\mathbb{P}$ int. symmetric trust
1 day	19.00%	8.3%
1 week	18.93%	13.2%
2 week	18.75%	41.0%
1 month	19.41%	50.3%
3 months	19.19%	15.0%
12 months	20.22%	42.3%

Annualizing a volatility - Confidence intervals

Brent, annualized volatility 1983-2012: **37.35%** based on max-likelihood estimation.

Sampling	Annualized Vol	$\mathbb{P}$ int. symmetric confidence
1 day	37.49%	34.4%
1 week	36.77%	60.3%
2 week	36.92%	34.3%
1 month	37.19%	9.1%
3 months	40.72%	83.0%
12 months	32.83%	63.5%

Annualizing a volatility - Non-overlapping

Using a sliding window (overlapping time steps) creates an autocorrelation bias in the serial dependence of volatilities, but ensures a larger number of data and therefore convergence (but due to autocorrelation, the gain is lower than if we had added as many non-overlapping returns) and increased stability. In the following example, the minimum time step of 1 day (and not less) and the very liquid security (EURUSD) seem to limit the autocorrelation bias.

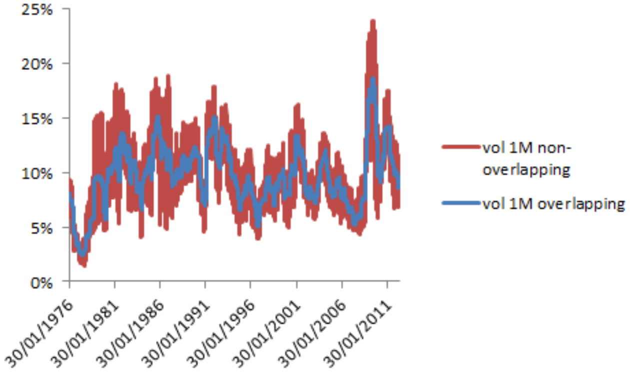
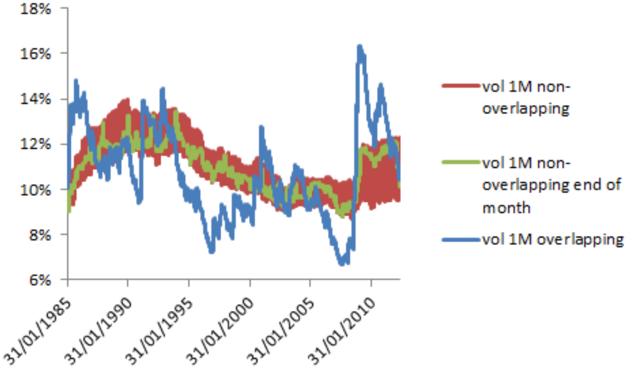


Figure: Annualized volatilities, taking 1 year of data, so 12 data in one case and 12×22 in the other.

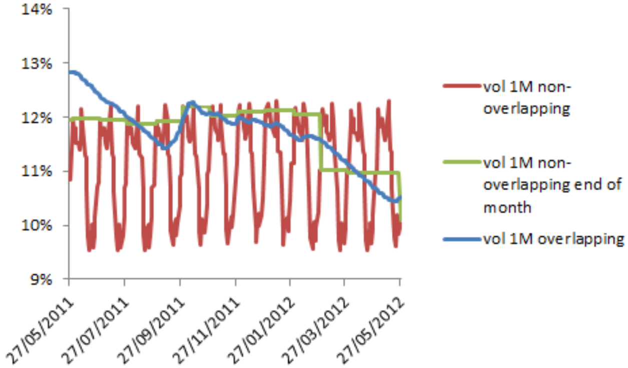
Annualizing a volatility - Non-overlapping

Increased data: 10-year sample, with exponential decay weighting, 8-month half-life.



Annualizing a volatility - Non-overlapping

Focus on the last year only.

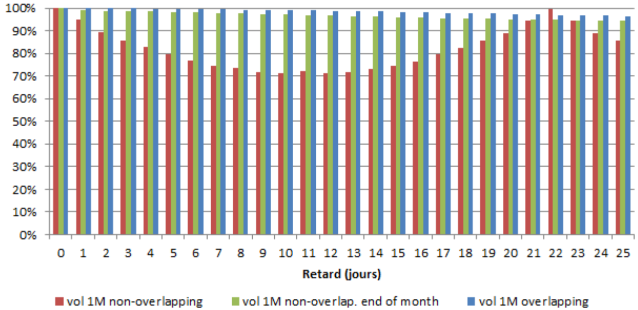


Annualizing a volatility - Non-overlapping

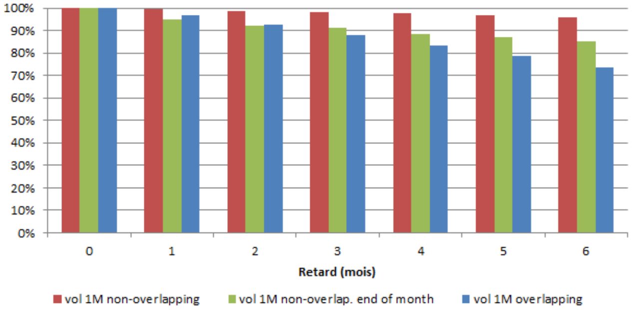
- Without overlap: high artificial **seasonality** (we change all the selected returns by shifting by one date, before falling back on them 22 days later) and high **variability**.
- Without overlap but with prices at the end of the month only: significant **jumps** at the end of the month (high weight allocated to the most recent observation: 8.3%).
- With overlap: not very different with 10 years and 1 year. More erratic than the two other coarse-scale estimators (but less than the red one at fine scale): bias (although not visible in 1 year of sample)? convergence for non-iid variables? ability to capture particularities?

Annualizing a volatility - Non-overlapping

Autocorrelogram of volatility series (10-year sample). A strong autocorrelation ensures a stable estimator. A decreasing then increasing autocorrelation indicates seasonality combined with variability.

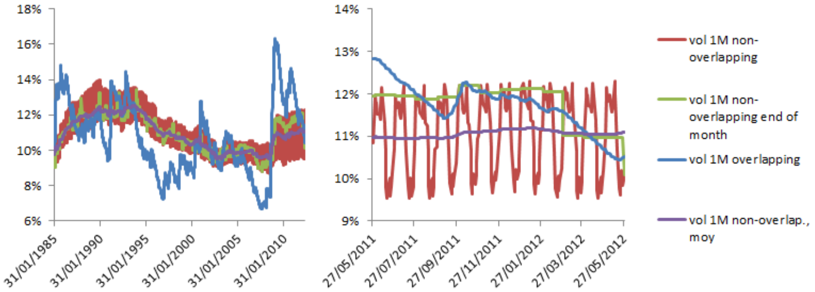


Annualizing a volatility - Non-overlapping



Annualizing a volatility - Non-overlapping

Alternative: average of 22 non-overlapping volatilities, in which the 22 volatilities each have a different observation time in a month.



Annualizing a volatility - Non-overlapping

Let H be the time horizon and T be the number of data.

- Let $\mathcal{J}(H)$ be the ratio of the non-overlapping variance (T/H observations) over $\hat{\sigma}^2$, the variance at a time step of 1.
- Let $\mathcal{M}(H)$ be the ratio of the overlapping variance over $\hat{\sigma}^2$.

For fixed H, the two volatility estimators converge to the same volatility.

Proposition (Lo and Mackinlay, 1988)

If the returns at the smallest time step are i.i.d. Gaussian, then, when $T \to +\infty$, we have:

- $\sqrt{T}(\mathcal{J}(H) - H)$ *converges in distribution to* $\mathcal{N}\left(0, 2H^2(H-1)\right)$.
- $\sqrt{T}(\mathcal{M}(H) - H)$ *converges in distribution to* $\mathcal{N}\left(0, \frac{2}{3}H(H-1)(2H-1)\right)$.

These approximations work well for small H and large T. Using overlapping data, the variance of the estimator vol is multiplied by $(2H-1)/3H$, or about $2/3$ for large enough H.

Annualizing a volatility - Non-overlapping

If H is large compared to T, we are interested in convergence for $T \to +\infty$ when the proportion $\delta = H/T$ is fixed, with $1/\delta \in \mathbb{N}$.

Proposition (Richardson and Stock, 1990)

For returns of the form $r_t = \mu + \varepsilon_t$, with ε_t of zero conditional expectation, and conditional heteroskedasticity (with finite mean conditional variance),[a]

- $\mathcal{J}(H)$ *converges in distribution to* $H\delta\chi^2(1/\delta)$.
- $\mathcal{M}(H)$ *converges in distribution to a function of a standard Brownian motion* W: $\frac{H}{\delta}\int_\delta^1 (W(\lambda) - W(\lambda - \delta))^2 d\lambda$.

[a] We limit ourselves to $\mu = 0$, but similar results exist without this assumption.

Indeed, up to a multiplicative factor, $\mathcal{J}(H)$ is equal to a number $1/\delta$ of sums of the type $(\sum_{t=kH+1}^{(k+1)H} r_t)^2$. H being infinite, the central limit theorem ensures that each sum is distributed according to a squared Gaussian distribution. The sum of these $1/\delta$ sums results in a χ^2 with $1/\delta$ degrees of freedom.

Annualizing a volatility - Non-overlapping

Remark

- *The expectation of $\mathcal{J}(H)$ is H and its variance $2H^2$.*
- *By expanding and inverting expectation and integral, we find for $\mathcal{M}(H)$ an expectation of $H(1-\delta)$. Moreover, by Jensen's inequality, the variance of the estimator is less than $H^2(1-\delta)(10-2\delta-2\delta^2+3\delta^3)/(3\delta^2)$. Numerically, this bound is less than $2H^2$ for $\delta \geq 68\%$.*

Warning

We can prefer overlapping returns to estimate a volatility if H is not too large compared to T, otherwise there is a bias. If we correct the bias, the estimator converges faster as soon as H is about 68% of T, which is not of great practical interest (and implies that $1/\delta$ is not an integer). The autocorrelation created will however be a problem if we use the series for other purposes: forecasts based on autocorrelations, seasonality, etc.

Annualizing a volatility - Non-overlapping

Example

If $H = 22$ and $T = 264$, we have:

- without overlapping, a volatility estimator of standard deviation $39.9\% \times \hat{\sigma}$ according to Lo and Mackinlay ($1.41 \times \hat{\sigma}$ for Richardson and Stock);
- with overlapping, a volatility estimator of standard deviation $32.2\% \times \hat{\sigma}$ ($20.79 \times \hat{\sigma}$ for Richardson and Stock! In addition, the bias is $8.3\% \times \hat{\sigma}$.).

In the Lo-Mackinlay approach, $T' = 1.53 \times T$ would be needed to have, without any overlapping, the same variance of the estimator as with overlapping on the sample of T dates.

Be careful about the validity of each estimator according to the values ??of H and T! For Richardson and Stock, if we multiply T by 10, the bias with overlapping decreases to $0.8\% \times \hat{\sigma}$ and the standard deviation of the estimator increases to $218 \times \hat{\sigma}$, but with $\delta = 0.8\%$ this estimator is no longer really valid.

Annualizing a volatility - HFT

- Instantaneous volatility models, absolute-return volatility:

$$\sigma_t = \left| \frac{y_t - \bar{y}}{\sigma_y} \right|,$$

  where (y) is the series, $\bar{y}$ its means, and σ_y its standard deviation.
- Realized volatility, Parkinson volatility.

Fractional Brownian motion

Figure: Google's tribute to Mandelbrot on November 20, 2020

Definition

A **fractional Brownian motion** *(fBm)* B_H *with Hurst exponent* $0 < H < 1$ *is a Gaussian process such that* $B_H(0) = 0$, $\mathbb{E}\{B_H(t)\} = 0$ *and*

$$\mathbb{E}\{(B_H(t) - B_H(s))^2\} = \sigma^2 |t-s|^{2H}.$$

Integral-based definition:

$$B_H(t) = \frac{1}{\Gamma\left(H + \frac{1}{2}\right)} \int_{-\infty}^{\infty} \left((t-s)_+^{H-1/2} - (-s)_+^{H-1/2}\right) dW_s.$$

Fractional Brownian motion

Proposition

- *$|B_H(t) - B_H(s)|$ is proportional to $|t - s|^H$, which corresponds to the definition of a function which is singular everywhere (it is therefore a homogeneous fractal) with (uniform) Hölder exponent H.*
- *The fBm is statistically self-similar since $B_H(t)$ and $c^{-H} B_H(ct)$ have the same distribution.*[a]
- *The fBm is not stationary because its covariance does not depend only on $(t - s)$:*

$$\mathbb{E}\{B_H(t)B_H(s)\} = \frac{\sigma^2}{2}(|t|^{2H} + |s|^{2H} - |t - s|^{2H}).$$

- *The increments of the fBm are stationary: $B_H(t) - B_H(s)$ has the same distribution as $B_H(t - s) - B_H(0)$.*
- *With probability 1, the graph of B_H has a fractal dimension equal to $2 - H$.*

[a] They are Gaussian, and we show that they have the same expectation (0) and the same variance.

Fractional Brownian motion

A classical Brownian motion has a Hurst exponent $H = 1/2$. The lower H, the more irregular the fBm. The fBm has properties depending on the value of H:

- If $H = 1/2$, the fBm is a classical Brownian motion, non-overlapping increments are **independent** to each other.
- If $H > 1/2$, the increments are **positively correlated** (erases singularities); this is a fractional primitive of a classical Brownian motion.
- If $H < 1/2$, the non-overlapping increments are **negatively correlated** (creates singularities); this is a fractional derivative of classical Brownian motion.

$$
\begin{aligned}
& Correl(B_H(n+1) - B_H(n), B_H(1) - B_H(0)) \\
= \quad & \mathbb{E}\{(B_H(n+1) - B_H(n))B_H(1)\} / \sqrt{\mathbb{E}\{(B_H(n+1) - B_H(n))^2\}\mathbb{E}\{B_H(1)^2\}} \\
= \quad & [\mathbb{E}\{B_H(n+1)B_H(1)\} - \mathbb{E}\{B_H(n)B_H(1)\}]/\sigma^2 \\
= \quad & [(n+1)^{2H} - 2n^{2H} + (n-1)^{2H}]/2 \\
\overset{\text{Taylor expansion}}{=} \quad & \frac{H(2H-1)}{n^{2(1-H)}} + \underset{n\to\infty}{o}\left(\frac{1}{n^{2(1-H)}}\right).
\end{aligned}
$$

Fractional Brownian motion

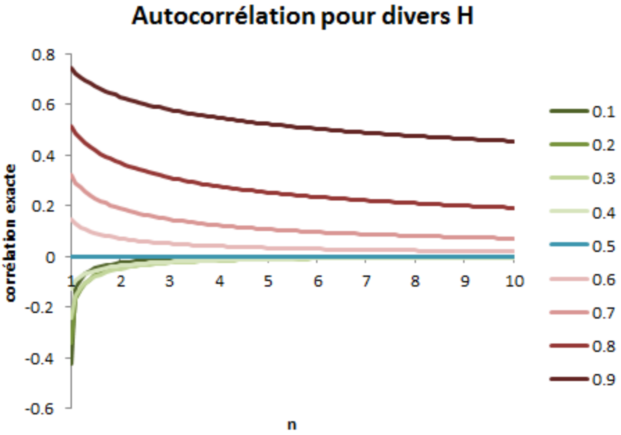
- The autocorrelation decreases faster with n, as H is small.
- The autocorrelation for $H > 1/2$ is stronger as H increases.
- For $H < 1/2$, the smallest H does not imply the strongest autocorrelation in absolute value. It depends on n.
- There is a long dependence for $H > 1/2$, which means that

$$\sum_{n=1}^{\infty} Correl(B_H(n+1) - B_H(n), B_H(1) - B_H(0)) = +\infty,$$

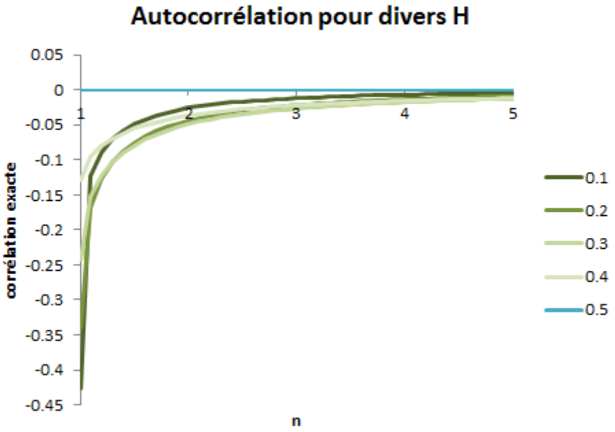
that is the autocorrelation becomes close to zero when n is large, but not negligible.

$$\begin{array}{ll} & \sum_{n=1}^{N} Correl(B_H(n+1) - B_H(n), B_H(1) - B_H(0)) \\ = & \frac{1}{2}\sum_{n=2}^{N+1} n^{2H} - \sum_{n=1}^{N+1} n^{2H} + \frac{1}{2}\sum_{n=0}^{N-1} n^{2H} \\ \sim & -\frac{1}{2} + HN^{2H-1} \overset{N\to+\infty}{\longrightarrow} +\infty. \end{array}$$

Fractional Brownian motion



Fractional Brownian motion



An estimate of H for a known Gaussian volatility

We define empirical absolute moments on our sample $[0, T]$ with $1/N$ spacings:

$$M_k = \frac{1}{NT}\sum_{i=1}^{NT}|X(i/N) - X((i-1)/N)|^k. \tag{2}$$

M_k is an estimate of $\mathbb{E}(|X(.) - X(. - 1/N)|^k)$, which value is:

$$\mathbb{E}(|X(.) - X(. - 1/N)|^k) = \frac{2^{k/2}\Gamma(\frac{k+1}{2})}{\Gamma(\frac{1}{2})}\sigma^k N^{-kH}. \tag{3}$$

It thus leads to the following estimate of H:

$$\check{H} = -\frac{\log(\sqrt{\pi}M_k/[2^{k/2}\Gamma(\frac{k+1}{2})\sigma^k])}{k\log(N)},$$

which converges towards H at the rate $O((\sqrt{\varepsilon_N}\log(N))^{-1})$, when N goes to infinity. Whatever k, it depends on σ.

A better estimate of H

From now $k = 2$ (estimator then stems from $\mathbb{E}\{(B_H(t) - B_H(s))^2\} = \sigma^2 |t-s|^{2H}$). We combine it with the same kind of statistic M_2' based on observations in the same window but with halved resolution:

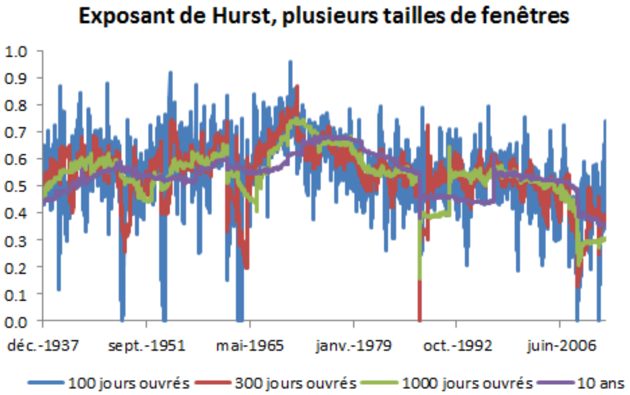
$$M_2' = \frac{2}{NT} \sum_{i=1}^{NT/2} |X(2i/N) - X(2(i-1)/N)|^2.$$

Then:

$$\hat{H} = \frac{1}{2} \log_2 \left(\frac{M_2'}{M_2} \right). \tag{4}$$

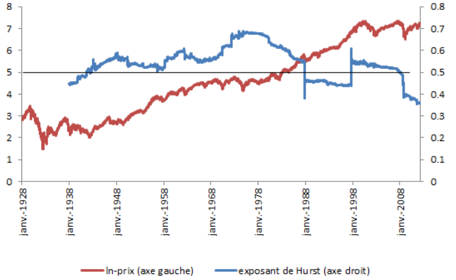
$\hat{H}$ converges almost surely toward H.

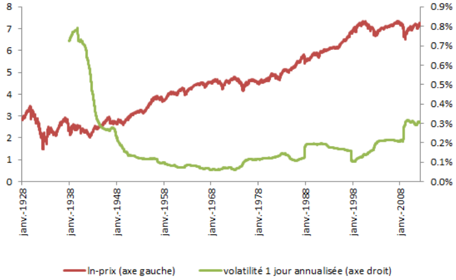
Fractional Brownian motion - Estimation



Estimate for the natural logarithm of the price for S&P 500 index, with a 10-year rolling window. The two peaks are due to the crisis of 10/19/1987 (worst day on a stock market after the crash of the Icelandic stock market in 2008, and which is at the origin of the establishment on Wall Street of circuit breakers to avoid too large falls).

Fractional Brownian motion - Estimation





Fractional Brownian motion - Estimation

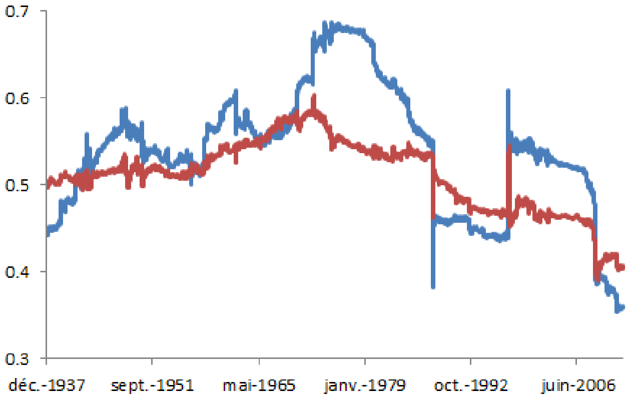


Figure: Estimator with 2-day and 1-day steps (in blue) vs. estimator with 10-day and 1-day steps (in red): changing these time steps allows for direct comparison of several scales and therefore provides additional information.

EMH vs FMH

Efficient market hypothesis (Eugène Fama, 1970) :

- Prices reflect available information, rational investors.
- Absence of arbitrage opportunities (martingales, random walk, etc.).
- Securities are substitutable for the same level of risk.

Fractal market hypothesis (Edgar Peters, 1991) :

- Noise affects the price (which is not equal to the fundamental value; according to Black (1986), the price evolves between half and double the value, which Bouchaud et alii (2017) explain by a positive autocorrelation in the short term and a mean reversion in the long term, which justifies the EMH in the long term only).
- There is (long) memory in the returns because the noise is not necessarily white noise. In Peters' theory, it is chaotic and therefore presents some regimes/periods with high predictability and others with low predictability.
- The presence of illiquidity and the adjustment of the investment horizon to market conditions (predictability, liquidity) prevent prices from being pure reflections of information.

Fractal dynamics (fBm) are a good model describing this theory.

EMH vs FMH

There are other criticisms against the EMH:

- **behavioural finance** (lack of rationality does not allow to reflect information);
- **noisy market hypothesis** (investors make decisions based on diversification goals, for liquidity or tax reasons and not only because of fundamental value);
- **adaptive market hypothesis** (Andrew Lo, 2004, tries to reconcile short-term EMH and behavioral biases, justified according to him by changing environment [number of competitors, possible profits, adaptability of investors]; consequently, the link between risk and return is unstable over time, there are arbitrage opportunities, any strategy will have good performances in a certain market environment and bad in another, which actually implies a good identification and adaptation to market conditions to succeed).

Nevertheless, a sort of fascination with the EMH persists... and consists of denaturing of fractal models to make them verify the EMH.

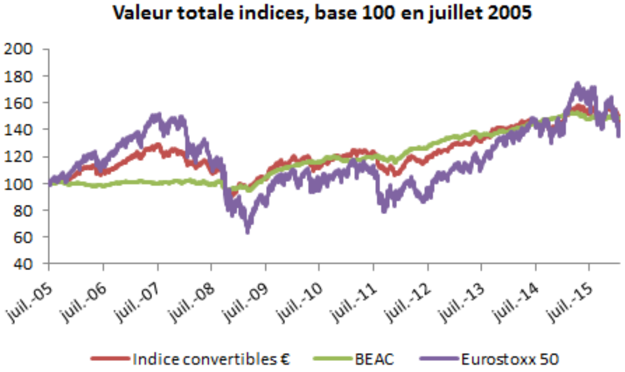
Risk measures

What does this have to do with risk measures?

- To calculate VaR, we can estimate a dynamic and make projections to the risk horizon. Fractal dynamics, more general than Brownian dynamics, offer a more precise estimate.
- The Hurst exponent is an indicator of a certain type of risk. It is interesting to compare the singularity spectra for different securities.
- Allows to annualize volatilities with the right annualization factor.

Risk measures - Hurst exponent

Example of comparison of singularities for the value of a convertible index, a stock index, and a corporate bond index, considered as multifractional Brownian motions.



Risk measures - Hurst exponent

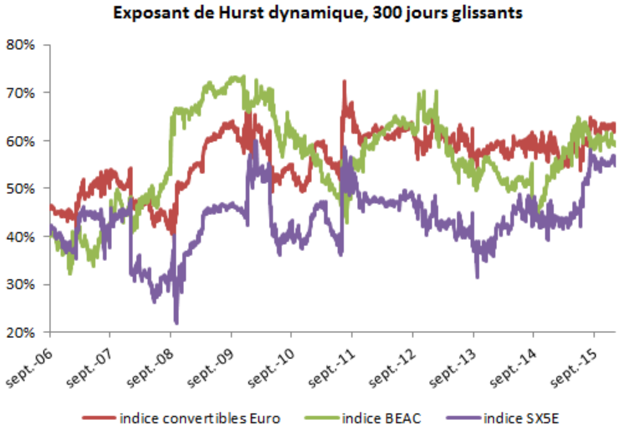
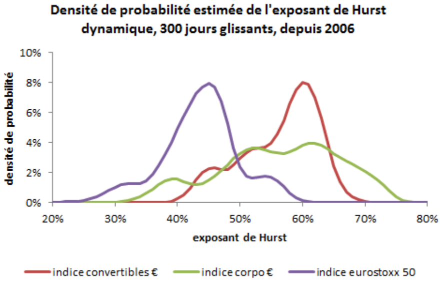


Figure: Estimator with 2 day and 1 day steps.

Risk measures - Hurst exponent



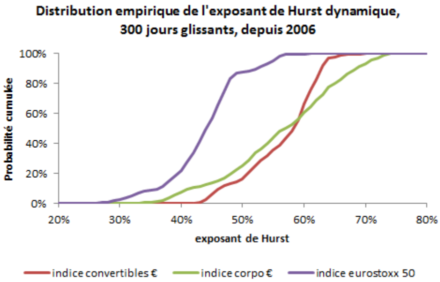


Figure: Estimated pdf with Gaussian kernel, empirical cdf. Stocks have more singularities, the spectrum of singularities of convertibles is narrower than that of bonds.

Risk measures - Hurst exponent

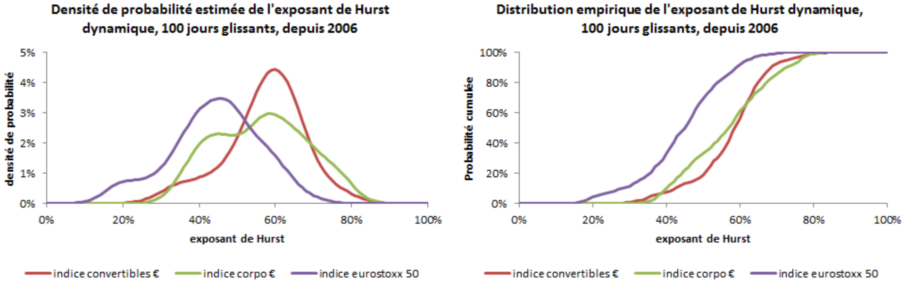


Figure: We reduce the window size from 300 to 100: more spread out distributions but identical interpretation.

Why?

Decomposition of a signal in several components of various frequencies. Useful in engineering but also in finance for:

- finding patterns at given frequencies, detect regime sitching;
- getting rid of noise (noise is predominant in price series);
- parsimonious representation and compression (jpeg, mp3...).

But also, in finance: making predictions (one at each scale and then aggregating).

Why not something else?

From the most smoothing method to the one respecting local behaviours of time series: Fourier transform, then windowed Fourier transform, and finally wavelets.

Fourier transform

$$\hat{z}^{\varepsilon}(\omega) = \int_{\mathbb{R}} z^{\varepsilon}(x) e^{-2i\pi\omega x} dx.$$

- Inverse transform (if $\hat{z}^{\varepsilon}$ integrable): $f(x) = \sum_{\omega\in\mathbb{N}} \hat{z}^{\varepsilon}(\omega) e^{2i\pi\omega x}$ almost everywhere.
- Riemann-Lebesgue lemma: $\hat{z}^{\varepsilon}(\omega) \to 0$ as $|\omega| \to 0$.
- Parseval-Plancherel theorem: $\int_{\mathbb{R}} |z^{\varepsilon}(x)|^2 = \sum_{\omega\in\mathbb{N}} |\hat{z}^{\varepsilon}(\omega)|^2$.
- For a finite discrete-time signal, the coefficient is $\frac{1}{N}\sum_{n=0}^{N-1} z(n) e^{-2i\pi\omega n/N}$ for the same inverse transform than in continuous time.

Fourier transform

Proposition

Effect of the Fourier transform of:

- *a linear equation:* $f(x) + \lambda g(x)$ *is transformed in* $\hat{f}(\omega) + \lambda g(\omega)$*;*
- *contraction:* $f(ax)$ *is transformed in* $\hat{f}(\omega/a)/|a|$*;*
- *translation:* $f(x+T)$ *is transformed in* $\hat{f}(\omega)e^{2i\pi\omega T}$*;*
- *convolution product:* $(f \star g)(x)$ *is transformed in* $\hat{f}(\omega)\hat{g}(\omega)$*;*
- *differentiation :* $f'(x)$ *is transformed in* $2i\pi\omega\hat{f}(\omega)$*;*
- *differentiation in the frequency domain:* $d\hat{f}(\omega)/d\omega$ *is the transform of* $-2i\pi x f(x)$*.*

Windowed Fourier transform

$$\hat{z}^{\varepsilon}(\omega, k) = \int_{\mathbb{R}} z^{\varepsilon}(x) e^{-2i\pi\omega x} \delta_{[k-\Delta k, k+\Delta k]}(x) dx.$$

Wavelet transform

$$\hat{z}^{\varepsilon}_{j,k} = \int_{\mathbb{R}} z^{\varepsilon}(x)\psi_{j,k}(x)dx.$$

- The wavelet function (dilatation and translation of the mother wavelet): $\psi_{j,k} : t \in \mathbb{R} \mapsto 2^{j/2}\Psi\left(2^j t - k\right)$.
- Empirical wavelet coefficients: $\langle z^{\varepsilon}, \psi_{j,k} V\rangle = \sum_{n=1}^{N} z^{\varepsilon}(u_n)\psi_{j,k}(u_n)V_n$.

Prophet is a procedure for ***forecasting time series data*** *based on an* ***additive model*** *where non-linear trends are fit with yearly, weekly, and daily* ***seasonality, plus holiday effects****.*

Accurate and fast.

Prophet is used in many applications across Facebook for producing reliable forecasts for planning and goal setting. We've found it to perform better than any other approach in the majority of cases. We fit models in Stan so that you get forecasts in just a few seconds.

Fully automatic.

Get a reasonable forecast on messy data with no manual effort. Prophet is robust to outliers, missing data, and dramatic changes in your time series.

Figure: Application of Fourier analysis to time series predictions by Facebook (source: medium.com, The math of prophet).

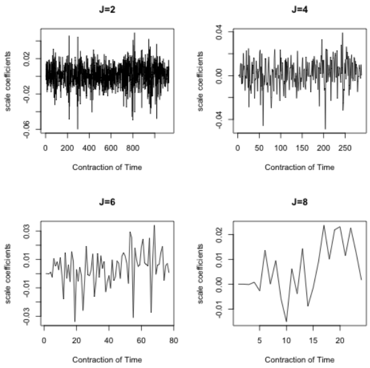


Figure: Example of projection on a wavelet basis. Scale coefficients of a signal for different resolution dimensions.

How to define a wavelet?

We choose a mother wavelet as a function Ψ from the space $\mathcal{L}^1(\mathbb{R}) \cap \mathcal{L}^2(\mathbb{R})$:

$$\begin{cases} \int_{\mathbb{R}} |\Psi(t)| dt < \infty \\ \int_{\mathbb{R}} |\Psi(t)|^2 dt < \infty. \end{cases}$$

Moreover, this function is chosen with zero mean and square norm one:

$$\begin{cases} \int_{\mathbb{R}} \Psi(t) dt = 0 \\ \int_{\mathbb{R}} |\Psi(t)|^2 dt = 1. \end{cases}$$

From a mother wavelet, one can derive wavelet functions:

$$\psi_{j,k} : t \in \mathbb{R} \mapsto 2^{j/2} \Psi\left(2^j t - k\right),$$

where $j \in \mathbb{R}$ is the resolution level and $k \in \mathbb{R}$ is the translation parameter. For a discrete wavelet analysis, we restrict to integer parameters: $j \in \mathbb{Z}$ and $k \in \mathbb{Z}$.

The wavelet is said to be orthogonal if the corresponding discrete wavelet basis is orthogonal:

$$\forall (j, j', k, k') \in \mathbb{R}^4, \ \int_{\mathbb{R}} \psi_{j,k}(t)\psi_{j',k'}(t) = \mathbf{1}_{j=j'}\mathbf{1}_{k=k'}.$$

In our work, we are mostly interested in orthogonal wavelets defined on a compact support, the Daubechies wavelets. The number of vanishing moments allows to define various types of Daubechies mother wavelets: it is defined as the maximal integer m such that

$$\int_{\mathbb{R}} t^m \Psi(t) dt = 0.$$

For example, the Haar wavelet corresponds to the Daubechies wavelet with zero vanishing moments:

$$\Psi^{Haar} : t \in \mathbb{R} \mapsto \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t).$$

More generally, for Daubechies wavelets, we define first a scaling function[1] ϕ by successive dyadic interpolations in a cascade algorithm:

$$\phi(x) = \sum_{k=0}^{2N-1} a_k \phi(2x - k),$$

for a well-chosen sequence of coefficients a_k and N the number of vanishing moments. Then, the mother wavelet can also be interpolated using the scaling function:

$$\Psi(x) = \sum_{k=0}^{2N-1} (-1)^k a_{1-k} \phi(2x - k).$$

[1] Also called father wavelet.

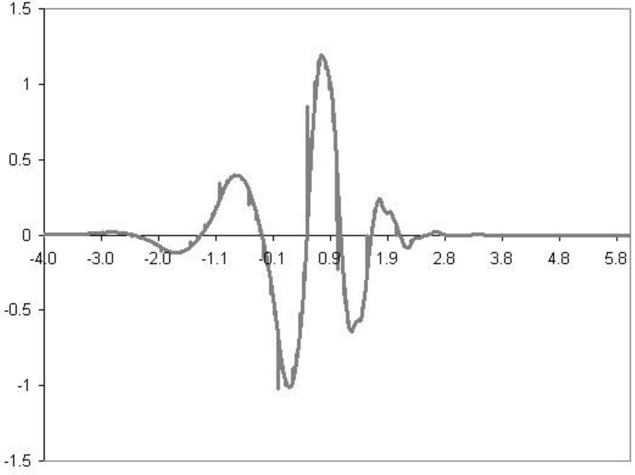


Figure: Daubechies mother wavelet with 5 vanishing moments (more parsimonious representation when this number increases).

Wavelet transform

Let z be a squared-integrable function: $z \in \mathcal{L}^2(\mathbb{R})$. It is the input signal. We decompose it into a discrete dyadic real wavelet basis. Such a wavelet basis is a countable subset of the function space $\mathcal{L}^2(\mathbb{R})$.
The discrete wavelet analysis consists in decomposing the function z in the discrete wavelet basis. For $j, k \in \mathbb{Z}$, the projection of z on the subspace generated by the vector basis $\psi_{j,k}$ is the wavelet coefficient $\hat{z}_{j,k}$ of resolution level j and translation parameter k:

$$\hat{z}_{j,k} = \int_{\mathbb{R}} z(t)\psi_{j,k}(t).$$

In most cases, we are only able to calculate an empirical wavelet coefficient, that is with an empirical integral defined on a discrete grid $u_0 < u_1 < ... < u_N$:

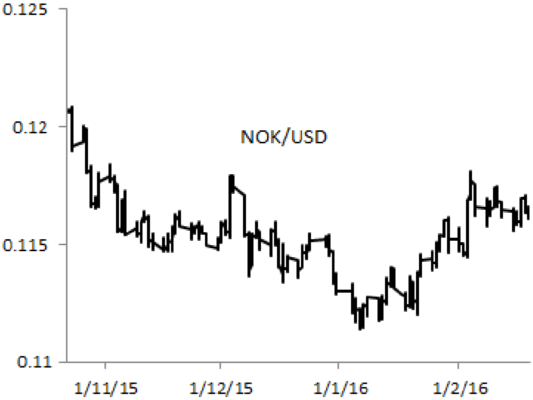
$$\hat{z}_{j,k}^{empirical} = \sum_{n=1}^{N} z(u_n)\psi_{j,k}(u_n)(u_n - u_{n-1}).$$

Interpreting coefficients

- Autocorrelogram: autocorrelation (Y axis) as a function of the number of lags (X axis); peaks indicate a kind of average cycle.
- 2D Spectrogram: Fourier coefficients (Y axis) as a function of the frequency (X axis); peaks also indicate seasonality (but not average: projection of a signal on a seasonality, therefore ignores what is related to other frequencies), but be careful, because the sum of sinusoids of different frequencies can lead to signals that are not very seasonal.
- 3D Spectrogram: windowed Fourier coefficients (represented by color), therefore adding a time component (X axis) to the frequency (Y axis): this is a dynamic spectrogram and can therefore depict seasonality variations if they have a period smaller than the size of the window.
- Scalogram: wavelet coefficients (represented by color) as a function of time (X axis) and scale (Y axis); not really seasonality (a single oscillation of the wavelet, unlike Fourier by windows), but projection of the signal at different time scales (monthly, weekly, daily, hourly returns, etc.); high coefficients rather give an idea of breakpoints (loss of differentiability of the signal).

Interpreting coefficients

Example of autocorrelogram, 2D- and 3D-spectrogram, and scalogram for the NOKUSD exchange rate, hourly observations between October 22, 2015 and February 19, 2016.



Interpreting coefficients

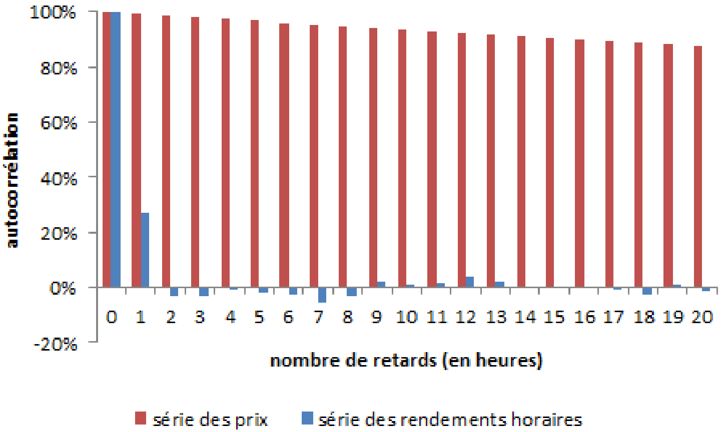


Figure: Simple autocorrelogram (partial autocorrelogram if we replace the correlation between X_t and X_{t-h} by that between $X_t - \mathcal{X}_{t,t-1}$ and $X_{t-h} - \mathcal{X}_{t-h,t-1}$, where $\mathcal{X}_{t,t-1}$ [resp. $\mathcal{X}_{t-h,t-1}$] is the linear regression of X_t [resp. X_{t-h}] on $X_{t-1}, \dots, X_{t-h+1}$) of hourly prices and returns of NOKUSD.

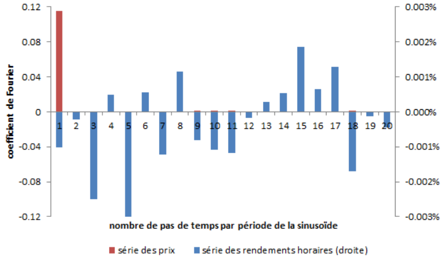
Interpreting coefficients

- The power spectral density of x_t associates with a frequency ω the square of the modulus of the Fourier coefficient $\hat{x}(\omega)$. We denote $\Gamma(\omega) = |\hat{X}(\omega)|^2$.
- By the **Wiener-Khintchine theorem**, $\Gamma(\omega)$ is the Fourier transform of the autocorrelation $\gamma(h)$ at the frequency ω.
- The Fourier transform of the autocorrelogram is called a **periodogram**. In a realistic framework where we do not integrate the signals over the entire $\mathbb{R}$, it is (only) an estimator of the power spectral density.
- Indeed, by denoting $x^\star$ the complex conjugate:

$$\begin{array}{rcl} \hat{\gamma}(\omega) & = & \int_{\mathbb{R}} \gamma(h) e^{-i\omega h} dh \\ & = & \int_{\mathbb{R}} \int_{\mathbb{R}} x^\star(t) x(t+h) e^{-i\omega h} dh dt \\ & = & \int_{\mathbb{R}} (\int_{\mathbb{R}} x(u) e^{-i\omega u} du) x^\star(t) e^{+i\omega t} dt \\ & = & \hat{x}(\omega) \int_{\mathbb{R}} x^\star(-v) e^{-i\omega v} dv = \hat{x}(\omega)\hat{x}(\omega)^\star = \Gamma(\omega). \end{array}$$

with two successive substitutions.

Interpreting coefficients



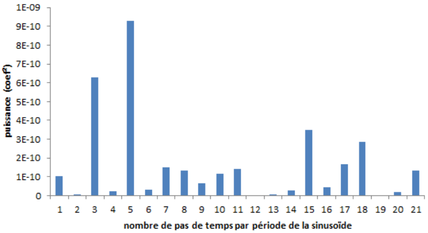


Figure: Spectrum for NOKUSD. White noise has a flat spectrum: all frequencies contribute equally to the signal. Peaks correspond to seasonality.

Interpreting coefficients

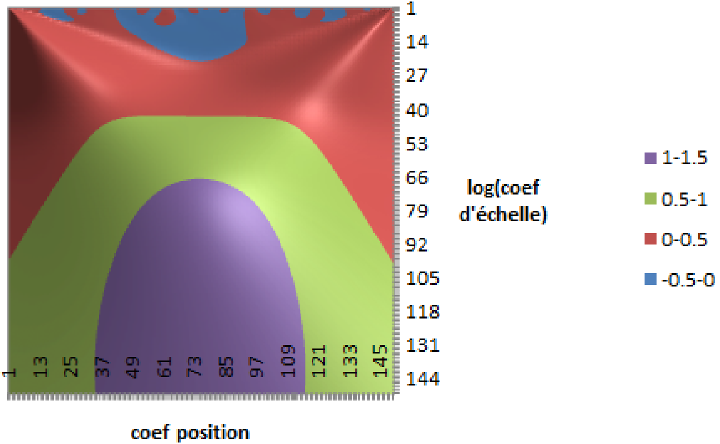


Figure: Scalogram on the first 150 dates of NOKUSD (about 10% of the sample).

Interpreting coefficients

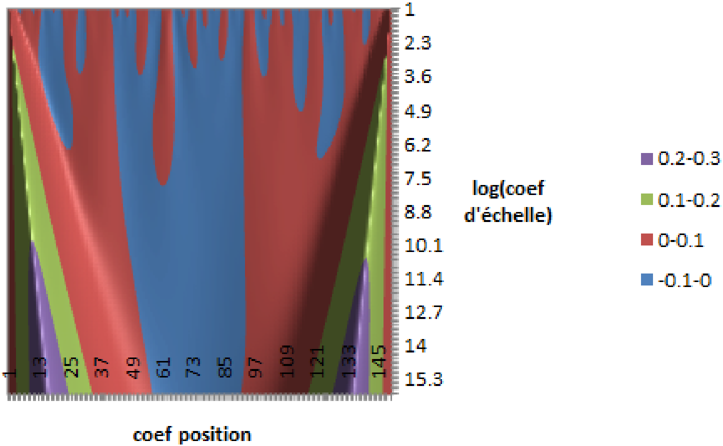


Figure: Scalogram, detail. The maxima lines correspond to singularities.

Interpreting coefficients

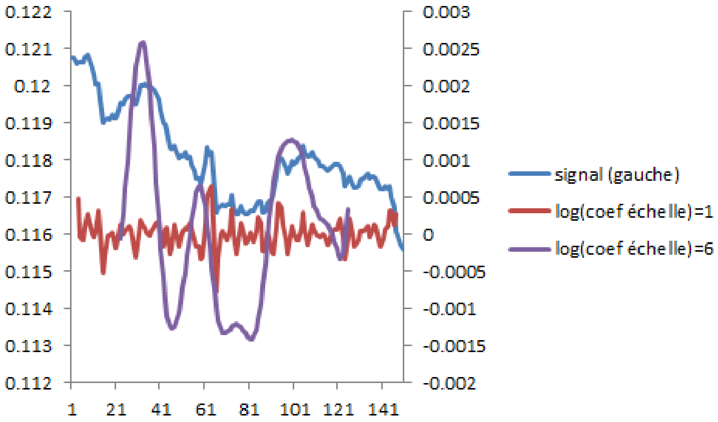


Figure: Scalogram, for two selected scales (edge ??effects eliminated) and NOKUSD signal.

Inverse transform

The wavelet inverse transform consists in reconstructing the observed signal, z, according to each translation and resolution level. The reconstructed signal for a given resolution level $j \in \mathbb{Z}$ is called the detail signal:

$$D_j : t \in \mathbb{R} \mapsto \sum_{k \in \mathbb{Z}} \hat{z}_{j,k} \psi_{j,k}(t).$$

The whole reconstructed signal is the sum of all the details:

$$z = \sum_{j \in \mathbb{Z}} D_j,$$

in which the details with a higher j correspond to a finer resolution. The finer details can also be reduced to a finite sum thanks to the scaling function:

$$z = \sum_{j \leq j_0} D_j + \sum_{k \in \mathbb{Z}} c_{j_0,k} \phi_{j_0,k},$$

where

$$c_{j_0,k} = \int_{\mathbb{R}} z(t) \phi_{j_0,k}(t) dt.$$

What limitations?

- Limitations on the number of data? Only for fast algorithms.
- The interpretation of the coefficients is different from that of the Fourier coefficients.
- Border effects (reduced by mirroring, but according to an arbitrary choice).
- Does not offer a forecasting model (on the other hand, can allow to estimate predictive models of the kind $y_{t+1} = z(y_t) + \varepsilon_t$).

What limitations?

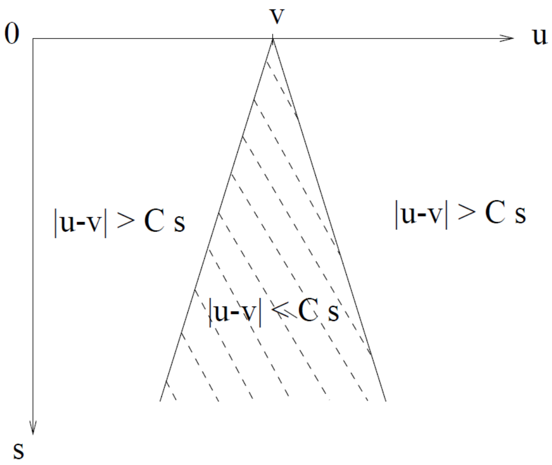


Figure 6.2: The cone of influence of an abscissa v consists of the scale-space points (u, s) for which the support of $\psi_{u,s}$ intersects $t = v$.

Figure: Impact across the scales of a border effect: it is contained in the cone of influence. Source: Stéphane Mallat

Multiresolution analysis

The orthogonal projection, $P_{V_j} f$, onto the set of scaling functions, $(\phi_{j,k})_{k\in\mathbb{Z}}$, of resolution 2^{-j} provides an approximation of a signal f at scale 2^j. The residual of the projection is the decomposition on wavelets, $(\psi_{i,k})_{i\leq j,k\in\mathbb{Z}}$, of scale less than or equal to 2^j. This is called **multiresolution analysis**.

V_j is the set of possible approximations at scale 2^j. $(\phi_{j,k})_{k\in\mathbb{Z}}$ is a basis for it. $(\psi_{j,k})_{k\in\mathbb{Z}}$ is a basis of W_j defined by the direct orthogonal sum (any element of V_{j-1} is written as the unique sum of two orthogonal elements of V_j and W_j):

$$V_{j-1} = V_j \oplus W_j.$$

Other properties:

- $V_{j+1} \subset V_j$;
- $\lim\limits_{j\to+\infty} V_j = \bigcap_{j=-\infty}^{+\infty} V_j = \{0\}$;
- $\lim\limits_{j\to-\infty} V_j = \text{Adhérence}\left(\bigcup_{j=-\infty}^{+\infty} V_j\right) = L^2(\mathbb{R})$.

Multiresolution analysis

Useful in finance to calculate the correlation between two stock price returns, without the noise represented by small-scale variations. We will see later that the detail, i.e. what is contained in small scales, can also reveal singularities and not just noise; it will then be necessary to remove the noise in another way than by truncating the wavelet series as is done by the projection on V_j.

Why not simply returns from various horizons?

Calculating correlations (or volatilities) at different scales is already within our reach without wavelets by considering monthly, daily, returns, etc. But :

- How to manage interval overlaps? (what is the monthly return in the middle of the month? see the chapter introduction) This is difficult without a basis.
- How to have a smooth approximation, without the jumps of the low-scale returns?
- How to define the residual or what is specific to the addition of information by changing scale (W_j), without wavelets?

Wavelets answer all these questions for us.

Multiresolution analysis

The Haar wavelet allows an intuitive understanding of wavelets:

- It is a Daubechies wavelet with a single zero moment, so very rudimentary to approximate regular functions.
- The scaling function is $\phi = \mathbf{1}_{[0,1)}$ and the wavelet:

$$\psi(t) = \begin{cases} -1 & \text{if } 0 \leq t < 1/2 \\ 1 & \text{if } 1/2 \leq t < 1 \\ 0 & \text{otherwise.} \end{cases}$$

- The decomposition on ϕ and ψ of a function $a\mathbf{1}_{[0,1/2]} + b\mathbf{1}_{[1/2,1]}$ allows us to understand multiresolution analysis.
- The approximation of daily returns r_t at the 2^j scale, i.e. $\sum_{k\in\mathbb{Z}} \langle r, \phi_{j,k}\rangle \phi_{j,k}(t)$, is therefore based on returns of horizon 2^j calculated each day, with overlap.
- The detail is easily calculated on a date as a difference in returns. It is zero if the returns follow a trend, non-zero if they oscillate around an average.

Multiresolution covariance

1. We calculate the scale coefficients for several adjacent discrete scales $j \in J$ for two series, by treating or eliminating the border effects: $(c^1_{j,k})_{j\in J, k\in[0,T]}$ and $(c^2_{j,k})_{j\in J, k\in[0,T]}$
2. We calculate the covariance of the scale coefficients for a given scale j:

$$Cov(j) = \frac{1}{T}\sum_{k=1}^{T}\left(c^1_{j,k} - \frac{1}{T}\sum_{l=1}^{T} c^1_{j,l}\right)\left(c^2_{j,k} - \frac{1}{T}\sum_{l=1}^{T} c^2_{j,l}\right).$$

Remark

- *If we can calculate covariances, we can also calculate variances.*
- *There is a bias in these covariances since their amplitude decreases with the scale. We can correct it:*
  - *by normalizing with $2^{j/2}$ (which amounts to calculating the covariance of the reconstructed series at this scale $\sum_{k=1}^{T} c^1_{j,k}\psi_{j,k}$ and $\sum_{k=1}^{T} c^2_{j,k}\psi_{j,k}$);*
  - *by limiting ourselves to correlations, and therefore by normalizing with the variances.*

Multiresolution covariance and Epps effect

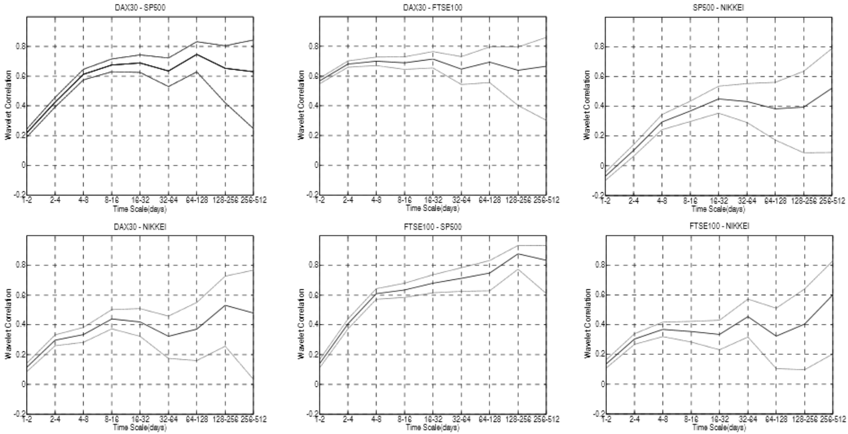


Figure: Wavelet correl, May 1988-January 2007, with 95% confidence intervals. Source: Mikko Ranta.

Multiresolution covariance

Table 3. Correlation diversification ranking on four different time scales. Time scales used are a day(scale 1, 1-2 days), a week(scale 3, 4-8 days), a month(scale 5, 16-32 days) and a year(scale 9, 256-512 days). Ranking is formed on the basis of portfolio diversification efficiency. Correlations are in the parentheses. 5% significance level is marked with *.

Rank	Day	Week	Month	Year
1	SP500 - NIKKEI (-0,071*)	SP500 - NIKKEI (0,294*)	FTSE100 - NIKKEI (0,334*)	DAX30 - NIKKEI (0,480*)
2	DAX30 - NIKKEI (0,114*)	DAX30 - NIKKEI (0,334*)	DAX30 - NIKKEI (0,420*)	SP500 - NIKKEI (0,521*)
3	FTSE100 - NIKKEI (0,135*)	FTSE100 - NIKKEI (0,369*)	SP500 - NIKKEI (0,449*)	FTSE100 - NIKKEI (0,602*)
4	FTSE100 - SP500 (0,139*)	FTSE100 - SP500 (0,609*)	FTSE100 - SP500 (0,681*)	DAX30 - SP500 (0,630*)
5	DAX30 - SP500 (0,218*)	DAX30 - SP500 (0,612*)	DAX30 - SP500 (0,689*)	DAX30 - FTSE100 (0,665*)
6	DAX30 - FTSE100 (0,567*)	DAX30 - FTSE100 (0,701*)	DAX30 - FTSE100 (0,714*)	FTSE100 - SP500 (0,834*)

Figure: Ideas for strategy based on the time horizon. Source: Mikko Ranta.

Multiresolution covariance

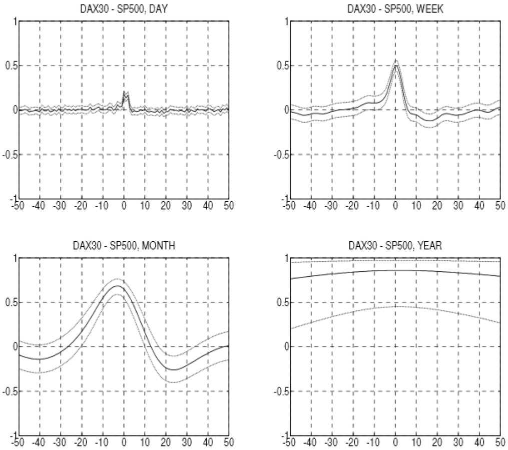


Figure: Day lag on the abscissa, correlation at several scales on the ordinate. Skew on the left: leading DAX30. Skew on the right: leading SP500. On the daily scale, SP500 is the leader, but on the monthly scale it is the DAX30. Source: Mikko Ranta.

Multiresolution covariance

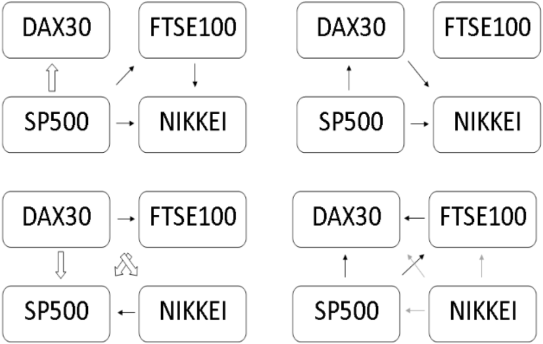


Figure 5. Volatility spillover flow charts for four different time scales. Flows have been visually estimated using the wavelet cross-correlation functions. From the upper-left corner the time scales are a day(1-2 days), a week(4-8 days), a month(16-32 days) and a year(256-512 days). Black arrows were statistically significant at a 5% level, while grey arrows were not. Big arrows describe a very strong volatility spillover.

Figure: Lead-lag effect at multiple scales. Ideas for strategies? Source: Mikko Ranta.

Stylized facts of volatility

- Well-known facts to econometricians: heteroskedasticity (GARCH), fat tails, persistence (FIGARCH);
- Well-known facts to econophysicists: fractality (fBm), multifractality (various scaling rules: MMAR, MSM);
- A new fact concerning multifrequency aspects: the **volatility cascade**.

Definition

The **volatility cascade** *reflects the asymmetry of volatility propagation: the volatility calculated for a coarse sample has a stronger influence on that calculated for a fine sample, than the reverse.*

Possible economic explanation: Long-term volatility can influence the investment decisions of all traders (long and short term), while short-term volatility does not influence the decisions of long-term traders.

Stylized facts of volatility – heterogeneous market

Heterogeneous market hypothesis (Müller, Dacorogna, Davé, Pictet, Olsen, Ward, 1995): agents do not interpret decisions in the same way and do not react in unison.
Characterized by:

- different investment horizons;
- positive correlation (stylized fact) between volatility and number of investors; for a homogeneous market, an increase in the number of investors increases liquidity and reduces volatility (traders all want the same price);
- geographical differences (not all traders sleep at the same time...).

Cascade of heterogeneous components of realized volatility

Three components of volatilities, e.g.: daily, weekly, monthly (Corsi). Gives HAR-RV model (heterogeneous AR model in the realized volatility [realized volatility is calculated each day from intraday data: good proxy for 1-day volatility without having to average with previous days; but what horizon to take for intraday data?]):

$$\log \sigma_{t+d}^{(d)} = c + \beta^{(d)} \log \sigma_t^{(d)} + \beta^{(w)} \log \sigma_t^{(w)} + \beta^{(m)} \log \sigma_t^{(m)} + \varepsilon_{t+d}.$$

This model is consistent with the following cascade:

- monthly forecast from monthly volatility: $\log \sigma_{t+m}^{(m)} = c^{(m)} + b^{(m)} \log \sigma_t^{(m)} + \varepsilon_{t+m}^{(m)}$;
- weekly forecast from past weekly volatility and monthly volatility forecast: $\log \sigma_{t+w}^{(w)} = c^{(w)} + b^{(w)} \log \sigma_t^{(w)} + a^{(w)} \mathbb{E}_t[\log \sigma_{t+m}^{(m)}] + \varepsilon_{t+w}^{(w)}$;
- daily forecast from past daily volatility and weekly volatility forecast:$\log \sigma_{t+d}^{(d)} = c^{(d)} + b^{(d)} \log \sigma_t^{(d)} + a^{(d)} \mathbb{E}_t[\log \sigma_{t+w}^{(w)}] + \varepsilon_{t+d}^{(d)}$.

Simulated HAR-RV

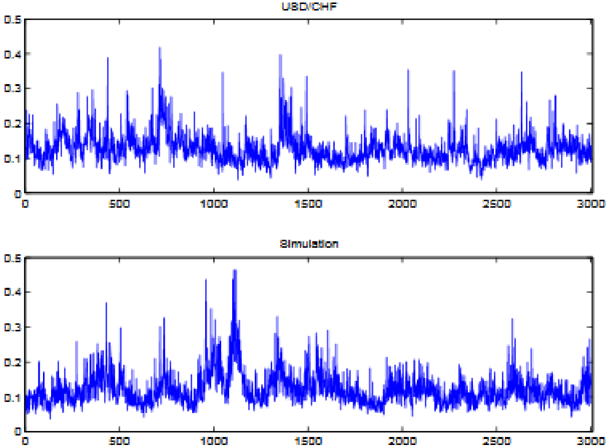
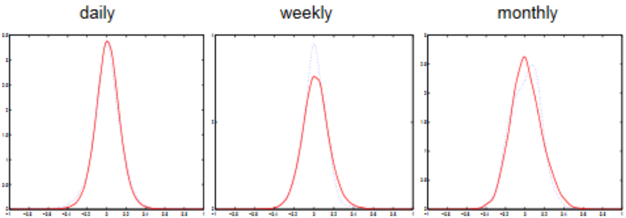


Figure: Source: Fulvio Corsi.

Simulated HAR-RV



Kurtosis	daily returns	weekly returns	monthly returns
USD/CHF	4.72	3.78	3.04
HAR-RV model	4.89	3.90	3.50

Figure: Source: Fulvio Corsi.

Optimal sampling

Consequence of this volatility forecasting model (Bandi, Russell, 2007): possibility of determining HF horizon (often arbitrarily taken equal to 5 minutes) minimizing conditional quadratic error of forecasting realized volatility. Below this horizon, microstructure noise disrupts the signal too much; beyond that, part of the information is neglected (and therefore it allows a less good volatility forecast).
Dynamics:

- classical diffusion (unobserved): $\ln(x_t) = \ln(x_0) + \int_0^t \sigma_s dW_s$, with possibly an additional trend;
- price observed in discrete time perturbed by centered i.i.d. microstructure noise: $\ln(p_{t_i}) = \ln(x_{t_i}) + \varepsilon_i$.

Optimal sampling

This is an optimization problem... but we can define a first approximation of the optimal frequency (inverse of the horizon, number of observations in a day), $M^\star$ (Bandi, Russell):

$$M^\star \approx \left(\frac{hQ}{(\mathbb{E}[\varepsilon^2])^2}\right)^{1/3},$$

where:

- ε is the microstructure noise;
- h is the time unit in seconds (1 trading day $=$ 23,400 seconds);
- $Q = \int_0^h \sigma_s^4 ds$ is the quarticity, estimated by averaging yields to the power of 4, for a sampling coarse enough not to be too exposed to microstructure noise.

In this formula, $M^\star$ can be seen as a signal-to-noise ratio: the stronger the signal (σ^2), the higher the sampling frequency should be.

Optimal sampling

Can be done for several volatility forecasting models (for HAR-RV, example Guerrier (2013), who finds approximately 9 minutes for Chesapeake Energy Corp stock). Below, approximately 2 minutes for IBM (Bandi, Russell), for other volatility models:

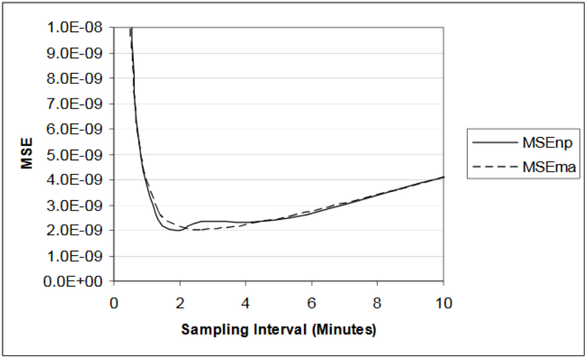


Figure 4. For IBM, we plot the nonparametric MSE for the case of dependent noise (in Theorem 5) and the MA(1) MSE (in Theorem 3).

Figure: Source: Bandi, Russell.